Model Theory

SWAT+ Documentation

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Overview

These pages describe how SWAT+ represents the processes that move water, sediment, nutrients and other constituents through a watershed. Each page covers one group of processes: the equations the model solves, the input files and columns that control them, and where SWAT+ 62.0.1 differs from the SWAT 2009 theoretical documentation.

The landscape processes run once per day for every hydrologic response unit (HRU): weather, snow, evapotranspiration, surface runoff, soil water, erosion, plant growth and the soil nutrient cycles. Their results leave the HRU as surface runoff, lateral flow, tile flow and percolation, and pass to the routing objects: channels, aquifers, reservoirs and wetlands, and recall points. Decision tables apply management and reservoir release rules across both groups.

Pages

Group	Page	Contents
Landscape water	Overview and water balance	HRU water balance, the order of daily calculations, where each term is reported
Landscape water	Climate and weather	Measured and generated weather, day length, sub-daily precipitation, elevation lapse, potential evapotranspiration

Group	Page	Contents
Landscape water	Hydrology	Canopy interception, surface runoff, lateral flow, percolation, tile flow, lag to the channel
Landscape water	Soil water and temperature	Soil layers, water retention, initial soil water, soil evaporation, soil temperature
Landscape water	Erosion and sediment	Peak runoff rate, MUSLE, sub-daily erosion, sediment in lateral flow, urban loads, filter strips, grassed waterways and user BMPs
Routing	Channels	Channel routing, sediment transport and in-stream water quality
Routing	Aquifers	Aquifer objects and the gwflow groundwater module
Routing	Reservoirs and wetlands	Reservoir water balance and release rules, wetlands
Routing	Recall and point sources	Recall time series and point source loads
Routing	Decision tables	Condition evaluation and actions
Plants and chemistry	Plant growth and land use management	Heat units, biomass, leaf area, stresses, harvest, scheduled management
Plants and chemistry	Nutrient cycling	Soil nitrogen and phosphorus pools, transformations, losses; algae, CBOD and oxygen in runoff; septic systems
Plants and chemistry	Carbon	Static and CENTURY-based soil carbon
Plants and chemistry	Pesticides	Application, partitioning, decay and transport
Plants and chemistry	Salts and constituents	Salt ion chemistry, other constituents, pathogens and heavy metals

SWAT 2009 processes not in SWAT+ 62.0.1

The SWAT 2009 theoretical documentation describes some processes that SWAT+ 62.0.1 does not simulate, or simulates in a different way:

- Elevation bands within a subbasin. SWAT+ corrects precipitation and temperature for the elevation of each object instead (see Climate and weather).
- Monthly climate change adjustments of precipitation, temperature, solar radiation and humidity. SWAT+ 62.0.1 has no input for them; adjust the weather files instead.
- Hourly air temperature and the distribution of solar radiation over the hours of the day.
- The adjustment of the curve number for slope.
- Transmission losses from tributary channels within a subbasin. SWAT+ has no tributary channels; losses are computed for each channel object.
- Bank storage in the main channel.
- The SWAT 2009 channel sediment routing (simplified Bagnold stream power equation, channel erodibility and cover factors, downcutting and widening). SWAT+ uses the channel sediment model described in Channels.
- Potholes and other depressional areas, replaced by wetlands and paddies in HRUs (see Reservoirs and wetlands).
- Reservoir release from measured monthly outflow or a target storage. SWAT+ uses release decision tables and measured releases from recall files.
- Chlorophyll-a and Secchi depth relations for lakes and reservoirs.
- The stream and dam reaeration formulas of QUAL2E; SWAT+ uses the reaeration rate given in `nutrients.cha`.
- The two-population bacteria model (persistent and less persistent bacteria), replaced by the pathogen routines described in Salts and constituents.
- Continuous fertilization and automatic fertilization by nitrogen stress. Decision tables can fertilize in response to conditions instead (see Decision tables).
- Consumptive water use and water transfer between objects, replaced by water allocation (see Water allocation).

Page layout

Every symbol is defined under its equation. Where a symbol is a user input, the input file and the column header the SWAT+ Editor writes are named with it, and the basin switches and parameters that change the result are listed together on each page.

The last section of every page, Sources, is for developers. It links to the Fortran routines at the 62.0.1 tag, maps symbols and columns to Fortran variable names, and records implementation details.

Sources

- Daily HRU sequence: `hru_control.f90`
- Object loop that calls HRUs, channels, aquifers and reservoirs: `command.f90`

The pages in this section were checked against SWAT+ 62.0.1 (commit `6f51536`). Where a page and the Fortran source disagree, the source describes what the model does.

Overview

The water balance drives everything else SWAT+ simulates: to predict the movement of sediment, nutrients, pesticides or salts, the model first has to reproduce the movement of water. SWAT+ splits the hydrology of a watershed into two parts. The land phase, covered here and in the pages that follow, controls how much water, sediment and chemical load each hydrologic response unit (HRU) delivers. The routing phase moves those loads through channels, aquifers, reservoirs and wetlands to the outlet.

Each HRU is computed once per day (or per sub-daily time step for runoff when Green-Ampt infiltration is used). The HRU keeps track of four water stores: the plant canopy, the snowpack, the soil profile and, where one is defined, a ponded surface such as a wetland or rice paddy. Water that leaves the soil profile at the bottom becomes recharge to an aquifer object or to the gwflow module.

Soil water balance

The land-phase balance for the soil profile of an HRU over a period of t days is

$$SW_t = SW_0 + \sum_{i=1}^{t} \left(R_{day} + I - Q_{surf} - E_a - w_{perc} - Q_{lat} - Q_{tile} \right)$$

Symbol	Meaning	Units
SW_t	Soil water content at the end of day t	mm H2O
SW_0	Soil water content at the start of the period	mm H2O
R_{day}	Rain and snowmelt reaching the soil surface, after canopy interception and snowfall	mm H2O
I	Irrigation and run-on from upslope objects	mm H2O
Q_{surf}	Surface runoff generated on the day	mm H2O
E_a	Evaporation from the soil and transpiration by plants	mm H2O
w_{perc}	Percolation from the bottom of the soil profile	mm H2O
Q_{lat}	Lateral subsurface flow leaving the soil layers	mm H2O
Q_{tile}	Tile drain flow	mm H2O

All terms are depths over the HRU area. Water in the canopy and in the snowpack is stored outside the soil profile: precipitation first fills canopy storage, then falls as snow on cold days, and only the remainder (plus snowmelt) is available for runoff and infiltration. Total evapotranspiration reported for the HRU is the sum of canopy evaporation, plant transpiration and soil evaporation, with snow sublimation and evaporation from ponded water counted in soil evaporation.

Surface runoff, lateral flow and tile flow do not all reach the channel on the day they are generated. Each passes through a lag store (see Hydrology), so the HRU water yield on a day is

$$Q_{yld} = Q_{surf,lag} + Q_{lat,lag} + Q_{tile,lag}$$

where the subscript lag marks the part of each store released that day.

Daily order of calculations

The order matters because each process sees the state left by the processes before it. For each HRU on each day SWAT+ 62.0.1 runs:

1. Elevation lapse corrections to precipitation and temperature, when `lapse` is on in `codes.bsn`.
2. Decision table operations and water allocation demands for the HRU.
3. Base-zero heat unit accumulation, albedo, salt and constituent soil chemistry.
4. Soil temperature for every layer.
5. Canopy interception.
6. Snowfall, snowpack temperature and snowmelt.
7. Run-on from upslope objects: surface runoff, lateral flow, tile flow and aquifer flow routed onto the HRU.
8. Soil crack volume, when `crack` is on in `codes.bsn`.
9. Potential evapotranspiration, canopy evaporation and soil evaporation.
10. Scheduled management operations from the HRU's management schedule.
11. Surface runoff, peak runoff rate and erosion. HRUs with a ponded surface (wetland or paddy) produce no surface runoff or erosion here; their water is handled in the next step.
12. Wetland and paddy water balance.
13. Infiltration and soil water routing: percolation, lateral flow, tile flow and saturation excess.
14. Grazing, residue decomposition, mineralisation, nitrification and volatilisation, and mineral phosphorus transformations.
15. Plant growth, including plant water and nutrient uptake (transpiration).
16. Pesticide wash-off, uptake, decay and leaching; nutrient and pesticide losses with runoff and sediment; atmospheric deposition; nitrate and soluble phosphorus movement.
17. Urban build-up and wash-off, sediment in lateral flow, and the surface lag store for sediment and chemicals.
18. Lag of lateral and tile flow, filter strips, grassed waterways and user-defined BMPs.
19. Lag of surface runoff.
20. Water yield and routing of the HRU's outflow to the objects it is connected to.

Switches and parameters

Input	File	Effect
`pet`	`codes.bsn`	Potential evapotranspiration method. See Climate and weather.
`gampt`	`codes.bsn`	Curve number (`0`) or Green-Ampt (`1`) infiltration. See Hydrology.
`lapse`	`codes.bsn`	Elevation lapse of precipitation and temperature.
`crack`	`codes.bsn`	Bypass flow through soil cracks.
`surq_lag`	`parameters.bsn`	Surface runoff lag coefficient.
`sw_init`	`parameters.bsn`	Initial soil water as a fraction of field capacity. See Soil water and temperature.
`can_max`, `esco`, `epco`, `lat_ttime`, `perco`, `cn3_swf`, `latq_co`, `pet_co`	`hydrology.hyd`	HRU-level hydrology parameters, described on the pages for each process.

Output

The terms of the HRU balance are written to the water balance files at HRU, landscape unit and basin level (`hru_wb_*`, `lsunit_wb_*`, `basin_wb_*`). `precip`, `snofall` and `snomlt` give the inputs; `surq_gen`, `latq`, `qtile`, `perc`, `et`, `eplant`, `esoil` and `ecanopy` the losses; `sw_init`, `sw_final` and `snopack` the stores; and `lagsurf` and `laglatq` the water still in the lag stores. See Water balance output.

Related pages

- Climate and weather for precipitation, temperature and potential evapotranspiration.
- Hydrology for interception, runoff, lateral flow, percolation and tile flow.

- Soil water and temperature for the soil profile and soil evaporation.
- Aquifers for the fate of percolation below the soil profile.
- Reservoirs and wetlands for ponded HRUs.

Sources

- Daily HRU sequence: `hru_control.f90`
- Evapotranspiration totals: `et_act.f90`
- Soil water routing: `swr_percmain.f90`
- Lateral and tile lag: `swr_substor.f90`; surface runoff lag: `sq_surfst.f90`
- Water balance output type: `type output_waterbal` in `output_landscape_module.f90`

Symbol or output	Fortran variable
SW	`soil(j)%sw`
R_{day} (before infiltration)	`precip_eff`
Infiltration into layer 1	`inflpcp`
Q_{surf} (generated, lagged)	`surfq(j)`, `qday`
Q_{lat}	`latq(j)`
Q_{tile}	`qtile`
w_{perc}	`sepbtm(j)`
Canopy evaporation, transpiration, soil evaporation	`canev`, `ep_day`, `es_day`
Total ET (`et` output)	`etday = ep_day + es_day + canev`
Water yield	`qdr(j) = qday + latq(j) + qtile`
Lag stores (`lagsurf`, `laglatq`, `lagsatex`)	`surf_bs(1,j)`, `bss(1,j)`, `bss_ex(1,j)`

Overview

Climate supplies the moisture and energy that drive the water balance and sets the relative size of its parts. SWAT+ needs daily precipitation, maximum and minimum air temperature, solar radiation, relative humidity and wind speed. Each weather station in `weather-sta.cli` takes each variable either from a measured data file or from the weather generator, and every HRU, channel and other spatial object uses the station assigned to it in its connection file. Potential evapotranspiration (PET) is computed in the model by one of three methods. A fourth option is meant to read PET from a file, but in 62.0.1 it returns the Hargreaves estimate.

Weather inputs

Measured and generated data

Each row of `weather-sta.cli` names a weather generator station from `weather-wgn.cli` and one data set for each variable: precipitation, temperature, solar radiation, relative humidity and wind speed, followed by a PET data set and an atmospheric deposition station. The value `sim` tells SWAT+ to generate that variable from the weather generator statistics.

Measured values come from the files listed in `pcp.cli`, `tmp.cli`, `slr.cli`, `hmd.cli`, `wnd.cli` and `pet.cli` (see `*.cli`, `*.pcp`, `*.tmp` and `*.tem`). The names in `weather-sta.cli` are matched to these lists with a search that assumes the lists are sorted by name; the `*.cli` page describes what happens when they are not. A value of -97 or less (for example -99) marks a missing record, and SWAT+ fills that day with a generated value. Days outside the period covered by a measured file are also generated.

Weather generator

The generator produces daily weather from the monthly statistics of a station in `weather-wgn.cli`. Precipitation is generated first, because temperature, radiation and humidity are adjusted for wet and dry days.

Whether a day is wet follows a first-order Markov chain. The probability of rain is `pr_wd` (a wet day following a dry day) or `pr_ww` (a wet day following a wet day) for the month. On a wet day the amount comes from a skewed distribution:

$$R_{day} = \mu_{mon} + 2\,\sigma_{mon}\left(\frac{\left[\left(SND_{day} - \frac{g_{mon}}{6}\right)\frac{g_{mon}}{6} + 1\right]^3 - 1}{g_{mon}}\right)$$

Symbol	Meaning	Input
R_{day}	Generated precipitation, at least 0.1 mm	mm H2O
μ_{mon}	Mean daily precipitation on wet days in the month	from `pcpmm` and `pcpd` in `weather-wgn.cli`
σ_{mon}	Standard deviation of daily precipitation	`pcpstd`, mm
g_{mon}	Skew coefficient of daily precipitation	`pcpskw`
SND_{day}	Standard normal deviate for the day	generated

The result is scaled by a monthly correction factor so that the long-term mean of generated wet-day amounts equals μ_{mon}.

Maximum and minimum temperature are drawn around the monthly means `tmpmx` and `tmpmn` with standard deviations `tmpstdmx` and `tmpstdmn`. The maximum is lowered on wet days and raised on dry days by half the monthly temperature range, weighted by the fraction of wet days. Relative humidity is drawn from a triangular distribution around the monthly mean, which comes from the dew point (`dewpt`) and mean temperature, and is raised on wet days. Wind speed follows a modified exponential distribution around the monthly mean `windav`.

The generator in 62.0.1 returns the monthly mean solar radiation `solarav` for every generated day. The wet and dry day adjustment is computed but then replaced by the monthly mean, so generated radiation does not vary from day to day within a month.

`igen` in `parameters.bsn` sets the random number seeds. With `igen = 0` every run uses the same seeds and the same generated weather. Any other value offsets the seeds, so different values give different weather sequences.

Maximum half-hour rainfall

The fraction of daily rain that falls in the wettest half hour, $\alpha_{0.5}$, is used by the peak runoff rate and erosivity calculations (see Erosion and sediment). It is generated every day from a triangular distribution between 0.02083 and $1 - \exp(-125/(R_{day} + 5))$, with its mode set from the monthly maximum half-hour rainfall `rainhmx` in `weather-wgn.cli`. The mode is multiplied by `adj_pkrt` in `parameters.bsn` and limited to between 0.1 and 0.95.

Day length and maximum radiation

Each day SWAT+ computes the day length and the maximum possible solar radiation for every weather generator station from its latitude. The solar declination δ and the relative distance between the earth and the sun d_r on day of year d are

$$\delta = \sin^{-1}\left[0.4\,\sin\left(\frac{d - 82}{58.09}\right)\right], \qquad d_r = 1 + 0.033\,\cos\left(\frac{d}{58.09}\right)$$

and the hour angle of sunset h, the day length T_{DL} and the maximum solar radiation H_{max} are

$$h = \cos^{-1}(-\tan\delta\,\tan\phi), \qquad T_{DL} = 7.6394\,h, \qquad H_{max} = 30\,d_r\,(h\,\sin\phi\,\sin\delta + \cos\phi\,\cos\delta\,\sin h)$$

Symbol	Meaning	Units
ϕ	Latitude of the weather generator station, `lat` in `weather-wgn.cli`	radians
h	Hour angle of sunset; 0 during polar night and π during polar day	radians
T_{DL}	Day length	h
H_{max}	Maximum possible solar radiation reaching the ground on a cloudless day	MJ/m²/day

H_{max} is used by net radiation and Hargreaves PET. The day length controls plant dormancy (see Plant growth and land use management). SWAT+ also computes how the day's radiation is spread over the hours of the day, but no process in 62.0.1 uses that distribution.

Sub-daily precipitation

With more than one time step per day (`step` in `time.sim`), a day of generated precipitation is spread over the day's time steps with a double exponential intensity curve. The peak rate is

$$i_{pk} = 2\, R_{day}\, \alpha_{0.5}$$

where R_{day} is the day's precipitation (mm) and $\alpha_{0.5}$ the fraction falling in the wettest half hour, so i_{pk} is in mm/h. The time to the peak, as a fraction v of the storm duration, is drawn from a triangular distribution between 0.05 and 0.95 with a mode of 0.25. The storm duration is

$$T_{dur} = \frac{R_{day}}{i_{pk}\,(k_1 + k_2)}, \qquad k_1 = \frac{v}{4.605}, \qquad k_2 = \frac{1 - v}{4.605}$$

limited to 24 hours, with i_{pk} recomputed when the limit applies. The cumulative rainfall at time t (h) from the start of the storm is

$$R(t) = R_{pk} - i_{pk}\, T_{dur}\, k_1 \left[1 - \exp\left(\frac{t - t_{pk}}{T_{dur}\, k_1}\right)\right] \qquad (t \le t_{pk})$$

$$R(t) = R_{pk} + i_{pk}\, T_{dur}\, k_2 \left[1 - \exp\left(\frac{t_{pk} - t}{T_{dur}\, k_2}\right)\right] \qquad (t > t_{pk})$$

with $R_{pk} = v\, R_{day}$ and $t_{pk} = v\, T_{dur}$. The storm starts at the beginning of the day, the rainfall of each time step is the increase of $R(t)$ over the step, and any amount left at the end is added to the last step that received rain.

The same curve fills a day of a measured sub-daily record that contains a missing value. Daily precipitation records are not split into time steps: in a sub-daily run a station with daily records gives no rainfall in its time steps. Use sub-daily records, or generated precipitation, for sub-daily runs.

Elevation lapse

When `lapse` is 1 in `codes.bsn`, precipitation and temperature are corrected for the difference in elevation between each object and the gauge (or the weather generator station, for generated data) that supplies it:

$$R_{obj} = \max\left(0,\ R_{gage} + \frac{\text{plaps}\,(z_{obj} - z_{gage})}{1000}\right)$$

$$T_{obj} = T_{gage} + \frac{\text{tlaps}\,(z_{gage} - z_{obj})}{1000}$$

Symbol	Meaning	Units
plaps	Precipitation lapse rate, `plaps` in `parameters.bsn`	mm/km
tlaps	Temperature lapse rate, `tlaps` in `parameters.bsn`	°C/km
z_{obj}, z_{gage}	Elevation of the object and of the gauge or generator station	m

The temperature correction is applied to the maximum, minimum and mean temperature. A positive `tlaps` makes objects above the gauge cooler. The precipitation correction is a fixed depth per day, added on dry days as well as wet ones, and the result is never negative.

Carbon dioxide

The CO2 concentration used by plant growth and Penman-Monteith PET is set for each year of the run. If the file `co2_yr.dat` is present, yearly values are read from it; years before the record use the first value and years after it the last. Without the file every year uses `co2` from `parameters.bsn`. The values used are written to `co2.out`.

Snow

On days when the mean air temperature is at or below `fall_tmp`, precipitation that passes the canopy is added to the snowpack. Snow melts on days when the maximum temperature is above `melt_tmp`, at a rate that varies through the year between `melt_min` and `melt_max` and is reduced by the fraction of the HRU covered by snow. The equations and the parameters are on the `snow.sno` page. Snow sublimation is taken from the soil evaporation demand (see Soil water and temperature).

Potential evapotranspiration

Potential evapotranspiration is the rate of evapotranspiration from a large area covered completely by a short, actively growing crop with plenty of water. The method is chosen with `pet` in `codes.bsn`:

`pet`	Method
0	Priestley-Taylor
1	Penman-Monteith
2	Hargreaves
3	Station PET (see PET read from file)

The three computed methods share these terms:

Symbol	Meaning	Calculation
$\bar{T}$	Mean air temperature, °C	from the weather inputs
P	Atmospheric pressure, kPa	$101.3 - z\,(0.01152 - 0.544 \times 10^{-6}\,z)$ with HRU elevation z in m
λ	Latent heat of vaporization, MJ/kg	$2.501 - 2.361 \times 10^{-3}\,\bar{T}$
γ	Psychrometric constant, kPa/°C	$1.013 \times 10^{-3}\,P/(0.622\,\lambda)$
e^0, e	Saturation and actual vapour pressure, kPa	$e = e^0 \times$ relative humidity
Δ	Slope of the saturation vapour pressure curve, kPa/°C	$4098\,e^0/(\bar{T} + 237.3)^2$

Net radiation

Priestley-Taylor and Penman-Monteith use net radiation for a reference crop:

$$H_{net} = (1 - \alpha)\,H_{day} - (0.34 - 0.139\sqrt{e})\left(0.9\,\frac{H_{day}}{H_{max}} + 0.1\right)\sigma\,T_K^4$$

where H_{day} is the solar radiation reaching the ground (MJ/m²/day), H_{max} the maximum possible solar radiation for the day, $\sigma = 4.903 \times 10^{-9}$ MJ m^{-2} K^{-4} day^{-1}, T_K the mean air temperature in kelvin, and α the albedo: 0.23 for the reference crop, or 0.8 when the snowpack contains more than 0.5 mm of water.

Priestley-Taylor

$$PET = 1.28\,\frac{\Delta}{\Delta + \gamma}\,\frac{H_{net}}{\lambda}$$

Priestley and Taylor (1972) replaced the aerodynamic term of Penman's equation with the coefficient 1.28, which suits humid areas. The method tends to underestimate PET in arid and semi-arid areas, where advection is a large part of the energy balance.

Penman-Monteith

$$PET = \frac{\Delta\, H_{net} + \gamma\, \rho_K\, (e^0 - e)/r_a}{\lambda\, [\Delta + \gamma\, (1 + r_c/r_a)]}$$

For the reference crop SWAT+ uses the aerodynamic resistance $r_a = 114/u_{1.7}$ (s/m), where $u_{1.7} = u_{10}\,(0.17)^{0.2}$ is the wind speed at 1.7 m estimated from the input wind speed u_{10} (taken as measured at 10 m), and a canopy resistance that responds to CO2:

$$r_c = \frac{49}{1.4 - 0.4\, CO_2/330}$$

ρ_K combines air density, the latent heat of vaporization and unit conversions, and is computed as $1710 - 6.85\,\bar{T}$. The method needs measured or generated relative humidity and wind speed.

Penman-Monteith also computes a crop-specific maximum transpiration from the plant canopy height, leaf area and stomatal conductance (`stcon_max`, `vpd` and `frac_stcon` in `plants.plt`). In 62.0.1 that value is replaced in the actual evapotranspiration step by the leaf area based estimate used with the other methods (see Soil water and temperature), so the choice of PET method changes PET but not the way PET is split into transpiration and soil evaporation.

Hargreaves

$$PET = 0.0023\, \frac{H_0}{\lambda}\, (\bar{T} + 17.8) \sqrt{T_{max} - T_{min}}$$

where H_0 is extraterrestrial radiation (MJ/m²/day), taken as $H_{max} \times 37.59/30$. PET is zero on days when T_{max} is not above T_{min}. Hargreaves needs only temperature, which makes it a common choice where humidity, wind and radiation records are poor.

PET read from file

With `pet = 3` the HRU takes the daily PET of its weather station. SWAT+ 62.0.1 first loads the station's measured PET (filling missing days with Hargreaves), but later in the same day it recomputes the station PET with the Hargreaves equation for every station, for use in the climatic moisture index. The HRU therefore receives the Hargreaves value, and `pet = 3` gives the same PET as `pet = 2`.

The same station PET (Hargreaves in every case) is the evaporative demand for channel and reservoir evaporation and for aquifer revap, whatever the `pet` setting.

HRU adjustment

Whatever the method, PET is multiplied by `pet_co` from `hydrology.hyd`:

$$PET_{HRU} = \text{pet_co} \times PET$$

A blank or zero `pet_co` is taken as 1. A value below 0.01 is read as a Hargreaves coefficient and converted to a multiplier by dividing it by 0.0023.

Switches and parameters

Input	File	Effect
pet	codes.bsn	PET method
lapse	codes.bsn	Elevation lapse of precipitation and temperature
plaps, tlaps	parameters.bsn	Lapse rates
adj_pkrt	parameters.bsn	Multiplier on the generated maximum half-hour rainfall fraction
co2	parameters.bsn	CO2 concentration when co2_yr.dat is absent, ppm
igen	parameters.bsn	Random number seed selection for the generator
pet_co	hydrology.hyd	HRU multiplier on PET
petco_pmpt	parameters.bsn	Read and converted to a fraction, but not applied to PET in 62.0.1

Related pages

- *.cli files, *.pcp, *.tmp for the weather input formats.
- snow.sno for snowfall and snowmelt.
- Soil water and temperature for actual evapotranspiration.
- Overview and water balance for where climate enters the daily sequence.

Sources

- Daily weather for all stations, maximum half-hour rainfall: climate_control.f90, cli_precip_control.f90
- Station file reader: cli_staread.f90; generator statistics: cli_wgnread.f90, cli_initwgn.f90
- Generators: cli_pgen.f90, cli_pgenhr.f90, cli_tgen.f90, cli_slrgen.f90, cli_rhgen.f90, cli_wndgen.f90
- Day length, declination, maximum radiation and hourly radiation fractions: cli_clgen.f90, called from climate_control.f90. The hourly fractions frad are computed and not used.
- Sub-daily precipitation curve: cli_pgenhr.f90, called from cli_precip_control.f90 for generated days and for sub-daily records with missing values; daily records leave wst%weat%ts unfilled.
- Lapse: cli_lapse.f90, applied in hru_control.f90
- CO2: co2_read.f90
- PET: et_pot.f90; pet_co setup in topohyd_init.f90
- Basin codes and parameters: basin_module.f90, basin_prm_default.f90

Symbol or column	Fortran variable
pet (codes.bsn)	bsn_cc%pet
lapse (codes.bsn)	bsn_cc%lapse
plaps, tlaps	bsn_prm%plaps, bsn_prm%tlaps; per object ob(iob)%plaps, ob(iob)%tlaps
adj_pkrt	bsn_prm%adj_pkr
co2, yearly CO2	bsn_prm%co2, co2y(time%yrs)
petco_pmpt	bsn_prm%petco_pmpt
pet_co	hru(j)%hyd%pet_co
$\alpha_{0.5}$	wst(iwst)%weat%precip_half_hr
H_{day}, H_{max}	w%solrad, w%solradmx
δ, d_r, h, T_{DL}	sd, dd, h, wst(iwst)%weat%daylength
i_{pk}, v, T_{dur}	pkrr, vv, dur
H_{net} (reference crop)	rn_pet
Δ, γ, λ	dlt, gma, xl
r_a, r_c	rv, rc
PET	pet_day

Overview

Precipitation reaching an HRU may be caught by the plant canopy or fall on the soil surface. Water on the surface infiltrates into the soil profile or runs off. Runoff moves quickly towards a channel and gives the short-term response

of a stream. Infiltrated water is stored in the soil and later evaporated or transpired, or it moves slowly through the profile: laterally towards the channel, into tile drains, or down past the bottom of the profile to recharge groundwater.

Evaporation from the soil is described in Soil water and temperature. What happens to the water once it reaches a channel is on the Channels page.

Canopy interception

The plant canopy intercepts part of the rain, and the amount stored depends on how much canopy there is. The storage capacity on a day is

$$can_{day} = can_{mx} \frac{LAI}{LAI_{mx}}$$

Symbol	Meaning	Units
can_{day}	Maximum canopy storage on the day	mm H2O
can_{mx}	Maximum canopy storage when the canopy is fully developed, `can_max` in `hydrology.hyd`	mm H2O
LAI	Leaf area index of the plant community on the day	m²/m²
LAI_{mx}	Maximum leaf area index of the plant community	m²/m²

Precipitation first fills the empty part of the canopy store, and only the rest reaches the ground. In sub-daily runs the store is filled from the first time steps of the day. Water stored in the canopy is the first water evaporated (see Soil water and temperature). Interception applies with both the curve number and the Green-Ampt method. HRUs with no leaf area have no interception.

Surface runoff

Surface runoff, or overland flow, occurs when water reaches the ground faster than it can infiltrate. SWAT+ has two methods, chosen with `gampt` in `codes.bsn`: the SCS curve number method (`0`) and the Green and Ampt infiltration method (`1`). Runoff is computed only on days when more than 0.1 mm of water reaches the surface. Runoff from irrigation is added to it afterwards.

SCS curve number

The SCS runoff equation (SCS, 1972) is an empirical model that came into common use in the 1950s, built from more than 20 years of rainfall and runoff studies on small rural watersheds across the U.S.:

$$Q_{surf} = \frac{(R_{day} - I_a)^2}{R_{day} - I_a + S} \quad \text{for } R_{day} > I_a, \qquad Q_{surf} = 0 \text{ otherwise}$$

with the initial abstraction $I_a = 0.2\,S$, so that

$$Q_{surf} = \frac{(R_{day} - 0.2\,S)^2}{R_{day} + 0.8\,S}$$

Symbol	Meaning	Units
Q_{surf}	Accumulated runoff or rainfall excess	mm H2O
R_{day}	Rain and snowmelt reaching the surface on the day, after interception	mm H2O

Symbol	Meaning	Units
I_a	Initial abstraction: surface storage, interception and infiltration before runoff	mm H2O
S	Retention parameter for the day	mm H2O

The retention parameter is linked to the curve number for the day, CN:

$$S = 25.4 \left(\frac{1000}{CN} - 10 \right)$$

Curve number for average conditions. Each HRU takes its moisture condition II curve number, CN_2, from the row of `cntable.lum` named in the `cn2` column of its `landuse.lum` row, using the column for the soil's hydrologic group (`cn_a` to `cn_d`, from `hyd_grp` in `soils.sol`). SWAT+ 62.0.1 does not adjust CN_2 for slope. The curve number for dry conditions (wilting point) and for wet conditions are

$$CN_1 = CN_2 - \frac{20\,(100 - CN_2)}{100 - CN_2 + \exp\left[2.533 - 0.0636\,(100 - CN_2)\right]}, \qquad CN_1 \geq 0.4\,CN_2$$

$$CN_3 = CN_2 \exp\left[0.006729\,(100 - CN_2)\right]$$

Daily curve number from soil water. The retention parameter varies with the water content of the soil profile:

$$S = S_{max} \left(1 - \frac{SW}{SW + \exp(w_1 - w_2\,SW)} \right), \qquad S \geq 3 \text{ mm}$$

Symbol	Meaning	Units
S_{max}	Retention parameter for CN_1	mm H2O
SW	Soil water content of the whole profile, excluding water below wilting point	mm H2O
w_1, w_2	Shape coefficients	-

The shape coefficients are fixed at the start of the run so that S equals S_{max} for a dry profile, equals the CN_3 retention parameter when the profile water content is SW_3, and equals 2.54 mm (a curve number of 99) when the profile is saturated:

$$SW_3 = FC + \text{cn3_swf}\,(SAT - FC)$$

where FC and SAT are the profile water contents at field capacity and saturation, and `cn3_swf` is the column in `hydrology.hyd`. With `cn3_swf = 0` the wet condition is reached at field capacity; values towards 1 move it towards saturation and reduce runoff from wet soils. For HRUs with a tile drain SWAT+ sets `cn3_swf` to 0.95.

SWAT 2009 offered a second method that ties the curve number to accumulated plant evapotranspiration. SWAT+ 62.0.1 has only the soil water method.

Frozen soil. When the second soil layer is at or below 0 °C, infiltration is reduced:

$$S_{frz} = S_{max}\,[1 - \exp(-\text{cn_froz}\ S)]$$

where `cn_froz` is the frozen soil parameter in `parameters.bsn`.

Urban areas. For an HRU with an urban land use, runoff from the directly connected impervious fraction (`frac_dc_imp` in `urban.urb`) is computed with a curve number of 98 and mixed with the runoff from the rest of the HRU by area. The `urb_cn` column of `urban.urb` is read but not used.

Green and Ampt infiltration

The Green and Ampt method (Green and Ampt, 1911) needs precipitation at a sub-daily time step, set in `time.sim` and supplied in the precipitation files. It assumes a sharp wetting front that separates saturated soil above from soil at its initial water content below. SWAT+ uses the Mein and Larson (1973) form with the modification of Kim et al., where the infiltration rate at the end of a time step is

$$f_{inf} = K_e \left(1 + \frac{\Psi_{wf}\,\Delta\theta_v}{F_{inf}}\right)$$

Symbol	Meaning	Units
f_{inf}	Infiltration rate	mm/hr
K_e	Effective hydraulic conductivity	mm/hr
Ψ_{wf}	Wetting front matric potential	mm
$\Delta\theta_v$	Change in volumetric water content across the wetting front	mm/mm
F_{inf}	Cumulative infiltration since the start of the day	mm H2O

Within each time step, while the rainfall intensity is below the infiltration rate all rain infiltrates. Otherwise the cumulative infiltration at the end of the step is solved by iteration from

$$F_{inf,t} = F_{inf,t-1} + K_e\,\Delta t + \Psi_{wf}\,\Delta\theta_v\,\ln\!\left(\frac{F_{inf,t} + \Psi_{wf}\,\Delta\theta_v}{F_{inf,t-1} + \Psi_{wf}\,\Delta\theta_v}\right)$$

and the rain that does not infiltrate becomes runoff for that step.

The effective conductivity is estimated from the saturated conductivity of the top soil layer, K_{sat}, and the curve number (Nearing et al., 1996):

$$K_e = \frac{56.82\,K_{sat}^{0.286}}{1 + 0.051\,\exp(0.062\,CN)} - 2, \qquad K_e \geq 0.001$$

The curve number used here is updated every time step from the soil water plus the water infiltrated so far, with the same relationship as the curve number method. The moisture deficit is

$$\Delta\theta_v = 0.95\,\phi_{soil}\left(1 - \frac{SW}{FC}\right)$$

where ϕ_{soil} is the porosity of the top layer and SW/FC is limited to 0.999. If more than 12 mm of rain fell on the previous day, the soil is assumed to be nearly saturated at the start of the day and the previous day's final infiltration rate is used as the starting rate. Ψ_{wf} is estimated once for each HRU from the porosity and the sand and clay content of the top layer (Rawls and Brakensiek, 1985).

On urban HRUs the directly connected impervious fraction produces runoff from the rain left after an initial abstraction store. The store is depleted by rain and recovers on dry time steps up to `urb_init_abst` in `parameters.bsn`.

Crack flow

When `crack` is 1 in `codes.bsn`, SWAT+ computes a crack volume for each layer from the soil's `perc_crk` value in `soils.sol` and its current water deficit. Cracks develop slowly as the soil dries and close as it wets. Runoff up to the total crack volume is moved into the cracks, where it fills layers below field capacity from the bottom of the profile upward, and any remainder percolates from the bottom of the profile.

Surface runoff lag

In large HRUs, or HRUs with a long time of concentration, only part of the runoff generated on a day reaches the channel that day. SWAT+ keeps a surface runoff store and releases a fixed fraction of it each day:

$$Q_{surf,lag} = \left(Q'_{surf} + Q_{stor,i-1}\right)\left[1 - \exp\left(\frac{-\text{surq_lag}}{t_{conc}}\right)\right]$$

Symbol	Meaning	Units
$Q_{surf,lag}$	Surface runoff released to the channel on the day	mm H2O
Q'_{surf}	Surface runoff generated on the day	mm H2O
$Q_{stor,i-1}$	Runoff stored or lagged from the previous day	mm H2O
surq_lag	Surface runoff lag coefficient, **surq_lag** in parameters.bsn	-
t_{conc}	Time of concentration of the HRU	hr

For a given time of concentration, a smaller **surq_lag** keeps more water in storage. In sub-daily runs the time of concentration is expressed in time steps, and the same fraction is released every step. Sediment, nutrients and other constituents in surface runoff are lagged in the same way.

The time of concentration is the sum of the overland flow time and a channel flow time within the HRU:

$$t_{ov} = \frac{0.0556\,(L_{slp}\,n)^{0.6}}{slp^{0.3}}, \qquad t_{ch} = \frac{0.31\,L\,n^{0.75}}{A^{0.125}\,(0.5\,slp + 0.001)^{0.375}}$$

Symbol	Meaning	Units
L_{slp}	Slope length, **slp_len** in topography.hyd	m
slp	Average slope, **slp** in topography.hyd	m/m
n	Manning's n for overland flow, from the **ov_mann** column of landuse.lum	-
L	Flow length, taken as $0.5\sqrt{\text{HRU area in ha}/2}$	km
A	HRU area	km²

Soil water routing

Water that infiltrates enters the top soil layer and is routed down the profile layer by layer. A layer can lose water by percolation to the layer below and by lateral flow, but only the water stored above field capacity can move.

Percolation

Percolation from a layer is computed with a storage routing method:

$$w_{perc} = SW_{ex}\left[1 - \exp\left(\frac{-24}{TT_{perc}}\right)\right], \qquad TT_{perc} = \frac{SAT - FC}{K_{sat}}$$

Symbol	Meaning	Units
w_{perc}	Water percolating to the next layer on the day	mm H2O
SW_{ex}	Water in the layer above field capacity	mm H2O
TT_{perc}	Travel time for percolation, at least 2 hours	hr
SAT, FC	Water content of the layer at saturation and at field capacity	mm H2O
K_{sat}	Saturated hydraulic conductivity of the layer, **soil_k** in soils.sol	mm/hr

No water percolates or flows laterally from a layer whose temperature is at or below 0 °C. Percolation from the bottom layer leaves the soil profile as recharge, and is multiplied by a factor derived from `perco` in `hydrology.hyd`:

$$f_{perco} = \min\left(1,\ \exp\left[-\left(1.0052\ \ln(-\ln \text{perco}) + 5.6862\right)\right]\right)$$

`perco` close to 1 gives a factor of 1, so the bottom layer drains freely. Smaller values keep water in the profile, and `perco = 0` stops deep percolation. For HRUs with a tile drain, SWAT+ sets `perco` to 0.1. Above the bottom layer, percolation is not restricted.

Lateral flow

Lateral subsurface flow is computed with a kinematic storage model (Sloan et al., 1983), applied to every layer below the top one:

$$Q_{lat} = \text{latq_co} \times 0.024\ \frac{2\,SW_{ex}\,K_{sat}\,slp}{\phi_d\,L_{hill}}$$

Symbol	Meaning	Units
Q_{lat}	Lateral flow from the layer	mm H2O/day
latq_co	Lateral flow coefficient, `latq_co` in `hydrology.hyd`	-
ϕ_d	Drainable porosity of the layer, $(SAT - FC)$ divided by layer thickness	mm/mm
slp	Average slope of the HRU	m/m
L_{hill}	Slope length for lateral flow, `lat_len` in `topography.hyd`; 50 m when blank or zero	m

If percolation and lateral flow together exceed the water above field capacity, both are reduced in proportion. The top soil layer, which is the 10 mm surface layer for most soils (see Soil water and temperature), produces no lateral flow in SWAT+.

Saturation excess

After percolation, any layer with more water than it can at saturation passes the excess to the layer above. Excess in the top layer is pushed down to the next layer with room, and what cannot be stored becomes surface runoff (or, on HRUs with a ponded surface, flows into the wetland or over its weir).

Tile drainage

Tile drains take water from the soil when a perched water table rises above the drain. Tile flow is computed for HRUs with a tile drain (`tile` in `landuse.lum`, pointing to a row of `tiledrain.str`) when the second soil layer is above freezing. The method is set by `tiledrain` in `codes.bsn`: `0` uses the drawdown equation below, `1` the DRAINMOD equations.

With `wtable = 0` in `codes.bsn`, the height of the water table above the bottom of the profile is estimated from the water stored above field capacity:

$$h_{wtbl} = \frac{SW - FC}{SAT_{90} - FC}\,depth_{prof}, \qquad SAT_{90} = \max(0.9\,SAT,\ 1.1\,FC)$$

where SW, FC and SAT are profile totals and $depth_{prof}$ is the profile depth. When the water table is above the drain, the drawdown method gives

$$Q_{tile} = \frac{h_{wtbl} - h_{drain}}{h_{wtbl}}\,(SW - FC)\left[1 - \exp\left(\frac{-24}{t_{drain}}\right)\right]$$

Symbol	Meaning	Units
Q_{tile}	Tile flow on the day, at most `drain` in tiledrain.str	mm H2O
h_{drain}	Height of the drain above the bottom of the profile, profile depth minus `dp`	mm
t_{drain}	Time to drain the soil to field capacity, `t_fc` in tiledrain.str; all excess drains in one day when it is under 1 hour	hr

Tile flow is taken from the layers with water above field capacity, starting at the top.

Lag of lateral and tile flow

In large HRUs with a mild slope, lateral and tile flow can take more than a day to reach the channel. Both pass through a store, and the part released on a day is

$$Q_{lat,lag} = (Q'_{lat} + Q_{latstor,i-1}) \left[1 - \exp\left(\frac{-1}{TT_{lag}}\right)\right]$$

TT_{lag} is `lat_ttime` in hydrology.hyd, in days. When `lat_ttime` is zero, SWAT+ estimates it from the slope length for lateral flow and the highest conductivity in the profile, $TT_{lag} = 10.4\, L_{hill}/K_{sat,mx}$ (at least one day). Tile flow uses $1 - \exp(-24/lag)$, with `lag` in hours from tiledrain.str. Nitrate in lateral and tile flow is lagged with the water.

Channel routing

Water yield from the HRUs enters channels, where it is routed downstream by the variable storage method or the Muskingum method, chosen with `rte_cha` in codes.bsn. See Channels.

Switches and parameters

Input	File	Effect
`gampt`	codes.bsn	`0` curve number, `1` Green-Ampt
`crack`	codes.bsn	`1` computes crack flow
`tiledrain`	codes.bsn	`0` drawdown tile equation, `1` DRAINMOD
`wtable`	codes.bsn	`0` water table from profile water, `1` DRAINMOD water table
`surq_lag`	parameters.bsn	Surface runoff lag coefficient
`cn_froz`	parameters.bsn	Frozen soil infiltration factor
`urb_init_abst`	parameters.bsn	Maximum initial abstraction on impervious urban area, Green-Ampt only
`can_max`, `cn3_swf`, `perco`, `latq_co`, `lat_ttime`	hydrology.hyd	Canopy storage, wet-condition curve number, deep percolation, lateral flow, lateral flow lag
`slp`, `slp_len`, `lat_len`	topography.hyd	Slope, slope length, slope length for lateral flow
`cn2`, `ov_mann`, `tile`	landuse.lum	Curve number table row, overland Manning's n, tile drain
`dp`, `t_fc`, `lag`, `drain`	tiledrain.str	Drain depth, drainage time, tile lag, drainage coefficient
`soil_k`, `awc`, `bd`, `perc_crk`	soils.sol	Layer conductivity and water retention, crack volume

Related pages

- Overview and water balance for the order of these calculations within the day.
- Soil water and temperature for soil layers and soil evaporation.

- Erosion and sediment for the peak runoff rate and soil loss.
- Aquifers for recharge below the soil profile.
- Channels for channel routing.
- Water balance output.

Sources

- Daily sequence: `hru_control.f90`, `surface.f90`
- Interception: `sq_canopyint.f90`
- Curve number: `sq_volq.f90`, `sq_daycn.f90`, `sq_dailycn.f90`, `curno.f90`, `cn2_init.f90`
- Green-Ampt: `sq_greenampt.f90`; wetting front suction in `soils_init.f90`
- Cracks: `sq_crackvol.f90`, `sq_crackflow.f90`, `swr_percmacro.f90`
- Runoff lag and time of concentration: `sq_surfst.f90`, `time_conc_init.f90`
- Soil water routing: `swr_percmain.f90`, `swr_percmicro.f90`, `swr_satexcess.f90`
- Tile drainage: `swr_origtile.f90`, `swr_drains.f90`
- Subsurface lag: `swr_substor.f90`; lag factors in `hydro_init.f90`
- HRU hydrology parameters: `topohyd_init.f90`

Symbol or column	Fortran variable
`gampt`, `crack`, `tiledrain`, `wtable` (codes.bsn)	`bsn_cc%gampt`, `bsn_cc%crk`, `bsn_cc%tdrn`, `bsn_cc%wtdn`
`surq_lag`, `cn_froz`, `urb_init_abst`	`bsn_prm%surlag`, `bsn_prm%cn_froz`, `bsn_prm%urb_init_abst`
`can_max`	`hru(j)%hyd%canmx`
`cn3_swf`, `perco`, `latq_co`, `lat_ttime`	`hru(j)%hyd%cn3_swf`, `hru(j)%hyd%perco`, `hru(j)%hyd%latq_co`, `hru(j)%hyd%lat_ttime`
f_{perco}	`hru(j)%hyd%perco_lim`
`slp`, `slp_len`, `lat_len`	`hru(j)%topo%slope`, `hru(j)%topo%slope_len`, `hru(j)%topo%lat_len`
`ov_mann`	`hru(j)%luse%ovn`
`dp`, `t_fc`, `lag`, `drain`	`sdr%depth`, `sdr%time`, `sdr%lag`, `sdr%drain_co`
CN, S_{max}, w_1, w_2	`cnday(j)`, `smx(j)`, `wrt(1,j)`, `wrt(2,j)`
Ψ_{wf}	`wfsh(j)`
t_{conc}, lag fraction	`tconc(j)`, `brt(j)`
SW_{ex}, w_{perc}, Q_{lat} (layer)	`sw_excess`, `sepday`, `latlyr`
TT_{perc}	`soil(j)%phys(ly)%hk`
h_{wtbl}	`wt_shall`
Lag stores	`surf_bs(1,j)`, `bss(1,j)` (lateral), `bss(3,j)` (tile)

Overview

The soil profile is the main water store of an HRU. Water that infiltrates is stored in the soil layers, taken up by plants, evaporated from the soil surface, or moved down and sideways as described in Hydrology.

Soil layers

Each HRU uses one soil from `soils.sol`. The soil can have any number of layers, each defined by the depth to its bottom (`dp`) and its physical properties.

If the first layer of the soil is deeper than 19.5 mm, SWAT+ adds a 10 mm surface layer with the properties of the first layer, so most HRUs have one more layer than the soil in `soils.sol`. The thin surface layer is used for soil evaporation, surface nutrient processes and erosion. When a file named `soil_lyr_depths.sol` is present in the project folder, SWAT+ instead rebuilds the layers of every soil at the depths listed in it, with a 10 mm surface layer on top. HRUs with an active septic system get extra layer boundaries at the top and bottom of the septic biozone.

Missing or out-of-range layer properties are replaced when the soil is read:

Property	Replacement
Bulk density `bd`	1.3 Mg/m³ when zero; at most 2.0 Mg/m³

Property	Replacement
Available water capacity `awc`	0.005 mm/mm when zero; at most 0.8 mm/mm
Saturated conductivity `soil_k`	When zero: 50, 20, 5 or 2 mm/hr for hydrologic group (`hyd_grp`) A, B, C or D, 10 mm/hr otherwise
Rock fragments `rock`	at most 98%
Soil albedo `alb`	at least 0.1
Calcium carbonate `caco3` and `ph`	2.8% and 6.5 when zero
Anion exclusion `anion_excl`	0.5 when zero; at most 0.99

Water retention

SWAT+ stores soil water as the depth of water above the wilting point. The volumetric water contents of each layer are

$$\theta_{wp} = 0.40\,\frac{m_c\,\rho_b}{100}, \qquad \theta_{fc} = \theta_{wp} + AWC, \qquad \phi = 1 - \frac{\rho_b}{2.65}$$

Symbol	Meaning	Units
θ_{wp}	Water content at wilting point (−1.5 MPa)	mm/mm
θ_{fc}	Water content at field capacity (−0.033 MPa)	mm/mm
ϕ	Porosity	mm/mm
m_c	Clay content, `clay` in `soils.sol`	%
ρ_b	Moist bulk density, `bd` in `soils.sol`	Mg/m³
AWC	Available water capacity, `awc` in `soils.sol`	mm/mm

If field capacity reaches the porosity, it is set to $\phi - 0.05$ and the wilting point is moved down with it. The water a layer can store above wilting point is then

$$FC_{ly} = (\theta_{fc} - \theta_{wp})\,depth_{ly}, \qquad SAT_{ly} = (\phi - \theta_{wp})\,depth_{ly}$$

where $depth_{ly}$ is the layer thickness in mm. FC_{ly} is the water that plants can use, and the difference $SAT_{ly} - FC_{ly}$ is the drainable water that can percolate or flow laterally.

Initial soil water

Each layer starts the run with the same fraction of its field capacity water. The fraction is `sw_init` in `parameters.bsn`. When `sw_init` is zero, SWAT+ estimates it from the average annual precipitation of the HRU's weather generator station:

$$f_{init} = \frac{P_{an}}{P_{an} + \exp(9.043 - 0.002135\,P_{an})}$$

with P_{an} in mm. Dry climates start near zero and wet climates near field capacity. The water content then adjusts during the warm-up years skipped in `print.prt`.

Actual evapotranspiration

Evapotranspiration covers every process that turns liquid or solid water at or near the land surface into water vapour: evaporation from the canopy, transpiration, sublimation and evaporation from the soil and from ponded water. SWAT+ starts from the HRU's potential evapotranspiration (PET, see Climate and weather) and removes water in this order.

Canopy evaporation

Water stored on the canopy (see Hydrology) evaporates first. If PET is smaller than the canopy store, the canopy loses PET and there is no transpiration or soil evaporation that day. Otherwise the canopy is emptied and the remaining demand is $E'_0 = PET - E_{can}$.

Transpiration and soil evaporation demand

The maximum transpiration on the day depends on the leaf area of the plant community:

$$E_t = \frac{E'_0 \, LAI}{3} \quad \text{for } LAI \le 3, \qquad E_t = E'_0 \quad \text{for } LAI > 3$$

The soil evaporation demand is reduced by shading from plants and residue:

$$E_s = E'_0 \, cov_{sol}, \qquad cov_{sol} = \exp(-5.0 \times 10^{-5} \, CV)$$

where CV is the above-ground biomass plus surface residue (kg/ha). When the snowpack contains 0.5 mm of water or more, cov_{sol} is 0.5. If $E_t + E_s$ is larger than the demand, E_s is reduced so that the two together do not exceed it. The transpiration demand is met by plant water uptake (see Plant growth and land use management).

The same leaf area rule sets the transpiration demand for all three PET methods. SWAT+ 62.0.1 does not use the crop transpiration computed by the Penman-Monteith method.

Sublimation and ponded water

On days with a mean air temperature above 0 °C, the soil evaporation demand is taken first from the snowpack as sublimation. On HRUs with ponded water (wetlands and paddies), the evaporation demand is

$$E_s = \text{evap} \left(1 - \frac{LAI}{4}\right) E'_0 \quad \text{for } LAI \le 4, \qquad E_s = 0 \text{ otherwise}$$

where `evap` is the evaporation coefficient in `hydrology.wet`. It is taken from the ponded water, and the soil itself does not evaporate while ponded water remains.

Soil evaporation by layer

The remaining demand is distributed with depth. The demand down to depth z is

$$E_{soil,z} = E''_s \, \frac{z}{z + \exp(2.374 - 0.00713 \, z)}$$

where E''_s is the remaining demand and z is in mm. Half of the demand comes from the top 10 mm and 95% from the top 100 mm. The demand on a layer with its top at z_u and bottom at z_l is

$$E_{soil,ly} = E_{soil,z_l} - E_{soil,z_u} \, (1 - \text{esco})$$

Symbol	Meaning	Input
esco	Soil evaporation compensation factor	`esco` in `hydrology.hyd`, 0 to 1

With `esco = 0` each layer meets only its share of the demand. As `esco` increases, deeper layers can make up for demand that the layers above could not meet, so more water can be evaporated from lower in the profile. This direction is the opposite of SWAT 2009, where lowering `esco` allowed more compensation.

When a layer is drier than field capacity its demand is reduced:

$$E'_{soil,ly} = E_{soil,ly} \exp\left(\frac{2.5\,(SW_{ly} - FC_{ly})}{FC_{ly}}\right)$$

and a layer never gives up more than 80% of its plant-available water on one day. Only layers whose top is shallower than 500 mm evaporate. As water evaporates from the second layer, a small amount of nitrate moves up with it into the surface layer.

Soil temperature

Soil temperature affects water movement and the rate of residue decomposition and nutrient transformations. SWAT+ computes the temperature at the centre of each layer every day from the soil surface temperature, the annual average air temperature and a damping depth.

Damping depth

The maximum damping depth depends on the average bulk density of the profile, ρ_b:

$$dd_{max} = 1000 + \frac{2500\,\rho_b}{\rho_b + 686\,\exp(-5.63\,\rho_b)}$$

The damping depth on a day is scaled by the soil water content:

$$dd = dd_{max}\,\exp\left[\ln\left(\frac{500}{dd_{max}}\right)\left(\frac{1-wc}{1+wc}\right)^2\right], \qquad wc = \frac{SW}{(0.356 - 0.144\,\rho_b)\,z_{tot}}$$

where SW is the profile water content (mm) and z_{tot} the profile depth (mm).

Surface temperature

The temperature of a bare soil surface is

$$T_{bare} = \bar{T} + \frac{T_{max} - T_{min}}{2} \cdot \frac{H_{day}(1-\alpha) - 14}{20}$$

where H_{day} is the solar radiation and α the albedo of the day. Biomass, residue and snow insulate the surface. The cover weighting factor is

$$bcv = \max\left(\frac{CV}{CV + \exp(7.563 - 1.297 \times 10^{-4}\,CV)},\ \frac{SNO}{SNO + \exp(6.055 - 0.3002\,SNO)}\right)$$

with CV the above-ground biomass and residue (kg/ha) and SNO the snow water content (mm); the snow term is 1 above 120 mm. The surface temperature is

$$T_{srf} = \frac{1}{2}\left[T_{bare} + bcv\,T_{ly2} + (1 - bcv)\,T_{bare}\right]$$

where T_{ly2} is the previous day's temperature of the second soil layer.

Layer temperature

$$T_{ly} = \ell\, T_{ly,prev} + (1-\ell)\left[df\,(\bar{T}_{AA} - T_{srf}) + T_{srf}\right]$$

$$df = \frac{zd}{zd + \exp(-0.8669 - 2.0775\,zd)}, \qquad zd = \frac{z_{mid}}{dd}$$

Symbol	Meaning	Units
ℓ	Lag coefficient, 0.8	-
$T_{ly,prev}$	Layer temperature on the previous day	°C
$\bar{T}_{AA}$	Average annual air temperature of the weather generator station	°C
z_{mid}	Depth to the centre of the layer	mm

Layers start the run at the average December maximum temperature of the weather generator station. In septic HRUs, layers at and below the biozone are kept from falling far below 10 °C.

Switches and parameters

Input	File	Effect
`sw_init`	`parameters.bsn`	Initial soil water as a fraction of field capacity; zero estimates it from precipitation
`esco`	`hydrology.hyd`	Soil evaporation compensation
`epco`	`hydrology.hyd`	Plant uptake compensation (see Plant growth)
`can_max`	`hydrology.hyd`	Maximum canopy storage
`evap`	`hydrology.wet`	Evaporation coefficient for ponded water
`dp`, `bd`, `awc`, `soil_k`, `clay`, `rock`, `alb`	`soils.sol`	Layer depth, bulk density, water retention, conductivity, texture, albedo
`lai_noevap`	`parameters.bsn`	Read but not used: the leaf area index of 3.0 in the transpiration rule is fixed

Related pages

- `soils.sol` for the soil input file.
- Hydrology for infiltration, percolation and lateral flow.
- Plant growth and land use management for transpiration and root water uptake.
- Nutrient cycling for the effect of soil water and temperature on nutrient transformations.
- Water balance output for `esoil`, `eplant`, `ecanopy` and soil water.

Sources

- Soil database and layering: `soil_db_read.f90`, `soils_init.f90`
- Physical properties: `soil_phys_init.f90`
- Initial soil water and temperature: `hydro_init.f90`
- Actual evapotranspiration: `et_act.f90`
- Soil temperature: `stmp_solt.f90`
- Wetland evaporation coefficient: `type wetland_hyd_data` in `reservoir_data_module.f90`

Symbol or column	Fortran variable
`sw_init`	`bsn_prm%ffcb`; per soil `soil(j)%ffc`
`lai_noevap`	`bsn_prm%evlai`
`esco`, `epco`	`hru(j)%hyd%esco`, `hru(j)%hyd%epco`
`evap` (hydrology.wet)	`wet_hyd(ihyd)%evrsv`
θ_{wp}, θ_{fc}, ϕ	`soil(j)%phys(ly)%wp`, `%up`, `%por`

Symbol or column	Fortran variable
FC_{ly}, SAT_{ly}, SW_{ly}	`soil(j)%phys(ly)%fc`, `%ul`, `%st`
Profile totals	`soil(j)%sumfc`, `soil(j)%sumul`, `soil(j)%sw`
E'_0, E_t, E_s	`pet`, `ep_max`, `es_max`
E_{can}, soil evaporation, sublimation	`canev`, `es_day`, `snoev`
dd_{max}, dd, ρ_b	`dp`, `dd`, `soil(j)%avbd`
T_{srf}, T_{ly}	`soil(j)%tmp_srf`, `soil(j)%phys(ly)%tmp`

Overview

Erosion is the detachment, transport and deposition of soil particles by raindrop impact and flowing water. Raindrops detach particles on unprotected ground between rills, and the particles move to rills, then to larger channels and rivers, and can be deposited anywhere along the way. Erosion matters for reservoirs, which lose storage to deposited sediment, and for the soil, which loses the organic matter and nutrients in its surface layer.

SWAT+ computes soil loss from each HRU with the Modified Universal Soil Loss Equation (MUSLE) (Williams, 1975) on daily runs, or with a splash and overland flow erosion model on sub-daily runs. MUSLE replaces the rainfall energy factor of the Universal Soil Loss Equation (USLE) (Wischmeier and Smith, 1965, 1978) with a runoff factor, so it applies to single storms, reflects antecedent moisture through runoff, and needs no delivery ratio. Sediment is then lagged on its way to the channel together with surface runoff. Erosion and deposition in channels are covered on the Channels page.

Erosion is computed only on days with surface runoff and a positive peak runoff rate. HRUs with a ponded surface (wetlands and paddies) produce no surface runoff and no erosion.

Peak runoff rate

The peak runoff rate is the largest flow rate during a runoff event. It sets the erosive power of the storm in MUSLE. The method is chosen with `sed_det` in `codes.bsn`.

NRCS peak rate equation (`sed_det = 0`):

$$q_{peak} = \frac{prf}{6578.6} \cdot \frac{A\,Q_{surf}}{t_{conc}} \cdot \frac{1}{35.3}$$

Symbol	Meaning	Units
q_{peak}	Peak runoff rate	m^3/s
prf	Peak rate factor, `adj_pkrt_sed` in `parameters.bsn`	-
A	HRU area	ha
Q_{surf}	Surface runoff generated on the day	mm H2O
t_{conc}	Time of concentration of the HRU (see Hydrology)	hr

This is the NRCS triangular unit hydrograph peak, $q = 484\,A\,Q/t_p$ in U.S. units, with the time of concentration used as the time to peak. The NRCS value of the peak rate factor is 484, which is also the built-in value. A zero in `parameters.bsn` is replaced by 1.0, and the SWAT+ reference data sets use 1.0, which gives peak rates 484 times smaller than the NRCS equation. Users of this method should check the value against their calibration.

Rational formula with half-hour rainfall (`sed_det = 1`):

$$q_{peak} = \frac{\alpha_{tc}\,Q_{surf}\,A_{km}}{3.6\,t_{conc}}, \qquad \alpha_{tc} = 1 - \exp\left[2\,t_{conc}\,\ln(1-\alpha_{0.5})\right]$$

where α_{tc} is the fraction of daily rain that falls in the time of concentration, $\alpha_{0.5}$ the fraction that falls in the wettest half hour (generated daily, see Climate and weather) and A_{km} the HRU area in km^2.

MUSLE

SWAT+ computes daily sediment yield as

$$sed = 11.8\,(V_{surf}\,q_{peak})^{0.56}\,K_{USLE}\,C_{USLE}\,P_{USLE}\,LS_{USLE}\,CFRG$$

Symbol	Meaning	Units
sed	Sediment yield on the day	metric tons
V_{surf}	Surface runoff volume, $10\,Q_{surf}\,A$	m^3
K_{USLE}	Soil erodibility factor, `usle_k` of the first layer in soils.sol	0.013 t m^2 hr/(m^3 t cm)
C_{USLE}	Cover and management factor, computed daily	-
P_{USLE}	Support practice factor, `usle_p` in cons_practice.lum	-
LS_{USLE}	Topographic factor	-
$CFRG$	Coarse fragment factor	-

The runoff factor is the original MUSLE form of Williams (1975), with the runoff volume in m^3. The SWAT 2009 documentation writes the same factor with runoff in mm and area in ha, without the factor 10, so SWAT+ sediment yields for the same inputs are larger by $10^{0.56}$, about 3.6.

Soil erodibility

Some soils erode more easily than others even when all other factors are the same. K_{USLE} is the soil loss per unit of erosion index on a standard plot 22.1 m long on a 9% slope, kept in continuous fallow. Values are entered for each soil in soils.sol (see that page for estimation methods).

Cover and management

The cover and management factor is the ratio of soil loss from land under given conditions to soil loss from clean-tilled, continuous fallow. Plant cover changes through the year, so SWAT+ computes C_{USLE} every day from the surface residue and the plant canopy:

$$C_{USLE} = \exp\left[-\text{rsd_cover}\,\frac{rsd + 1}{1000}\right]\left[1 - f_{gc}\,\exp(-0.328\,h_c)\right]$$

$$f_{gc} = \frac{B}{B + \exp(1.175 - 1.748\,B)}$$

Symbol	Meaning	Units
rsd_cover	Residue cover factor, `rsd_cover` in parameters.bsn (0.75 when zero)	-
rsd	Surface residue	kg/ha
B	Above-ground biomass of the plant community	t/ha
h_c	Canopy height of the plant community	m

The minimum C factor of each plant (`usle_c_min` in plants.plt) is not used by this daily calculation. It is used only by the sub-daily erosion model below. The daily C factor and its two parts are written to the erosion output.

Support practice

The support practice factor is the ratio of soil loss with a practice such as contouring, strip cropping or terracing to soil loss with up-and-down-slope culture. It comes from the row of cons_practice.lum named in the `cons_prac` column of landuse.lum.

Topographic factor

$$LS_{USLE} = \left(\frac{L_{hill}}{22.1}\right)^{m} \left(65.41 \sin^2 \alpha_{hill} + 4.56 \sin \alpha_{hill} + 0.065\right), \qquad m = 0.6\left[1 - \exp(-35.835\, slp)\right]$$

where L_{hill} is the slope length (`slp_len` in `topography.hyd`, m), slp the slope (`slp`, m/m) and $\alpha_{hill} = \arctan(slp)$. `cons_practice.lum` also has a maximum slope length for each practice (`slp_len_max`); SWAT+ 62.0.1 reads it but the LS factor uses the full slope length.

Coarse fragments

$$CFRG = \exp(-0.053\, rock)$$

where $rock$ is the rock content of the first soil layer (`rock` in `soils.sol`, %).

Snow cover

The SWAT 2009 documentation reduces MUSLE sediment yield under a snowpack. In SWAT+ 62.0.1 that reduction is not reached for any snow depth, so snow cover does not change the sediment yield. Snowmelt runoff still produces erosion through Q_{surf} and q_{peak}.

USLE soil loss for output

Alongside MUSLE, SWAT+ computes the USLE soil loss from rainfall energy for output (`usle` in the HRU loss files):

$$A_{USLE} = 1.292\, EI_{USLE}\, K_{USLE}\, C_{USLE}\, P_{USLE}\, LS_{USLE}\, CFRG$$

$$EI_{USLE} = R_{day}\left[12.1 + 8.9\left(\log_{10} I - 0.434\right)\right] \frac{I_{30}}{1000}$$

where $I_{30} = 2\,\alpha_{0.5}\, R_{day}$ is the maximum 30-minute intensity (mm/hr) and $I = -2\ln(1-\alpha_{0.5})\, R_{day}$ the peak intensity. The USLE value is reported only. It does not add to the sediment yield.

Sub-daily erosion

When the simulation uses a sub-daily time step with Green-Ampt infiltration, sediment for each time step comes from raindrop splash and overland flow erosion instead of MUSLE.

Splash erosion. Rain is split into direct throughfall and leaf drip by the canopy cover, $\min(LAI, 1)$. Their kinetic energies are

$$KE_{direct} = 8.95 + 8.44 \log_{10} I, \qquad KE_{leaf} = 15.8\sqrt{0.5\, h_c} - 5.87$$

in J/m²/mm, where I is the rainfall intensity of the step (mm/hr). Splash detachment is

$$sed_{spl} = k_d\, KE \exp\left(-\text{splash}\, \frac{h_{flow}}{1000}\right) A_{km}$$

with k_d a soil detachability (g/J) set from the texture of the top layer (1.2 for silt up to 3.0 for loamy sand), KE the total kinetic energy of the step, h_{flow} the runoff depth of the step (mm) and `splash` in `parameters.bsn`. Splash stops when the runoff depth is at least three times the median raindrop diameter, or is negligible.

Overland flow erosion.

$$sed_{ov} = 11.02\, \text{rill}\, K_{USLE}\, \text{cov_mgt}\, C\, \tau^{\text{surq_exp}}, \qquad \tau = 9807\, \frac{h_{flow}}{1000}\, slp$$

in kg/hr/m², converted to tons for the step and the HRU area. `rill`, `cov_mgt` and `surq_exp` are in `parameters.bsn`; C is the SWAT 2009 cover factor computed from `usle_c_min` and the soil cover. All three default to zero. The rate is proportional to `rill` and `cov_mgt`, so overland flow erosion is zero unless both are set; a zero `surq_exp` makes the shear stress term equal to 1.

On urban HRUs both terms apply only to the pervious fraction. The sediment of the day is the sum over the steps after the lag described below.

With a sub-daily time step and the curve number method, the splash and overland flow terms are not computed and MUSLE is not used either, so the HRU produces no lagged sediment from the surface.

Sediment in lateral and tile flow

Lateral and tile flow transport a small sediment load:

$$sed_{lat} = \frac{\text{lat_sed}\,(Q_{lat} + Q_{tile})\,A}{100000}$$

in metric tons, with `lat_sed` from `hydrology.hyd` treated as a concentration in mg/L, and Q_{lat}, Q_{tile} in mm. Sediment concentrations in subsurface flow are usually very low and matter only where return flow is large. Organic nitrogen and phosphorus in lateral flow are set by `lat_orgn` and `lat_orgp` in the same file.

Particle size

The sediment from the surface is split into five classes (sand, silt, clay, small aggregates and large aggregates) using fractions set from the sand, silt and clay content of the top soil layer (Foster et al., 1980):

Class	Fraction of detached sediment
Sand	$f_{sa}(1 - f_{cl})^{2.49}$
Silt	$0.13\,f_{si}$
Clay	$0.20\,f_{cl}$
Small aggregates	$2.0\,f_{cl}$ for $f_{cl} < 0.25$; $0.28\,(f_{cl} - 0.25) + 0.5$ up to $f_{cl} = 0.5$; 0.57 above
Large aggregates	the remainder

where f_{sa}, f_{si} and f_{cl} are the sand, silt and clay fractions. If the first four add up to more than 1, they are rescaled and large aggregates are set to zero.

Sediment lag

Sediment, the particle size classes and the nutrients attached to sediment pass through the same surface lag store as runoff, and the same fraction, $1 - \exp(-\text{surq_lag}/t_{conc})$, is released each day (see Hydrology).

Urban areas

For an HRU with an urban land use (the `urban` and `urb_ro` columns of `landuse.lum`, parameters in `urban.urb`), the sediment and nutrient loads of the impervious fraction f_{imp} (`frac_imp`) are computed separately and mixed with the loads of the pervious part:

$$L = L_{urb}\,f_{imp} + L_{perv}\,(1 - f_{imp})$$

where L_{perv} is the load from MUSLE and the soil nutrient routines. Urban loads are computed before the surface lag store, and sediment from impervious surfaces is all silt. `urb_ro` selects one of two methods; any other value gives no urban loads.

USGS regression equations (`usgs_reg`). On days with more than 0.1 mm of precipitation and of surface runoff, the storm load of each constituent (kg) is

$$Y = \frac{\beta_0}{2.205}\left(\frac{R_{day}}{25.4}\right)^{\beta_1}\left(\frac{A\,f_{imp}}{2.589}\right)^{\beta_2}\left(100\,f_{imp}+1\right)^{\beta_3}\beta_4$$

with R_{day} the precipitation (mm) and A the HRU area (km²). The coefficients (Driver and Tasker, 1988) depend on the mean annual precipitation of the HRU's weather generator station:

Constituent	Below 508 mm	508 to 1016 mm	1016 mm and above
Suspended solids	1778, 0.867, 0.728, 0.157, 2.367	812, 1.236, 0.436, 0.202, 1.938	97.7, 1.002, 1.009, 0.837, 2.818
Total N	20.2, 0.825, 1.070, 0.479, 1.258	4.04, 0.936, 0.937, 0.692, 1.373	1.66, 0.703, 0.465, 0.521, 1.845
Total P	1.725, 0.884, 0.826, 0.467, 2.130	0.697, 1.008, 0.628, 0.469, 1.790	1.618, 0.954, 0.789, 0.289, 2.247

Each cell lists β_0 to β_4. Total N is split into 70 % organic N and 30 % nitrate, and total P into 75 % sediment-bound and 25 % soluble P.

Build-up and wash-off (`buildup_washoff`). Solids build up on impervious surfaces between storms:

$$SED = \frac{SED_{mx}\,t}{t_{1/2}+t}$$

where SED_{mx} is `dirt_max` (kg per curb km), $t_{1/2}$ is `t_halfmax` (days) and t the number of days since the surface was last washed. On a day with more than 0.1 mm of surface runoff the solids left after wash-off are

$$SED_{after} = SED\,\exp\!\left(-k_w\,q_{pk}\,t_{ro}\right)$$

with k_w = `urb_wash` (1/mm), q_{pk} the peak runoff rate (mm/h) and t_{ro} the storm duration from the half-hour rainfall fraction plus the time of concentration (h, at most 24). The washed-off solids are $(SED - SED_{after})$ times `curb_den`, in kg/ha; their N, P and nitrate contents are `conc_totn`, `conc_totp` and `conc_no3n` (mg/kg). Organic N is total N minus nitrate, and total P is split 75 % sediment-bound and 25 % soluble. After the storm, t is reset to the time that would build up SED_{after}. Sub-daily runs apply the same method for each time step. Street sweeping has no effect in 62.0.1 (see `sweep.ops`).

Filter strips, grassed waterways and user BMPs

These practices are attached to a land use through the `vfs`, `grww` and `bmp` columns of `landuse.lum` (parameters in the structural practice files). They act on the loads released from the surface lag store each day. None of them changes the runoff volume, so the water balance is unchanged.

Filter strips

A vegetative filter strip at the field edge (White and Arnold, 2009) is split into two sections. Section 1, 90 % of the strip area, receives the fraction $1-c$ of the field runoff; section 2, 10 % of the strip area, receives the fraction $c\,(1-h)$. The rest, $c\,h$, is channelized and passes the strip unchanged. Here c is `con_vfs` and h is `cha_q`, and the strip area is the HRU area divided by `fld_vfs`.

For each section i, the runoff depth over the strip q_i (mm) and the sediment load per strip area s_i (kg/m²) give the percent reductions

$$RR_i = 75.8 - 10.8\,\ln q_i + 25.9\,\ln K_{sat}, \qquad SR_i = 79.0 - 1.04\,s_i + 0.213\,RR_i$$

with K_{sat} the saturated conductivity of the top soil layer (mm/h). The runoff reduction RR_i is used only to estimate nutrient removal. The other reductions follow from these:

Load	Reduction (%)
Sediment	SR_i
Organic N	$0.036\,SR_i^{1.69}$
Nitrate in surface runoff	$39.4 + 0.584\,RR_i$
Sediment-bound P	$0.903\,SR_i$
Soluble P	$29.3 + 0.51\,RR_i$
Pesticide in runoff and on sediment	RR and SR

Each reduction is limited to 0 to 100 % and weighted by the share of the HRU runoff each section receives. Pesticide removed by the strip is returned to the top soil layer. In 62.0.1 the soluble P reduction of section 2 uses the value computed for sediment-bound P, and pesticide loads are reduced a second time after the removed amount is returned.

Grassed waterways

A grassed waterway is a vegetated trapezoidal channel with 8:1 side slopes through the field. The peak flow is computed from the day's runoff with a triangular hydrograph, and the flow depth in the waterway from Manning's equation. Part of the runoff enters the waterway as sheet flow over its side slopes; 20 % of the wetted side slope area above the flow, at most 10 % of the HRU, acts as a filter, with reductions computed as for filter strips:

$$RR = 95.6 - 10.79\,\ln q_{sf}, \qquad SR = 79.0 - 1.04\,s_{sf} + 0.213\,RR$$

both limited to 95 %. The sediment left after this filtering is then routed through the waterway: where its concentration exceeds the transport capacity $C_{ch} = \text{sed_co}\; v^{1.5}$ (with v the flow velocity), the excess is deposited. Sediment, organic N and sediment-bound P are reduced by the fraction of sediment that leaves the waterway, and nitrate and soluble P in surface runoff by RR.

User BMPs

A user BMP multiplies the HRU loads by $1 - e$ for each removal efficiency e in `bmpuser.str`: sediment, sediment-bound P, soluble P, organic N, and nitrate in surface runoff and lateral flow.

Riparian buffers and saturated buffers

A riparian buffer reduction of nitrate in lateral flow is coded but depends on a buffer width that no input sets, so it has no effect in 62.0.1. A saturated buffer diverts part of the tile flow of one HRU into the soil of another HRU under a decision table (see `satbuffer.str`).

Channel erosion and deposition

Sediment that reaches a channel can be deposited on the floodplain, deposited in the channel, or joined by sediment eroded from the channel banks and bed. These processes use the channel properties in `hyd-sed-lte.cha` and are described on the Channels page.

Switches and parameters

Input	File	Effect
`sed_det`	`codes.bsn`	`0` NRCS peak rate, `1` rational formula with half-hour rainfall
`adj_pkrt_sed`	`parameters.bsn`	Peak rate factor for `sed_det = 0`
`adj_pkrt`	`parameters.bsn`	Multiplier on the generated half-hour rainfall fraction
`rsd_cover`	`parameters.bsn`	Residue term of the daily C factor
`splash`, `rill`, `surq_exp`, `cov_mgt`	`parameters.bsn`	Sub-daily splash and overland flow erosion
`usle_k`, `rock`, `sand`, `silt`, `clay`	`soils.sol`	Erodibility, coarse fragments, particle size

Input	File	Effect
usle_p	cons_practice.lum	Support practice factor
slp, slp_len	topography.hyd	Slope and slope length
lat_sed	hydrology.hyd	Sediment concentration in lateral and tile flow

Related pages

- Hydrology for surface runoff, time of concentration and the surface lag.
- Nutrient cycling for nitrogen and phosphorus attached to sediment.
- Channels for channel erosion and deposition.
- Channel, sediment and nutrient output.

Sources

- Daily call sequence: surface.f90, hru_control.f90
- Peak rate: ero_pkq.f90
- MUSLE: ero_ysed.f90; C factor: ero_cfactor.f90; erosivity: ero_eiusle.f90
- K, P and CFRG product: hydro_init.f90; LS and P: plant_init.f90
- Sub-daily erosion: ero_ovrsed.f90
- Lateral sediment: swr_latsed.f90
- Particle size fractions: soil_phys_init.f90
- Sediment lag: stor_surfstor.f90
- Urban loads: hru_urban.f90, hru_urbanhr.f90 (sub-daily), regression in regres.f90; precipitation region wgn_pms%ireg set in cli_initwgn.f90; days since wash-off twash. The regression for COD is computed but not used.
- Filter strips: smp_filter.f90 (section 2 soluble P assigns remove21 and then uses remove2); riparian buffer: smp_buffer.f90 (filterw is allocated as zero and never set). Grassed waterways: smp_grass_wway.f90. User BMPs: smp_bmpfixed.f90. Called from hru_control.f90 after stor_surfstor.
- References: Driver, N. E., and Tasker, G. D. (1988), Techniques for estimation of storm-runoff loads, volumes, and selected constituent concentrations in urban watersheds in the United States, USGS Open-File Report 88-191. White, M. J., and Arnold, J. G. (2009), Development of a simplistic vegetative filter strip model for sediment and nutrient retention at the field scale, Hydrological Processes 23, 1602–1616.
- Half-hour rainfall: climate_control.f90, cli_initwgn.f90
- Defaults for zero parameters: basin_prm_default.f90

Symbol or column	Fortran variable
sed_det	bsn_cc%sed_det
adj_pkrt_sed (prf), adj_pkrt	bsn_prm%prf, bsn_prm%adj_pkr
rsd_cover	bsn_prm%rsd_covco
splash, rill, surq_exp, cov_mgt	bsn_prm%eros_spl, bsn_prm%rill_mult, bsn_prm%eros_expo, bsn_prm%c_factor
usle_c_min	pldb%usle_c, community value cvm_com(j)
lat_sed	hru(j)%hyd%lat_sed
q_{peak}	qp_cms
sed	sedyld(j)
C_{USLE}	usle_cfac(j)
$KPLSCFRG \times 11.8$	hru(j)%lumv%usle_mult
$C\times$ the above	cklsp(j)
LS_{USLE}, P_{USLE}	hru(j)%lumv%usle_ls, hru(j)%lumv%usle_p
EI_{USLE}, A_{USLE}	usle_ei, usle
Sub-daily sediment	hhsedy(j,k)
Particle size fractions	soil(j)%det_san, det_sil, det_cla, det_sag, det_lag

Overview

Each HRU has a plant community: one or more plants listed in `plant.ini` and described in `plants.plt`. SWAT+ grows every living plant of the community each day with a simplified version of the EPIC plant growth model. Phenological development follows accumulated heat units. Potential growth comes from intercepted light and the radiation use efficiency of the plant, and is reduced by water, temperature, nitrogen, phosphorus, aeration and (when salts are simulated) salinity stress. Growth removes water and nutrients from the soil, and harvest, kill and senescence return plant material to the residue and soil pools.

Plants are planted, harvested and killed by the management schedule of the HRU and by decision tables (see Land use management).

Plant types

The `plnt_typ` column of `plants.plt` sets how a plant develops:

`plnt_typ`	Behaviour
`warm_annual`, `warm_annual_tuber`	Annual; leaf area declines towards the end of the season
`cold_annual`, `cold_annual_tuber`	Annual; goes dormant in winter if planted late enough
`perennial`	Grows over several years towards maturity (`yrs_mat`); goes dormant each year

The `gro_trig` column sets what restarts growth: `temp_gro` for plants driven by temperature and day length, `moisture_gro` for plants in climates with a wet and dry season, where growth follows the ratio of precipitation to PET.

Heat units

Plants develop at a rate set by temperature. Each day a plant above its base temperature accumulates heat units:

$$HU = \bar{T} - T_{base} \quad \text{when } \bar{T} > T_{base}$$

and the fraction of potential heat units reached is

$$fr_{PHU} = \sum_{i=1}^{d} \frac{HU_i}{PHU}$$

Symbol	Meaning	Input
$\bar{T}$	Mean air temperature of the day	°C
T_{base}	Minimum (base) temperature for growth	`tmp_base` in `plants.plt`, °C
PHU	Potential heat units needed to bring the plant to maturity	see below

fr_{PHU} starts at 0 at planting and reaches 1 at maturity. SWAT+ sets PHU for each plant at the start of the run from the long-term monthly temperatures of the weather generator station and `days_mat` in `plants.plt`:

- `days_mat` > 0: the heat units accumulated over `days_mat` days, starting from an estimated planting date. For annuals the planting date is taken where base-zero heat units reach 15% of their annual total.
- `days_mat` $= 0$: 95% of the heat units above T_{base} in a whole year, so the plant can stay active through the entire growing season.

For perennials a second counter runs over the years to maturity, with $PHU\times$ `yrs_mat` as its target. It sets canopy height and root depth.

Base-zero heat units ($T_{base} = 0$, divided by the annual total) are also accumulated each year from 1 January. They schedule operations that happen when no plant is growing.

Dormancy

Perennials and cold-season annuals go dormant when the day length falls below the minimum day length of the station plus a threshold. While dormant a plant does not grow. When a perennial goes dormant its seed mass is moved to residue and its heat unit counter is reset; it starts accumulating heat units again when the day length rises above the threshold. Cold-season annuals go dormant only if they have reached less than 75% of their heat units.

Light interception and biomass

Each growing plant intercepts photosynthetically active radiation (PAR), taken as half of incoming solar radiation, according to Beer's law:

$$H_{phosyn} = 0.5\, H_{day}\left[1 - \exp\left(-k_\ell\left(LAI + 0.05\right)\right)\right]$$

where H_{day} is the solar radiation (MJ/m^2), k_ℓ the light extinction coefficient (`ext_co` in `plants.plt`) and LAI the leaf area index of the plant. Each plant intercepts light as though it were alone: SWAT+ 62.0.1 has a routine for competition for light between taller and shorter plants, but it is switched off.

Radiation use efficiency

The radiation use efficiency (RUE) is the biomass produced per unit of intercepted light. It responds to CO2 (Stockle et al., 1992). When `co2` in `parameters.bsn` is above 350 ppm,

$$RUE = \frac{100\, CO_2}{CO_2 + \exp(r_1 - r_2\, CO_2)}$$

where r_1 and r_2 are fitted so that RUE equals `bm_e` at 330 ppm and `bm_e_hi` at `co2_hi`. Otherwise RUE is `bm_e`. CO_2 is the concentration for the year (see Climate and weather).

When the vapour pressure deficit vpd is above 1 kPa, RUE declines:

$$RUE = RUE - \Delta ruedcl_{vpd}\,(vpd - 1), \qquad RUE \geq 0.27\,\text{bm_e}$$

where $\Delta ruedcl_{vpd}$ is `ru_vpd` in `plants.plt`.

Biomass growth

Potential daily biomass growth is $\Delta bio = RUE \times H_{phosyn}$ (kg/ha). The actual growth is

$$\Delta bio_{act} = \Delta bio\, \gamma_{reg}, \qquad \gamma_{reg} = \min\left(\gamma_w,\ \gamma_t,\ \gamma_n,\ \gamma_p,\ \gamma_a,\ \gamma_s\right)$$

where each γ is a growth factor between 0 (no growth) and 1 (no stress) for water, temperature, nitrogen, phosphorus, aeration and salinity. When salinity is simulated with the salt tolerance option, the salinity factor multiplies the minimum of the others instead. Plant carbon is taken as 42% of biomass. `nostress` in `codes.bsn` turns stresses off: `1` sets all factors to 1 and `2` sets only the nitrogen and phosphorus factors to 1.

The new biomass is split into roots, leaves, stems and seed. The root fraction falls from `rt_st_beg` at the start of the season to `rt_st_end` at maturity (for perennials, over the first 20% of the years to maturity).

Leaf area and canopy

Leaf area development

Leaf area follows an optimal curve defined by two points from `plants.plt`: at `frac_hu1` of the heat units the plant reaches `lai_max1` of its maximum LAI, and at `frac_hu2` it reaches `lai_max2`. The fraction of maximum LAI for a given fr_{PHU} is

$$fr_{LAImx} = \frac{fr_{PHU}}{fr_{PHU} + \exp(\ell_1 - \ell_2\, fr_{PHU})}$$

with the shape coefficients ℓ_1 and ℓ_2 fitted to the two points. Until fr_{PHU} reaches `hu_lai_decl`, the leaf area added on a day is

$$\Delta LAI = \left(fr_{LAImx,i} - fr_{LAImx,i-1}\right) LAI_{mx} \left[1 - \exp\left(5\left(LAI_{i-1} - LAI_{mx}\right)\right)\right] \sqrt{\gamma_w}$$

and is shared among plants in proportion to their LAI times canopy height. LAI_{mx} is `lai_pot`. For perennials it grows with age:

$$LAI_{mx} = \text{lai_pot} \left(\frac{yr_{cur}}{\text{yrs_mat}}\right)^{\text{bm_tree_acc}}$$

Leaf senescence

After `hu_lai_decl`, annuals lose leaf area until maturity:

$$LAI = LAI_{mx} \left(\frac{1 - fr_{PHU}}{1 - fr_{PHU,sen}}\right)^{\text{dlai_rate}}$$

where $fr_{PHU,sen}$ is the heat unit fraction at which decline started. Temperature-driven perennials drop to `lai_min` over 15 days once `hu_lai_decl` is passed. Moisture-driven perennials shed leaves continuously at a turnover rate between `leaf_tov_mn` and `leaf_tov_mx` months, faster when the 30-day precipitation to PET ratio is below 0.5. Leaves that fall are added to the surface residue.

Canopy height and roots

$$h_c = h_{c,mx} \sqrt{fr_{LAImx}}$$

with $h_{c,mx}$ = `can_ht_max`. Root depth grows with heat units:

$$z_{root} = 2.5\, fr_{PHU}\, z_{root,mx}, \qquad 25.4 \text{ mm} \le z_{root} \le \text{profile depth}$$

with $z_{root,mx}$ = `rt_dp_max` (converted to mm). Perennials use the multi-year heat unit fraction, so their roots keep deepening until maturity.

Water uptake

The potential transpiration of the HRU (see Soil water and temperature) is shared among growing plants in proportion to their LAI. Each plant takes up water from the layers within its root zone. The potential uptake down to depth z is

$$w_{up,z} = \frac{E_t}{1 - \exp(-\beta_w)} \left[1 - \exp\left(-\beta_w \frac{z}{z_{root}}\right)\right]$$

with the water use distribution parameter β_w fixed at 10, so about half of the demand comes from the top 7% of the root zone. The uptake from a layer is the potential uptake to its bottom minus that to its top, with the shortfall of the layers above added back in part:

$$w'_{up,ly} = w_{up,z_l} - w_{up,z_u}\,(1 - \text{epco})$$

where `epco` is the plant uptake compensation factor in `hydrology.hyd`. With `epco` near 1 deeper layers can make up for dry layers above; near 0 they cannot. The second soil layer always compensates fully. Uptake is limited by the water in the layer, and SWAT+ does not reduce uptake as the soil dries below 25% of available water as SWAT 2009 did.

The water stress factor is the ratio of actual to potential transpiration:

$$\gamma_w = \frac{\sum w_{actualup}}{E_t}$$

Aeration stress

When the profile contains more water than field capacity, roots lack oxygen. With $sat_{co} = (SW - FC)/(SAT - FC)$ for the whole profile and $a =$ `aeration` in `plants.plt`,

$$\gamma_a = 1 - \frac{s}{s + \exp(2.9014 - 0.03867\,s)}, \qquad s = 100\,\frac{sat_{co} - a}{1.0001 - a}$$

for $s > 0$; otherwise $\gamma_a = 1$.

Temperature stress

In SWAT+ 62.0.1 temperature stress is

$$\gamma_t = \sin\left(\frac{\pi}{2} \cdot \frac{\bar{T} - T_{base}}{T_{opt} - T_{base}}\right), \qquad 0 \le \gamma_t \le 1$$

where T_{opt} is `tmp_opt`. The factor is 0 at or below the base temperature, 1 at the optimum, falls again above it, and reaches 0 at $2\,T_{opt} - T_{base}$. The SWAT 2009 exponential temperature stress and the cold-night rule (no growth when the minimum temperature is 15 °C below the annual mean) are computed first but replaced by this sine function.

Nitrogen and phosphorus

Demand

The optimal nitrogen fraction of the plant falls as it develops:

$$fr_N = (fr_{N,1} - fr_{N,3})\left[1 - \frac{x}{x + \exp(n_1 - n_2\,x)}\right] + fr_{N,3}$$

where $fr_{N,1}$, $fr_{N,2}$ and $fr_{N,3}$ are `frac_n_em`, `frac_n_50` and `frac_n_mat` (at emergence, at 50% of maturity and at maturity), n_1 and n_2 are fitted to them, and x is fr_{PHU} for annuals or the fraction of `yrs_mat` reached for perennials. The demand is the optimal nitrogen mass, $fr_N \times$ biomass, minus the nitrogen in the plant, and is limited to four times `frac_n_mat` times the day's potential growth. Phosphorus follows the same form with `frac_p_em`, `frac_p_50` and `frac_p_mat`.

Uptake

Nitrate is taken from the root zone with a depth distribution like water uptake, using `n_uptake` in `parameters.bsn` as the distribution parameter; labile phosphorus uses `p_uptake`. When several plants compete and the soil cannot meet the total demand, each plant gets a share in proportion to its demand.

Nitrogen fixation

Legumes (`nfix_co` > 0 in `plants.plt`) fix nitrogen when the soil cannot supply their demand. Fixation depends on growth stage, soil water and soil nitrate:

$$N_{fix} = \min\left(6,\ f_{gr}\ \min(1, f_{sw}, f_{no3})\,(N_{dem} - N_{up})\right)$$

Factor	Value
f_{gr}	0 before 15% of heat units; rises to 1 at 30%; 1 to 55%; falls to 0 at 75%
f_{sw}	$SW/(0.85\,FC)$ for the profile
f_{no3}	1 for profile nitrate up to 100 kg N/ha, falling linearly to 0 at 300 kg N/ha

The result is blended with the unmet demand by the legume coefficient, $N'_{fix} = \text{nfix_co}\,N_{fix} + (1 - \text{nfix_co})(N_{dem} - N_{up})$, and limited to the unmet demand and to `n_fix_max` in `parameters.bsn` (kg N/ha/day).

Nutrient stress

For non-legumes the nitrogen stress factor is

$$\gamma_n = \frac{\varphi}{\varphi + \exp(3.535 - 0.02597\,\varphi)}, \qquad \varphi = 200\left(\frac{N_{plant}}{N_{opt}} - 0.5\right)$$

which is 0 when the plant contains half its optimal nitrogen and close to 1 at the optimum. It is raised to the ratio of the day's uptake to demand when that is larger. Legumes have no nitrogen stress. Phosphorus stress uses the same form.

Harvest index and yield

The harvest index is the fraction of above-ground biomass removed as yield. It grows through the season towards the optimal value `harv_idx` in `plants.plt`:

$$HI = \text{harv_idx}\,\frac{100\,fr_{PHU}}{100\,fr_{PHU} + \exp(11.1 - 10\,fr_{PHU})}$$

After half the heat units are reached, water deficit lowers it. With $wu = 100\sum E_t / \sum PET$ accumulated from that point,

$$HI_{act} = (HI - HI_{min})\,\frac{wu}{wu + \exp(6.13 - 0.0883\,wu)} + HI_{min}$$

with $HI_{min} = \text{HI}/2$. The seed (yield) pool of the plant is kept at this fraction of the above-ground biomass as the plant grows. A grain harvest raises the plant's harvest index target to at least `harv_idx_ws`, but only after the seed pool has been built, so `harv_idx_ws` does not change the yield of that harvest.

Harvest operations are defined in `harv.ops` by `harv_typ`, `harv_idx`, `harv_eff` and `harv_bm_min`. A harvest takes place only when the plant biomass is above `harv_bm_min` (kg/ha).

harv_typ	Yield removed
grain, picker	harv_eff × seed mass; the rest of the seed goes to the soil's active humus
biomass, tree, stripper	harv_idx × harv_eff × (seed + leaf + stem)
tuber, peanuts	Below-ground yield
residue	A fraction of surface residue, given in the operation

Yield is reduced by any pesticide stress on the plant, and pesticide in the plant is removed with the yield.

Land use management

landuse.lum

Each HRU points to a row of landuse.lum, which names the plant community (plnt_com), the management schedule (mgt), the curve number row (cn2), the conservation practice (cons_prac), the urban land type, the overland Manning's n row (ov_mann), and any tile drain, septic system, filter strip, grassed waterway or user BMP.

Management schedules

A schedule in management.sch lists operations by date or by heat units, followed by the names of any decision tables that apply to the HRU. Each operation has a type, a month and day, a heat unit value and up to three data columns:

Operation	Action
plnt	Plant one plant or the whole community
harv	Harvest (see above)
kill	Kill the plant; biomass goes to residue
hvkl	Harvest and kill
till	Tillage, mixing residue and nutrients into the soil (tillage.til)
irrm	Irrigation by date (irr.ops)
fert	Fertiliser (fertilizer.frt)
manu	Manure application
pest	Pesticide application (chem_app.ops)
graz	Start grazing (graze.ops)
cnup	Change the curve number
burn	Prescribed burn (fire.ops)
swep	Street sweeping
dwm	Set drain depth for drainage water management
weir	Set or change a paddy or wetland weir height
irrp, irpm, pudl	Rice paddy irrigation and puddling
mons	Start or end of a monsoon planting window
skip	Skip to the end of the year

What each operation does to plants and soil, with its equations, is described with its parameters: harvest types and harvest efficiency, grazing, irrigation, burning and street sweeping in *.ops; fertilizer and manure placement in fertilizer.frt; tillage and biological mixing in tillage.til; and automatic irrigation and fertilization by decision table in *.dtl.

An operation with a heat unit value runs when the heat unit fraction passes it: fr_{PHU} of the plant named in the operation if that plant is growing, or the base-zero heat units otherwise. An operation without a heat unit value runs on its month and day. When all remaining operations of a year share one date, the scheduler moves on after one full pass.

Decision tables

Decision tables in lum.dtl are conditional rule sets: each table has conditions, alternatives that combine them, and actions that fire when an alternative is true. The tables named in an HRU's schedule are evaluated every day

before the other landscape processes, so they can plant, harvest, irrigate, fertilise or change land use in response to weather, soil water or plant growth. See Decision tables.

Switches and parameters

Input	File	Effect
`nostress`	`codes.bsn`	`0` all stresses, `1` no stress, `2` no nutrient stress
`co2`	`parameters.bsn`	CO2 concentration without `co2_yr.dat`, ppm
`n_uptake`, `p_uptake`	`parameters.bsn`	Depth distribution of N and P uptake
`n_fix_max`	`parameters.bsn`	Maximum daily N fixation, kg N/ha
`epco`	`hydrology.hyd`	Plant uptake compensation
`plnt_typ`, `gro_trig`, `days_mat`	`plants.plt`	Plant type, growth trigger, days to maturity
`bm_e`, `bm_e_hi`, `co2_hi`, `ru_vpd`, `ext_co`	`plants.plt`	Light interception and RUE
`lai_pot`, `frac_hu1`, `lai_max1`, `frac_hu2`, `lai_max2`, `hu_lai_decl`, `dlai_rate`, `lai_min`	`plants.plt`	Leaf area curve and decline
`can_ht_max`, `rt_dp_max`, `rt_st_beg`, `rt_st_end`	`plants.plt`	Canopy height, root depth, root fraction
`tmp_base`, `tmp_opt`	`plants.plt`	Temperature response
`frac_n_*`, `frac_p_*`, `nfix_co`	`plants.plt`	Nutrient demand and fixation
`harv_idx`, `harv_idx_ws`	`plants.plt`	Harvest index
`aeration`	`plants.plt`	Aeration stress threshold
`yrs_mat`, `bm_tree_acc`, `bm_tree_max`, `leaf_tov_mn`, `leaf_tov_mx`	`plants.plt`	Perennials
`bm_dieoff`	`plants.plt`	Read but not applied at dormancy in 62.0.1

Related pages

- `plants.plt`, `landuse.lum`, `management.sch` and operations, `lum.dtl`
- Soil water and temperature for the transpiration demand.
- Nutrient cycling for the soil nitrogen and phosphorus pools.
- Carbon for residue and soil carbon.
- Decision tables.

Sources

- Community, PAR and transpiration split: `pl_community.f90`
- Growth driver: `pl_grow.f90`; biomass: `pl_biomass_gro.f90`
- Heat units and nutrient demand: `pl_nut_demand.f90`, `pl_nupd.f90`, `pl_pupd.f90`; PHU setup in `plant_init.f90`
- Leaf area, roots, seed: `pl_leaf_gro.f90`, `pl_leaf_senes.f90`, `pl_root_gro.f90`, `pl_seed_gro.f90`
- Dormancy: `pl_dormant.f90`
- Stresses: `pl_waterup.f90`, `pl_tstr.f90`, `nuts.f90`
- Uptake and fixation: `pl_nup.f90`, `pl_pup.f90`, `pl_nfix.f90`
- Scheduling: `mgt_operatn.f90`, `mgt_sched.f90`, `mgt_read_mgtops.f90`; decision tables evaluated in `hru_control.f90`
- Harvest: `mgt_harvgrain.f90`, `mgt_harvbiomass.f90`, `mgt_harvtuber.f90`, `mgt_harvresidue.f90`; `type harvest_operation` in `mgt_operations_module.f90`
- Plant database type: `type plant_db` in `plant_data_module.f90`

`plants.plt` column	Fortran variable
`plnt_typ`, `gro_trig`	`pldb%typ`, `pldb%trig`
`bm_e`, `bm_e_hi`, `co2_hi`, `ru_vpd`	`pldb%bio_e`, `pldb%bioehi`, `pldb%co2hi`, `pldb%wavp`
`harv_idx`, `harv_idx_ws`	`pldb%hvsti`, `pldb%wsyf`
`lai_pot`, `frac_hu1`, `lai_max1`, `frac_hu2`, `lai_max2`	`pldb%blai`, `pldb%frgrw1`, `pldb%laimx1`, `pldb%frgrw2`, `pldb%laimx2`
`hu_lai_decl`, `dlai_rate`, `lai_min`	`pldb%dlai`, `pldb%dlai_rate`, `pldb%alai_min`

plants.plt column	Fortran variable
can_ht_max, rt_dp_max	pldb%chtmx, pldb%rdmx
tmp_opt, tmp_base	pldb%t_opt, pldb%t_base
frac_n_em, frac_n_50, frac_n_mat	pldb%pltnfr1, pldb%pltnfr2, pldb%pltnfr3
frac_p_em, frac_p_50, frac_p_mat	pldb%pltpfr1, pldb%pltpfr2, pldb%pltpfr3
stcon_max, vpd, frac_stcon	pldb%gsi, pldb%vpdfr, pldb%gmaxfr
usle_c_min	pldb%usle_c
plnt_decomp	pldb%rsdco_pl
bm_tree_acc, yrs_mat, bm_tree_max	pldb%laixco_tree, pldb%mat_yrs, pldb%bmx_peren
ext_co, leaf_tov_mn, leaf_tov_mx	pldb%ext_coef, pldb%leaf_tov_min, pldb%leaf_tov_max
rt_st_beg, rt_st_end	pldb%rsr1, pldb%rsr2
n_uptake, p_uptake, n_fix_max (parameters.bsn)	bsn_prm%n_updis, bsn_prm%p_updis, bsn_prm%nfixmx
harv_typ, harv_idx, harv_eff, harv_bm_min (harv.ops)	harvop_db%typ, %hi_ovr, %eff, %bm_min

Symbol	Fortran variable
fr_{PHU}, multi-year fraction	pcom(j)%plcur(ipl)%phuacc, %phuacc_p
PHU	pcom(j)%plcur(ipl)%phumat, %phumat_p
H_{phosyn}	par(ipl)
Δbio, γ_{reg}	bioday, pcom(j)%plstr(ipl)%reg
γ_w, γ_t, γ_n, γ_p, γ_a, γ_s	pcom(j)%plstr(ipl)%strsw, %strst, %strsn, %strsp, %strsa, %strss
E_t per plant	epmax(ipl)
β_w	uptake%water_dis
HI	pcom(j)%plg(ipl)%hi_adj

Overview

SWAT+ follows nitrogen and phosphorus through mineral, organic and plant pools in every soil layer. Fertiliser, manure, residue and atmospheric deposition add nutrients; plants take them up; and runoff, sediment, lateral flow, tile flow, percolation, denitrification and volatilisation remove them. The nutrients that leave the HRU are routed through channels, where in-stream processes are described on the Channels page.

The soil processes below are those of the static carbon option (`carbon = 0` in `codes.bsn`). With `carbon = 2` the residue and humus pools and their decomposition follow the CENTURY-based model on the Carbon page, while nitrification, volatilisation, mineral phosphorus, deposition and all the transport processes below stay the same. `carbon = 1` is reserved for C-FARM, which is not implemented; with it SWAT+ 62.0.1 skips residue decomposition and mineralisation altogether.

Plant uptake and nitrogen fixation are described in Plant growth and land use management.

Soil pools

Element	Pool	Contents
N	Nitrate	NO3-N in solution, mobile with water
N	Ammonium	NH4-N, adsorbed on the soil
N	Active humus N	The part of humus organic N that mineralises
N	Stable humus N	The rest of humus organic N
N	Residue N	Organic N in fresh residue, kept separately for each plant in each layer and on the surface
P	Labile (solution) P	Soluble mineral P available to plants and mobile with water
P	Active mineral P	Mineral P in rapid equilibrium with labile P
P	Stable mineral P	Mineral P in slow equilibrium with active P
P	Humus organic P	Organic P in humus
P	Residue P	Organic P in fresh residue

Manure applied to the soil has separate organic pools, which are also lost with sediment.

Initial values

The initial nutrient contents come from the row of nutrients.sol named for the HRU in soil_plant.ini, with built-in values when there is none:

Pool	Initial value
Nitrate	`nitrate` (7 mg/kg) at the surface, decreasing with depth as exp(−exp_co z), with z the depth to the bottom of the layer in mm
Labile P	`lab_p` (5 mg/kg) at the surface, decreasing with depth the same way
Active mineral P	Labile P × (1 − PSP)/PSP
Stable mineral P	4 × active P; with the Vadas and White P model, from the White et al. (2009) ratio of 1 to 7 times active plus labile P
Humus C, N and P	Humus C from the organic carbon of the layer (`carbon` in soils.sol); N and P from the humus C:N and C:P ratios `hum_c_n` (10) and `hum_c_p` (80)
Active and stable humus	`fr_hum_act` (0.02) of humus in the active pool

Concentrations in mg/kg are converted to kg/ha with the bulk density and thickness of each layer. The other soil test columns of nutrients.sol (`inorgp`, `watersol_p`, `h3a_p`, `mehlich_p`, `bray_strong_p`) are not used.

Temperature and water factors

Mineralisation, decomposition and denitrification depend on soil temperature and water through two factors:

$$\gamma_{tmp} = \max\left(0.1,\ 0.9\,\frac{T_{soil}}{T_{soil} + \exp(9.93 - 0.312\,T_{soil})} + 0.1\right), \qquad \gamma_{sw} = \max\left(0.05,\ 0.1 + 0.9\sqrt{\frac{SW_{ly}}{FC_{ly}}}\right)$$

where T_{soil} is the layer temperature (°C), SW_{ly} the layer water and FC_{ly} its field capacity water. They combine as $\gamma = \sqrt{\gamma_{tmp}\,\gamma_{sw}}$. None of these processes runs in a layer at or below 0 °C. For the 10 mm surface layer, SWAT+ uses the temperature and water content of the second layer.

Humus mineralisation

Nitrogen moves between the active and stable humus pools towards a fixed active fraction:

$$N_{trns} = 10^{-5}\left[orgN_{act}\left(\frac{1}{fr_{actN}} - 1\right) - orgN_{sta}\right]$$

where fr_{actN} is `n_act_frac` in carbon.bsn (0.02). A positive value moves N from active to stable humus.

Active humus N mineralises to nitrate:

$$N_{mina} = \beta_{min}\,\gamma\,orgN_{act}$$

with β_{min} the rate coefficient for humus mineralisation, `orgn_min` in parameters.bsn. Phosphorus is mineralised from humus in proportion:

$$P_{mina} = 1.4\,N_{mina}\,\frac{orgP_{hum}}{orgN_{act} + orgN_{sta}}$$

and added to the labile pool.

Residue decomposition

Fresh residue decomposes at a rate limited by its C:N and C:P ratios. For each plant's residue in each layer,

$$\gamma_{ntr} = \min\left(\exp\left[-0.693\,\frac{\varepsilon_{C:N} - 25}{25}\right],\ \exp\left[-0.693\,\frac{\varepsilon_{C:P} - 200}{200}\right],\ 1\right)$$

with $\varepsilon_{C:N}$ capped at 500 and $\varepsilon_{C:P}$ at 5000. The fraction decomposed on the day is

$$\delta_{ntr} = \min\left[1,\ \max\left(\text{rsd_decay},\ \beta_{rsd}\,\gamma_{ntr}\,\gamma\right)\right]$$

where β_{rsd} is the residue decomposition coefficient of the plant (`plnt_decomp` in `plants.plt`) and `rsd_decay` in `parameters.bsn` is a minimum daily rate. Of the N and P released, 80% goes to nitrate and labile P, and 20% to humus (active humus N; stable humus P for soil residue, active humus P for surface residue).

Surface residue decomposes the same way, using the temperature and water of the top layer.

Nitrification and ammonia volatilisation

Ammonium is nitrified and volatilised together in each layer (Reddy et al., 1979; Godwin et al., 1984). The temperature factor is

$$\eta_{tmp} = 0.41\,\frac{T_{soil} - 5}{10}$$

and both processes need $\eta_{tmp} \geq 0.001$. The water factor for nitrification rises linearly from 0 at wilting point to 1 at wilting point plus 25% of the field capacity water. Volatilisation decreases with depth through

$$\eta_{midz} = 1 - \frac{z_{mid}}{z_{mid} + \exp(4.706 - 0.0305\,z_{mid})}$$

with z_{mid} the depth to the middle of the layer (mm), and uses a cation exchange factor of 0.15. With $\eta_{nit} = \eta_{tmp}\,\eta_{sw}$ and $\eta_{vol} = 0.15\,\eta_{tmp}\,\eta_{midz}$, the ammonium lost to both is

$$N_{nit|vol} = NH_4\left[1 - \exp(-\eta_{nit} - \eta_{vol})\right]$$

and is split between nitrification and volatilisation in proportion to $1 - \exp(-\eta_{nit})$ and $1 - \exp(-\eta_{vol})$.

Denitrification

When the water factor reaches a threshold, nitrate in the layer is denitrified:

$$N_{denit} = NO_3\left[1 - \exp\left(-\beta_{denit}\,\gamma_{tmp}\,\frac{orgC}{100}\right)\right] \quad \text{for } \gamma_{sw} \geq \gamma_{sw,thr}$$

Symbol	Meaning	Input
β_{denit}	Denitrification rate coefficient	`denit_exp` in `parameters.bsn`
$\gamma_{sw,thr}$	Threshold of the water factor	`denit_frac` in `parameters.bsn`
$orgC$	Organic carbon of the layer	`carbon` in `soils.sol`, %

The septic biozone layer of an active septic system is excluded.

Atmospheric deposition

Nitrate and ammonium in rain and dry deposition are added to the surface layer every day from the station named in the `atmo_dep` column of `weather-sta.cli`. Rain deposition is the concentration in rain (mg/L) times the day's precipitation; dry deposition is spread evenly over the days of the month or year, depending on whether `atmodep.cli` gives monthly, yearly or average annual values.

Nitrate movement

Nitrate is mobile and moves with water. In each layer the water that moves is the percolation, lateral flow, surface runoff (top layer only) and tile flow (tile layer only), w_{mobile}. The nitrate concentration in the mobile water is

$$conc_{NO_3} = \frac{NO_3 \left[1 - \exp\left(\frac{-w_{mobile}}{(1-\theta_e)\, SAT_{ly}}\right)\right]}{w_{mobile}}$$

where θ_e is the fraction of porosity from which anions are excluded (`anion_excl` in `soils.sol`) and SAT_{ly} the saturation water of the layer. Then:

Path	Nitrate removed
Surface runoff	$\beta_{NO_3}\, conc_{NO_3}\, Q_{surf}$, top layer, with β_{NO_3} = `n_perc` in `parameters.bsn`
Lateral flow	$conc_{NO_3}\, Q_{lat,ly}$ for each layer
Tile flow	`nperco_lchtile` × concentration in the tile flow × Q_{tile}, with the concentration taken over the tile layer and all layers below
Percolation	$conc_{NO_3}\, w_{perc,ly}$, added to the next layer; from the bottom layer it leaves the profile

`n_perc` close to 0 gives little nitrate in surface runoff; close to 1 gives runoff the same concentration as percolate. As water evaporates from the second layer, a small amount of nitrate moves up with it into the surface layer.

Nitrate in lateral and tile flow passes through the same lag stores as the water (see Hydrology). When the gwflow module passes groundwater into the soil profile, it also adds the nitrate and phosphorus that the groundwater contains.

Organic N and P with sediment

Organic nitrogen leaving with sediment is

$$orgN_{surf} = 1000\, \frac{orgN_{ly1}}{M_{ly1}}\, \varepsilon_N\, \frac{sed}{A}$$

Symbol	Meaning	Units
$orgN_{surf}$	Organic N removed in sediment	kg N/ha
$orgN_{ly1}$	Active plus stable humus N in the surface layer	kg N/ha
M_{ly1}	Mass of the surface layer, $10000\, \rho_b\, depth$	kg/ha
ε_N	Nitrogen enrichment ratio	-
sed	Sediment yield	metric tons
A	HRU area	ha

Sediment-attached phosphorus is computed the same way from the humus, manure and active and stable mineral P of the surface layer, using ε_P, and is reported as organic P and mineral P.

The enrichment ratio accounts for the finer, nutrient-rich particles that erosion removes. It is `orgn_enrich` (N) or `orgp_enrich` (P) in `hydrology.hyd` when that is above 0.001. Otherwise it is computed from the sediment concentration in runoff (Menzel, 1980):

$$\varepsilon = 0.78\,(conc_{sed})^{-0.2468}, \qquad conc_{sed} = \frac{sed}{10\,A\,Q_{surf}}, \qquad \varepsilon \le 3$$

Soluble phosphorus

Labile phosphorus is strongly bound by the soil, so little of it moves with water. The amount removed in surface runoff is

$$P_{surf} = \frac{P_{lab,1}\,Q_{surf}}{\rho_b\,depth_1\,k_{d,surf} + 1}$$

where $P_{lab,1}$ is the labile P of the surface layer (kg/ha), ρ_b and $depth_1$ its bulk density and thickness (mm), and $k_{d,surf}$ the phosphorus soil partitioning coefficient, `p_soil` in `parameters.bsn` (m^3/Mg).

Leaching of labile P from each layer to the next is

$$P_{perc} = 0.001\,P_{lab}\left[1 - \exp\left(\frac{-w_{perc}}{0.01\,SW_{ly} + 0.1\,k_{d,perc}\,\rho_b}\right)\right]$$

with $k_{d,perc}$ the phosphorus percolation coefficient `p_perc` in `parameters.bsn`. The same expression with tile flow in place of percolation gives the P in tile flow from the tile layer. This form differs from the SWAT 2009 leaching equation.

Mineral phosphorus transformations

Original model

With `soil_p = 0` in `codes.bsn`, labile and active mineral P move towards the ratio set by the phosphorus availability index pai (`p_avail` in `parameters.bsn`):

$$P_{sol|act} = P_{lab} - P_{act}\,\frac{pai}{1 - pai}$$

Movement from labile to active P takes 10% of the imbalance per day; movement back takes 60%. Active and stable P move towards a stable pool four times the active pool:

$$P_{act|sta} = 0.01\,(4\,P_{act} - P_{sta})$$

with a tenth of that rate when stable P returns to the active pool.

Vadas and White model

With `soil_p = 1`, SWAT+ uses the phosphorus model of Vadas and White (2010). The availability index is computed for each layer from clay content, labile P and organic carbon, and averaged over 30 days. The rate of movement between labile and active P depends on how many days the imbalance has lasted. The active to stable equilibrium ratio follows the White et al. (2009) relationship, and the active to stable rate depends on the calcium carbonate content (`caco3` in `soils.sol`). Both rates are scaled by the water content of the layer relative to field capacity, with a lower limit of 0.25.

Lag of nutrients in surface runoff

Nitrate and labile P in surface runoff, organic N and P, and mineral P attached to sediment pass through the surface lag store with the runoff and sediment, and the same daily fraction is released (see Hydrology).

Algae, CBOD and dissolved oxygen in runoff

Each day SWAT+ also gives the water leaving an HRU a chlorophyll-a, carbonaceous BOD and dissolved oxygen load, which the channels then use as inflow concentrations. On days without water yield all three are zero. With water yield Q (surface runoff, lateral flow and tile flow, mm):

- Chlorophyll-a is 0.1 times the concentration of organic N in sediment plus nitrate in surface runoff, $100\,(N_{org,sed} + NO_{3,surq})/Q$.
- CBOD is 2.7 times the organic carbon in the day's sediment yield, divided by the water yield. The carbon is the soil carbon content of the top layer times the sediment yield and the enrichment ratio, or, with `carbon` = 2 in `codes.bsn`, the carbon the carbon model reports in the sediment.
- Dissolved oxygen is $O_{sat}\, e^{-0.1\,CBOD}$, limited to between 0 and O_{sat}, with O_{sat} the saturation concentration at a water temperature of $5 + 0.75\,\bar{T}_{air}$ (see Channels for the formula).

Septic systems

A land use with a septic system (the `sep` column of `landuse.lum`) adds a biozone layer to the soil, and the biozone routine runs on days when that layer is above freezing. In 62.0.1 the effluent volume is never computed, so septic systems add no water, nitrogen, phosphorus or BOD to the soil. See `septic.str` for the inputs.

In-stream processes

Nutrients entering channels are transformed by the QUAL2E water quality routines. With `i_fpwet = 1` in `codes.bsn` (the column read as the QUAL2E switch), a simplified exchange between dissolved and particulate N and P is added. See Channels.

Switches and parameters

Input	File	Effect
`carbon`	`codes.bsn`	`0` static pools, `2` CENTURY-based carbon, `1` reserved
`soil_p`	`codes.bsn`	`0` original mineral P model, `1` Vadas and White (2010)
`orgn_min`	`parameters.bsn`	Humus mineralisation rate coefficient
`rsd_decay`	`parameters.bsn`	Minimum daily residue decomposition
`denit_exp`, `denit_frac`	`parameters.bsn`	Denitrification rate and water threshold
`n_perc`	`parameters.bsn`	Nitrate in surface runoff relative to percolate
`nperco_lchtile`	`parameters.bsn`	Nitrate concentration coefficient for tile flow
`p_soil`, `p_perc`	`parameters.bsn`	Phosphorus partitioning in runoff and percolation
`p_avail`	`parameters.bsn`	Phosphorus availability index (original model)
`n_act_frac`	`carbon.bsn`	Active fraction of humus N
`orgn_enrich`, `orgp_enrich`	`hydrology.hyd`	Fixed enrichment ratios
`plnt_decomp`	`plants.plt`	Residue decomposition coefficient of each plant
`nitrate`, `lab_p`, `exp_co`, `fr_hum_act`, `hum_c_n`, `hum_c_p`	`nutrients.sol`	Initial nutrient contents
`carbon`, `anion_excl`, `caco3`	`soils.sol`	Organic carbon, anion exclusion, CaCO3

`orgn_min`, `n_perc`, `nperco_lchtile`, `p_soil`, `p_perc` and `p_avail` are copied to every HRU at the start of the run, so calibration can change them for single HRUs.

Related pages

- Plant growth and land use management for uptake and fixation.
- Carbon for the CENTURY-based pools.
- Erosion and sediment for the sediment yield that transports organic nutrients.

- fertilizer.frt and manure allocation for nutrient inputs.
- Channels for in-stream processes.

Sources

- Daily sequence and carbon switch: hru_control.f90
- Initial pools: soil_nutcarb_init.f90; `type soiltest_db` in soil_data_module.f90
- Mineralisation, soil residue decomposition, denitrification: nut_nminrl.f90; surface residue: rsd_decomp.f90
- Nitrification and volatilisation: nut_nitvol.f90
- Deposition: nut_nrain.f90
- Nitrate movement: nut_nlch.f90; upward nitrate with evaporation in et_act.f90
- Sediment-bound nutrients: nut_orgn.f90, nut_psed.f90; enrichment ratio: pest_enrsb.f90
- Soluble P: nut_solp.f90; mineral P: nut_pminrl.f90, nut_pminrl2.f90
- Chlorophyll-a, CBOD and dissolved oxygen in HRU runoff: swr_subwq.f90 (`chl_a`, `cbodu`, `doxq`; the code labels the organic N plus nitrate concentration `tp`), passed to the HRU outflow in hru_hyds.f90
- Septic biozone: sep_biozone.f90, called from `hru_control.f90`; set up in `soils_init.f90`
- Lag stores: swr_substor.f90, stor_surfstor.f90
- HRU copies of basin nutrient parameters: hrudb_init.f90
- `n_act_frac`: carbon_module.f90

Column or symbol	Fortran variable
`carbon`, `soil_p`, `i_fpwet` (codes.bsn)	`bsn_cc%cswat`, `bsn_cc%sol_p_model`, `bsn_cc%qual2e`
`orgn_min` (β_{min})	`bsn_prm%cmn`, per HRU `hru(j)%nut%cmn`
`rsd_decay`	`bsn_prm%decr_min`
`denit_exp`, `denit_frac`	`bsn_prm%cdn`, `bsn_prm%sdnco`
`n_perc`, `nperco_lchtile`	`hru(j)%nut%nperco`, `hru(j)%nut%nperco_lchtile`
`p_soil`, `p_perc`, `p_avail`	`hru(j)%nut%phoskd`, `hru(j)%nut%pperco`, `hru(j)%nut%psp`
`orgn_enrich`, `orgp_enrich`	`hru(j)%hyd%erorgn`, `hru(j)%hyd%erorgp`
`plnt_decomp`	`pldb%rsdco_pl`
Nitrate, ammonium	`soil1(j)%mn(ly)%no3`, `%nh4`
Active and stable humus	`soil1(j)%hact(ly)`, `soil1(j)%hsta(ly)`
Labile, active, stable mineral P	`soil1(j)%mp(ly)%lab`, `%act`, `%sta`
Residue in soil, on surface	`soil1(j)%pl(ipl)%rsd(ly)`, `pl_mass(j)%rsd(ipl)`
γ_{tmp}, γ_{sw}, γ	`cdg`, `sut`, `csf`
ε (computed)	`enratio`
Nitrate in runoff, lateral, tile, percolation	`surqno3(j)`, `latno3(j)`, `tileno3(j)`, `percn(j)`
Sediment-bound N and P	`sedorgn(j)`, `sedorgp(j)`, `sedminpa(j)`, `sedminps(j)`
Soluble P in runoff	`surqsolp(j)`

Overview

The `carbon` column of codes.bsn selects how SWAT+ handles soil organic matter:

`carbon`	Model
0	Static model. Residue decomposes and organic N and P mineralise as in SWAT; soil carbon is not tracked as a state. See Nutrient cycling.
1	Reserved for C-FARM, which is not implemented.
2	Carbon model based on CENTURY (Parton et al., 1993, 1994) as implemented in SWAT-C by Zhang (2018). Carbon, nitrogen and phosphorus move through litter and soil organic matter pools in every soil layer.

The carbon model is switched on by `carbon = 2`; earlier revisions used 1. With `carbon = 1` neither model runs, so residue does not decompose and organic N and P are not mineralised. Set the column to 0 or 2.

When `carbon = 2`, the files `carbon.bsn` (basin-wide settings) and `carbon_lyr.bsn` (rates and CO_2 fractions) are required, and the run stops if either is missing. The legacy `carb_coefs.cbn` file is no longer read.

Pools

Each soil layer has three litter pools and three soil organic matter pools. Every pool contains mass, carbon, nitrogen and phosphorus.

Pool	Content
Metabolic litter	Easily decomposed part of residue and manure
Structural litter	Resistant part of residue, of which a share is lignin
Lignin	Lignin within structural litter
Microbial biomass	Active microbes and their products
Slow humus	Organic matter with turnover times of years to decades
Passive humus	Stabilised organic matter with turnover times of centuries

Surface residue is kept separately for each plant until it decomposes into the litter pools of the top soil layer.

Carbon moves between pools as follows. Each transfer releases part of the carbon as CO_2.

- structural litter (non-lignin part) and metabolic litter to microbial biomass
- lignin to slow humus (70 %; the other 30 % is respired)
- microbial biomass to slow humus and passive humus
- slow humus to microbial biomass and passive humus
- passive humus to microbial biomass

Residue inputs

Surface residue

Each plant's surface residue decomposes when the top soil layer is above 0 °C:

$$\Delta R = \delta\, R, \qquad \delta = \min\left(1,\ \max\left(\delta_{min},\ k_{rsd}\ \gamma_{nut}\ \sqrt{\gamma_{tmp}\,\gamma_{sw}}\right)\right)$$

$$\gamma_{tmp} = \max\left(0.1,\ 0.9\,\frac{T}{T + e^{9.93 - 0.312\,T}} + 0.1\right), \qquad \gamma_{sw} = \max\left(0.05,\ 0.1 + 0.9\sqrt{SW/FC}\right)$$

$$\gamma_{nut} = \min\left(1,\ e^{-0.693\,(CN - CN_{ref})/CN_{ref}},\ e^{-0.693\,(CP - CP_{ref})/CP_{ref}}\right)$$

Symbol	Meaning	Input
R	surface residue of a plant (mass, C, N, P)	
k_{rsd}	residue decomposition coefficient of the plant	`plnt_decomp` in `plants.plt`
δ_{min}	minimum daily decomposition fraction	`rsd_decay` in `parameters.bsn`
T, SW, FC	top layer temperature (°C), soil water and field capacity (mm)	
CN, CP	residue C:N and C:P ratios, capped at `cnr_cap` and `cpr_cap`	`carbon.bsn`
CN_{ref}, CP_{ref}	ratios at which the factor is 1	`cnr_ref`, `cpr_ref` in `carbon.bsn`

The decomposed residue is added to the litter pools of the top soil layer. The plant's above-ground lignin fraction (`ab_lig_frac` in `plants.plt`) goes to lignin, the lignin fraction divided by 0.8 goes to structural litter (lignin included), and the rest goes to metabolic litter.

Before decomposition, a fraction `sfc_rsd_photodeg` (`carbon.bsn`) of each plant's surface residue is lost each day to photodegradation and counted as emitted carbon.

Roots and incorporated residue

Residue in the soil (dead roots and residue mixed in by tillage) is moved to the litter pools of its layer on each day the layer is above 0 °C, using the plant's below-ground lignin fraction (`bg_lig_frac` in `plants.plt`) in the same way.

Manure adds to the litter pools according to the manure partitioning settings in `carbon.bsn`.

Pool transformations

Environmental controls

The daily rate of every transformation in a layer is the optimal rate times a combined control factor:

$$c_s = \min\left(15,\ 0.9\sqrt{f_{tmp}\, f_{sw}}\ f_{ox}\ f_{till}\right)$$

Temperature f_{tmp} is chosen with `tmpf_eqn` in `carbon.bsn`:

`tmpf_eqn`	Function of layer temperature T (°C)
1	$T/(T + e^{5.058 - 0.2504T})$ up to 35 °C, then $1 - 0.04(T - 35)$ up to 60 °C, then 0
2	$\dfrac{(T - T_n)^q\,(T_x - T)}{(T_o - T_n)^q\,(T_x - T_o)}$ with $q = (T_n - T_o)/(T_o - T_x)$, and 0 outside $T_n < T < T_x$; T_n, T_o, T_x are `t_cbn_min`, `t_cbn_opt`, `t_cbn_max`
3	$0.9\,T/(T + e^{9.93 - 0.312T}) + 0.1$

Soil water f_{sw} is chosen with `watf_eqn`:

`watf_eqn`	Function
1	$0.1 + 0.9\sqrt{SW/FC}$, limited to 0.05 to 1
2	Product of a water factor (rising from 0 at no water to 1 at field capacity) and an aeration factor based on the air-filled pore space; reduces activity in both dry and waterlogged soil

Oxygen decreases with the depth z (mm) of the middle of the layer:

$$f_{ox} = 1 - \frac{0.8\,z}{z + e^{18.40961 - 0.023683632\,z}}$$

Tillage f_{till} depends on `idc_till` in `codes.bsn`:

`idc_till`	Effect
1	$f_{till} = 1.6$ in layers within the tillage depth for `till_eff_days` days after tillage (partly in the layer containing the tillage depth), otherwise 1
2, 4	No effect ($f_{till} = 1$)
3 (default)	f_{till} is 1 plus the layer tillage factor, which rises with each tillage operation and decays afterwards (Kemanian method)

Rates

Potential daily transformation of each pool is its carbon (and nitrogen) content times an optimal rate from `carbon_lyr.bsn`, times c_s:

Pool	Daily fraction transformed	Rate column
Structural litter	$k_{str}\, c_s\, e^{-3\,L}$, where L is the lignin share of structural litter (at most 0.8)	`str_rate`
Metabolic litter	$k_{meta}\, c_s$	`meta_rate`
Microbial biomass	$k_{mic}\, c_s\, x_{bm}$, with $x_{bm} = 1$ in the top layer and $0.25 + 0.0075\,\text{sand}$ below	`microb_top_rate` (top layer), `microb_rate`
Slow humus	$k_{hs}\, c_s$	`hs_rate`
Passive humus	$k_{hp}\, c_s$	`hp_rate`

`carbon_lyr.bsn` has two rows: row 1 applies to the top soil layer and row 2 to all other layers.

Allocation and CO_2

Transfer	Fraction respired as CO_2	Remainder goes to
Metabolic and non-lignin structural litter	`a1co2`	microbial biomass
Lignin	0.3	slow humus
Microbial biomass	`abco2` in the top layer; $0.17 + 0.0068\,\text{sand}$ below	passive humus ($0.003 + 0.00032\,\text{clay}$), rest to slow humus
Slow humus	`asco2`	passive humus (the larger of 0.001 and `hs_hp` minus $0.00009 \times$ clay), rest to microbial biomass
Passive humus	`apco2`	microbial biomass

Sand and clay are in percent.

Nitrogen

Each receiving pool has a target N:C ratio. In the top layer these depend on the N content of the residue; in deeper layers they depend on the mineral N concentration of the layer. When the N released by the decomposing pools is less than the N needed to build the receiving pools, the difference is immobilised from soil nitrate and ammonium. If mineral N cannot cover it, the transformations that need N are reduced in proportion. Surplus N is mineralised to ammonium.

Before the transformations, denitrification removes nitrate when the soil water factor exceeds `denit_frac` in `parameters.bsn`, at a rate set by `denit_exp`, the temperature factor and the layer organic carbon.

Phosphorus

Phosphorus is mineralised from passive humus at 1.4 times the fraction of humus N that was transformed on the day, and goes to the labile mineral pool. Phosphorus in decomposing litter is released in proportion to the litter carbon transformed: 80 % to labile mineral P and 20 % to passive humus.

Carbon in water and sediment

- Sediment: eroded soil removes organic carbon from the top layer in proportion to the share of the layer mass eroded (at most 90 %).
- Dissolved carbon from microbial biomass: in the top layer, microbial carbon is lost to water leaving the layer with $1 - \exp\left(-w/(SAT + 10^{-4}\, K_{oc}\, C)\right)$, where w is runoff plus lateral flow plus percolation, K_{oc} is `koc_c` and C the layer organic carbon. The concentration in surface runoff and lateral flow is `solc_ratio` times the concentration in percolation. In deeper layers, carbon moving with percolation and lateral flow is removed with $1 - \exp(-w/(\phi z - WP))$.
- Carbon leaving the bottom of the profile is reported as percolated carbon.

Initial pools

At the start of the run, the organic carbon of each layer (from soils.sol) is split into pools. A share `init_seq` is humus and microbial carbon, divided between microbial biomass (`init_microb`), slow humus (`init_slow`) and passive humus (`init_passive`). Humus pools start with C:N of 10 and C:P of 80; microbial biomass with C:N of 8 and C:P of 80. With `mathers_method = 1`, slow humus is initialised with the Mathers relation based on clay plus silt content.

Parameters

Parameter	File	Effect
`carbon`	codes.bsn	0 = static, 1 = reserved, 2 = carbon model
`idc_till`	codes.bsn	Tillage effect on decomposition (default 3)
`tmpf_eqn`, `watf_eqn`, `t_cbn_min`, `t_cbn_opt`, `t_cbn_max`	carbon.bsn	Temperature and soil water controls
`till_eff_days`, `bmix_*`, `tillmix_*`, `bio_consol`, `till_consol`	carbon.bsn	Duration and shape of tillage and biological mixing effects
`cnr_cap`, `cnr_ref`, `cpr_cap`, `cpr_ref`, `sfc_rsd_photodeg`	carbon.bsn	Surface residue decomposition
`koc_c`, `solc_ratio`	carbon.bsn	Dissolved carbon losses
`init_seq`, `init_microb`, `init_slow`, `init_passive`, `mathers_method`	carbon.bsn	Initial pool split
`hp_rate`, `hs_rate`, `microb_rate`, `microb_top_rate`, `meta_rate`, `str_rate`	carbon_lyr.bsn	Optimal transformation rates (1/day)
`hs_hp`, `a1co2`, `asco2`, `apco2`, `abco2`	carbon_lyr.bsn	Allocation and CO_2 fractions
`plnt_decomp`, `ab_lig_frac`, `bg_lig_frac`	plants.plt	Residue decomposition rate and lignin fractions

Carbon model parameters can be calibrated through the parameter change files; see Parameter change files.

Related pages

- carbon.bsn and carbon_lyr.bsn
- Carbon outputs
- Nutrient cycling for the static model
- Plant growth and land use management

Sources

Code at revision 62.0.1:

- Model switch: `cswat` in src/basin_module.f90; daily calls in src/hru_control.f90
- Inputs: src/carbon_bsn_read.f90 (reads `carbon.bsn` and `carbon_lyr.bsn`), types in src/carbon_module.f90
- Pool transformations: src/cbn_zhang2.f90; temperature option 2: src/fcgd.f90
- Residue: src/cbn_surfrsd_decomp.f90, src/cbn_rsd_transfer.f90; lignin partitioning in src/plant_parm_read.f90
- Carbon in runoff, sediment and percolation: src/nut_orgnc2.f90
- Initial pools: src/soil_nutcarb_init.f90
- Tillage: src/mgt_newtillmix_cswat1.f90, src/mgt_tillfactor.f90, src/mgt_biomix.f90

Symbols and variables:

Symbol or column	Variable
`carbon` in `codes.bsn`	`bsn_cc%cswat`
c_s, f_{tmp}, f_{sw}, f_{ox}, f_{till}	`org_con%cs`, `%cdg`, `%sut`, `%ox`, `%till_eff`
`tmpf_eqn`, `watf_eqn`, `t_cbn_min/opt/max`	`org_con%tmpf`, `%watf`, `%tn`, `%top`, `%tx`
`koc_c`, `solc_ratio`	`cb_wtr_coef%prmt_21`, `cb_wtr_coef%prmt_44`

Symbol or column	Variable
`init_seq`, `init_microb`, `init_slow`, `init_passive`	`org_frac%frac_seq`, `%frac_hum_microb`, `%frac_hum_slow`, `%frac_hum_passive`
rates in `carbon_lyr.bsn`	`carbdb(1:2)%hp_rate`, `%hs_rate`, `%microb_rate`, `%microb_top_rate`, `%meta_rate`, `%str_rate`, `%hs_hp`
CO_2 fractions	`org_allo(1:2)%a1co2`, `%asco2`, `%apco2`, `%abco2`
microbial to passive, slow to passive	`org_allo%abp`, `org_allo%asp`
pools	`soil1%meta`, `%str`, `%lig`, `%microb`, `%hs`, `%hp`
lignin fractions of residue	`cswat_1_part_fracs%lig_frac_abg`, `%lig_frac_blg` (from `pldb%res_part_fracs%str_frac` and `%lig_frac`)

Overview

SWAT+ simulates any number of pesticides, listed in the constituent setup and described in `pesticide.pes`. Each pesticide has a sorption coefficient, a solubility, half-lives on foliage, in soil, in water and in benthic sediment, and optional degradation products (metabolites). On land, a pesticide is applied by a management operation, split between plant foliage and the soil, washed from foliage by rain, decayed, taken up by plants, and moved by surface runoff, lateral flow, tile flow, percolation and eroded sediment. Pesticide that reaches channels, reservoirs and aquifers continues to react and move there.

The daily order in an HRU is washoff (on days with at least 2.54 mm of rain), plant uptake, decay, transport in water, then transport with sediment (on days with erosion).

Application

A pesticide application operation names the pesticide, the amount and an application method from `chem_app.ops`. The amount reaching the field is the applied amount multiplied by the application efficiency of the method:

$$P_{app} = P_{op}\,\varepsilon_{app}$$

The fraction intercepted by the canopy depends on the leaf area index of all plants in the HRU:

$$f_{gc} = \frac{1.99532 - \operatorname{erfc}(1.333\,LAI - 2)}{2.1}, \qquad f_{gc} \geq 0$$

Symbol	Meaning	Units	Input
P_{app}	pesticide reaching the field	kg/ha	
P_{op}	amount in the operation	kg/ha	application operation
ε_{app}	application efficiency	fraction	`app_eff` in `chem_app.ops`
f_{gc}	fraction intercepted by foliage	fraction	
LAI	sum of leaf area index of all plants	m^2/m^2	

The intercepted part, $f_{gc}P_{app}$, is divided among the plants in proportion to their leaf area. The rest reaches the soil: the surface fraction `surf_frac` of the application method goes to the top soil layer (10 mm) and the remainder to the second layer.

Washoff

On days with at least 2.54 mm of precipitation, a fixed fraction of the pesticide on each plant's foliage is washed to the top soil layer:

$$P_{wash} = f_{wash}\,P_{fol}$$

where f_{wash} is the washoff fraction `frac_wash` of the pesticide and P_{fol} the pesticide on the foliage (kg/ha).

Decay

Pesticide on foliage and in each soil layer decays by first-order kinetics. The daily survival factor is computed from the half-life:

$$P_i = P_{i-1} \exp\left(\frac{-0.693}{t_{1/2}}\right)$$

with $t_{1/2}$ the foliar half-life `hl_foliage` for foliage and the soil half-life `hl_soil` for soil (days). A half-life of 0 removes all of the pesticide in one day.

Temperature adjustment

When the temperature switch for pesticide decay in `parameters.bsn` is 1, the half-lives are treated as values at 25 °C and adjusted each day with a Q10 relation:

$$t_{1/2}(T) = t_{1/2,25}\, Q_{10}^{(25-T)/10}, \qquad Q_{10} = 2.58$$

Pool	Half-life used	Temperature T
Soil layer	`hl_soil`	temperature of that layer
Foliage	`hl_foliage`	temperature of the top soil layer
Channel water	`aq_hlife`	channel water temperature (at least 0.1 °C)
Channel benthic sediment	`ben_hlife`	channel water temperature
Aquifer	`aq_hlife`	0 °C; the aquifer temperature is not set in 62.0.1

Each 10 °C below 25 °C multiplies the half-life by 2.58, and each 10 °C above 25 °C divides it by 2.58. Reservoirs always use the unadjusted half-lives.

The temperature switch and the pesticide groundwater mixing factor are read as the 6th and 7th values of `parameters.bsn`. Files written by the SWAT+ Editor label these columns `lin_sed` and `exp_sed`, which older revisions did not use. A file with `exp_sed = 1` therefore turns temperature-adjusted decay on, and the `lin_sed` value becomes the groundwater mixing factor. Check both columns before running pesticides with an older `parameters.bsn`.

Metabolites

When a pesticide has degradation products, a share of the mass that decays is added to each daughter pesticide:

$$\Delta P_d = \Delta P_{parent}\, f_d\, \frac{M_d}{M_{parent}}$$

Symbol	Meaning	Units
ΔP_d	mass added to the daughter	kg/ha
ΔP_{parent}	parent mass decayed on the day	kg/ha
f_d	fraction of the parent that degrades to this daughter, set separately for foliage, soil, water and benthic sediment	fraction
M	molecular weight `mol_wt`	g/mol

Metabolites form on foliage, in soil, in channel water and benthic sediment, and in aquifers.

Plant uptake

Growing plants take up dissolved pesticide with the water they extract from each soil layer. Uptake from a layer is

$$P_{up} = f_{TSCF}\ \frac{w_{up}}{SW_{eff} + w_{up} + K_d\,\rho_b\,z}\ P_{ly}$$

$$SW_{eff} = SW + n\ \max(0,\ FC - SW)$$

Symbol	Meaning	Units	Input
P_{up}	pesticide taken up from the layer	kg/ha	
f_{TSCF}	transpiration stream concentration factor	-	`pl_uptake`
w_{up}	plant water uptake from the layer	mm	
SW, FC	layer soil water and field capacity	mm	
n	resistance blending factor, 0 uses actual soil water and 1 uses field capacity	0 to 1	`resistance_n` (default 1)
K_d	soil partition coefficient, $K_{oc} \cdot OC/100$	mL/g	`soil_ads`
ρ_b, z	bulk density and layer thickness	Mg/m³, mm	
P_{ly}	pesticide in the layer	kg/ha	

When `harvest_sink` is 1 (default), pesticide taken up is lost from the system. When it is 0, it is stored in the plant, removed with the harvested yield, and returned to the soil surface when the plant is killed.

Transport in water

In each soil layer, pesticide is shared between the soil water and sorbed phases with a linear partition coefficient based on the organic carbon content of the layer:

$$K_d = \frac{K_{oc}\ OC}{100}$$

where K_{oc} is `soil_ads` (mL/g) and OC the organic carbon content of the layer (%). The concentration of pesticide in water leaving the layer is

$$C = \min\left(\frac{P_{ly}\left[1 - \exp\left(\dfrac{-w}{SAT + K_d\,\rho_b\,z}\right)\right]}{w},\ \frac{S_w}{100}\right)$$

Symbol	Meaning	Units
C	pesticide concentration in mobile water	kg/ha per mm
w	water leaving the layer: percolation and lateral flow, plus surface runoff for the top layer and tile flow for the drained layer	mm
SAT	water content of the layer at saturation	mm
S_w	solubility `solub`	mg/L

Mass moved by each flow path is the concentration times the flow:

- surface runoff removes $\beta_{perc}\,C$ times the runoff depth, where β_{perc} is the pesticide percolation coefficient `pest_perc` in `parameters.bsn`;
- lateral flow from the top layer also uses $\beta_{perc}\,C$; lateral flow from deeper layers uses C;
- tile flow and percolation use C. Percolation moves pesticide to the next layer, and out of the bottom layer to the aquifer.

Transport with sediment

On days with erosion, pesticide sorbed to the top soil layer leaves with the sediment yield:

$$P_{sed} = 0.001\,\frac{C_{sorb}\;sed\;\varepsilon_{er}}{A}$$

Symbol	Meaning	Units
P_{sed}	pesticide removed with sediment	kg/ha
C_{sorb}	sorbed concentration in the top layer	g/t
sed	sediment yield	t
ε_{er}	enrichment ratio, `erorgn` in `hydrology.hyd` if above 0.001, otherwise computed from the sediment concentration	-
A	HRU area	ha

Channels

Pesticide entering a channel is split into dissolved and sorbed phases using the channel carbon content and the sediment concentration of the inflow. Over the travel time through the reach (as a fraction of a day, at most 1) the dissolved and sorbed mass in water decays with the aquatic half-life `aq_hlife`. Dissolved pesticide volatilises at rate `aq_volat` and sorbed pesticide settles to the bed at rate `aq_settle` (m/day, divided by flow depth). Bed pesticide is resuspended at rate `aq_resus`, exchanges with the water by diffusion (a mixing velocity derived from the molecular weight and bed porosity), is buried at rate `ben_bury` over the active bed depth `ben_act_dep`, and decays with the benthic half-life `ben_hlife`. Dissolved pesticide above the solubility limit is moved to the bed. The pesticide left in the water leaves the reach in proportion to outflow over outflow plus storage.

Reservoirs follow the same set of processes (decay, volatilisation, settling, resuspension, diffusion and burial) without temperature adjustment. Pesticide reactions in wetlands are not simulated in this revision.

Aquifers

Pesticide reaching an aquifer decays with the aquatic half-life and leaves in baseflow and deep seepage. Its concentration is the aquifer pesticide mass divided by a mixing volume made of baseflow, seepage, revap and a fraction of aquifer storage. That fraction is the pesticide groundwater mixing factor in `parameters.bsn` (default 0, so only the day's outflows dilute the pesticide). See Aquifers.

Parameters

Parameter	File	Effect
`soil_ads`	`pesticide.pes`	Sorption coefficient normalised for organic carbon, K_{oc} (mL/g)
`frac_wash`	`pesticide.pes`	Fraction washed off foliage by rain
`hl_foliage`, `hl_soil`	`pesticide.pes`	Half-lives on foliage and in soil (days)
`solub`	`pesticide.pes`	Water solubility (mg/L)
`aq_hlife`, `ben_hlife`	`pesticide.pes`	Half-lives in water and in benthic sediment (days)

Parameter	File	Effect
aq_volat, aq_resus, aq_settle, ben_bury	pesticide.pes	Channel volatilisation, resuspension, settling and burial velocities (m/day)
ben_act_dep	pesticide.pes	Depth of the active benthic layer (m)
mol_wt	pesticide.pes	Molecular weight (g/mol)
pl_uptake, harvest_sink, resistance_n	pesticide.pes	Plant uptake factor and its fate in the plant
app_eff, surf_frac	chem_app.ops	Application efficiency and share placed in the top 10 mm
pest_perc	parameters.bsn	Pesticide concentration in runoff relative to soil water
6th and 7th values	parameters.bsn	Groundwater mixing factor and temperature switch (columns labelled lin_sed and exp_sed by the Editor)

Related pages

- pesticide.pes and operation files
- parameters.bsn
- Plant growth and land use management
- Erosion and sediment
- Channels, Reservoirs and wetlands, Aquifers

Sources

Code at revision 62.0.1:

- Application: src/pest_apply.f90, amount and efficiency in src/mgt_sched.f90 and src/actions.f90
- Washoff: src/pest_washp.f90; call order and the 2.54 mm rain threshold: src/hru_control.f90
- Decay and metabolites: src/pest_decay.f90; Q10 function: src/pest_tempdecay_module.f90
- Plant uptake: src/pest_pl_up.f90; harvest and kill: src/mgt_harvtuber.f90, src/mgt_killop.f90
- Transport in water: src/pest_lch.f90; with sediment: src/pest_pesty.f90, src/pest_enrsb.f90
- Database and decay factors: src/pest_parm_read.f90, src/pesticide_data_module.f90
- Channels: src/ch_rtpest.f90, mixing velocity in src/sd_hydsed_init.f90; reservoirs: src/res_pest.f90; aquifers: src/aqu_1d_control.f90
- Basin parameters: src/basin_module.f90 (basin_parms), read list-directed in src/basin_read_prm.f90

Symbols and variables:

Symbol or column	Variable
f_{gc}	gc
app_eff, surf_frac	chemapp_db%app_eff, chemapp_db%surf_frac
soil_ads, frac_wash, hl_foliage, hl_soil	pestdb%koc, %washoff, %foliar_hlife, %soil_hlife
daily survival factors	pestcp%decay_f, %decay_s, %decay_a, %decay_b
f_d	pestcp%daughter%foliar_fr, %soil_fr, %aq_fr, %ben_fr
f_{TSCF}, n	pestdb%pl_uptake, pestdb%resistance_n
β_{perc}	bsn_prm%percop
temperature switch	bsn_prm%temp_decay
groundwater mixing factor	bsn_prm%pestgwfact
pesticide on foliage, in the plant, in soil	cs_pl%pl_on%pest, cs_pl%pl_in%pest, cs_soil%ly%pest

Overview

SWAT+ can simulate two optional groups of dissolved substances, both listed in constituents.cs:

- Salt ions: the eight major ions sulfate (SO_4), calcium (Ca), magnesium (Mg), sodium (Na), potassium (K), chloride (Cl), carbonate (CO_3) and bicarbonate (HCO_3). In soil and aquifers they take part in mineral precipitation and dissolution and in cation exchange, and soil salinity can reduce plant growth.

- Constituents: selenate (SeO_4), selenite (SeO_3) and boron. Selenium species react in the soil (reduction, release from shale), and all three sorb to soil.

Both groups move with water along the same paths as nitrate: surface runoff, lateral flow, tile flow and percolation in the soil profile, then through aquifers, channels, reservoirs and wetlands. Inputs come from initial conditions, rain and dry deposition, fertiliser, irrigation, urban runoff and, for salts, road salt.

Transport in soil water

Salt ions and constituents are transported in each layer with the nitrate leaching equations, including anion exclusion. For each ion or constituent m in a layer:

$$C_m = \frac{M_m}{w}\left[1 - \exp\left(\frac{-w}{(1-\theta_e)\,SAT}\right)\right]$$

Symbol	Meaning	Units	Input
C_m	concentration in water leaving the layer	kg/ha per mm	
M_m	mass in the layer	kg/ha	
w	percolation plus lateral flow, plus surface runoff for the top layer and tile flow for the drained layer	mm	
θ_e	fraction of porosity from which anions are excluded	fraction	`anion_excl` in `soils.sol`
SAT	water content at saturation	mm	

Percolation, tile flow and lateral flow below the top layer remove mass at concentration C_m. Surface runoff and lateral flow from the top layer remove mass at $\beta_N C_m$, where β_N is the nitrate percolation coefficient `n_perc` in `parameters.bsn` (adjusted per HRU by calibration). Percolation moves mass to the next layer and out of the bottom of the profile to the aquifer. When gwflow is active with solute transport, groundwater discharging into the soil profile also adds mass.

Salt chemistry

In each soil layer and in each aquifer, the dissolved ions are brought into equilibrium with five minerals and with the exchange sites of the soil:

- Precipitation and dissolution of calcite ($CaCO_3$), magnesite ($MgCO_3$), gypsum ($CaSO_4$), epsomite ($MgSO_4$) and halite (NaCl). A mineral dissolves when the ion activity product is below its solubility product and precipitates when above it.
- Activity coefficients from the ionic strength I of the solution: the extended Debye-Hückel equation for $I \leq 0.1$ mol/L and the Davies equation above. For $I \geq 5$ mol/L the Davies equation is evaluated at $I = 0.5$.
- Cation exchange among Ca, Mg, Na and K.

The calculation iterates until concentrations converge. The solubility products are fixed in the code:

Mineral	Solubility product
$CaCO_3$	3.0702×10^{-9}
$MgCO_3$	4.7937×10^{-6}
$CaSO_4$	7.888×10^{-5}
$MgSO_4$	7.244×10^{-3}
NaCl	37.3

Salinity stress on plants

When the stress option in `salt_plants` is on and salts are simulated, each rooted soil layer gets a reduction factor from the FAO crop salt tolerance relation:

$$f_{salt} = \begin{cases} 1 & EC_e \le a \\ \dfrac{100 - b\,(EC_e - a)}{100} & EC_e > a \end{cases}, \qquad 0 \le f_{salt} \le 1$$

$$EC_e = \frac{TDS}{k_{EC}}$$

Symbol	Meaning	Units	Input
EC_e	electrical conductivity of the soil water	dS/m	
TDS	sum of the salt ion concentrations in the soil water, scaled by water content	mg/L	
k_{EC}	TDS to EC conversion factor	(mg/L)/(dS/m)	`salt_plants`
a	salinity threshold of the plant	dS/m	`salt_plants`, per plant
b	yield loss per dS/m above the threshold	%/(dS/m)	`salt_plants`, per plant

For gypsiferous soils (soil type 1 in `salt_plants`) the threshold is raised by 2 dS/m. A plant with $a = 0$ has no salt stress. The smallest factor over the rooted layers is the plant's salt stress.

Salt stress reduces daily biomass growth. With the "after other stresses" option (`salt_effect = 1`), growth is multiplied by the salt stress after taking the smallest of the water, temperature, nitrogen, phosphorus and aeration stresses. With option 2, salt stress is included in that minimum.

Salt uptake by plants

If the file `salt_uptake` is present, each ion is removed from the soil in the amounts it specifies (kg/ha per day), distributed over the layers in proportion to root mass. The first plant of the HRU's plant community is used. Without the file, plants take up no salt.

Selenium reactions

Selenate, selenite and nitrate in each soil layer are solved together each day (fourth-order Runge-Kutta, one-day step). The processes are:

- reduction of selenate to selenite and of selenite to elemental selenium, both first order and slowed by nitrate through the inhibition term $I/(I + C_{NO3})$;
- oxidation of selenium-bearing shale by oxygen and by nitrate, which releases selenate and consumes nitrate, in proportion to the shale fraction of the soil.

Rate constants, the inhibition constant, oxygen content and shale fractions are given per soil group in `cs_reactions`. Boron does not react.

Sorption of constituents

Selenate, selenite and boron are split between soil water and the soil matrix with a linear partition coefficient:

$$M_{sol} = \frac{M_{tot}}{1 + \dfrac{K_d\,\rho_b\,V_{soil}}{V_w}}, \qquad M_{sorb} = M_{tot} - M_{sol}$$

Symbol	Meaning	Units	Input
M_{tot}, M_{sol}, M_{sorb}	total, dissolved and sorbed mass in the layer	kg	
K_d	partition coefficient for each constituent	L/kg	`cs_reactions`
ρ_b	bulk density	Mg/m^3	
V_{soil}, V_w	volumes of soil and of soil water in the layer	m^3	

Aquifers use the same reaction and sorption equations with the aquifer parameters in `cs_reactions`.

Inputs

Source	File
Initial soil and aquifer content	`salt_hru.ini`, `salt_aqu.ini`, `cs_hru.ini`, `cs_aqu.ini`; channels: `salt_channel.ini`, `cs_channel.ini`
Rain and dry deposition	`salt_atmo.cli`
Fertiliser	`salt_fertilizer.frt`, `fertilizer.frt_cs`
Irrigation water	`salt_irrigation`, `cs_irrigation`
Urban runoff	`salt_urban`, `cs_urban`
Road salt (monthly, daily or annual amounts added to the top layer)	`salt_road`
Plant tolerance and stress options	`salt_plants`
Plant uptake	`salt_uptake`, `cs_uptake`
Reaction and sorption parameters	`cs_reactions`
Point sources	recall files (see Recall and point sources)

Aquifers, channels and water bodies

In the 1D aquifer, recharge adds salt and constituent mass; the salt chemistry, constituent reactions and sorption are applied; and baseflow and seepage remove mass at the concentration of the aquifer water (mass divided by groundwater volume). Channels, reservoirs and wetlands mix incoming mass with the stored water and release it with outflow. With gwflow, salts and constituents are transported between grid cells.

Pathogens and heavy metals

`constituents.cs` can also list pathogens and heavy metals.

Pathogens (bacteria and other microorganisms) are followed on foliage and in the top soil layer. Each day their number changes by first-order die-off minus growth, corrected for temperature:

$$N_d = N_{d-1}\,\exp\!\left(-(k_{die} - k_{grow})\,\theta^{T-20}\right)$$

with the rates of `pathogens.pth` for foliage or soil, θ = `t_adj` and T the mean air temperature. Rain of at least 2.54 mm washes a fraction of the pathogens on foliage to the soil; pathogens then leave the top layer with percolation, surface runoff and sediment. In channels they die off over the travel time at `do_stream`, corrected to the water temperature. The pathogen routines of 62.0.1 do not form a complete model: the HRU routines use one `pathogens.pth` row for all pathogens, the runoff step removes all pathogens left in the top layer every day, and the channel die-off does not act on the routed water. `pathogens.pth` lists these points.

Heavy metals are not transformed. They enter the network only from export coefficient and delivery ratio objects and move with the water as conservative loads (see `metals` and Recall and point sources).

Related pages

- Salt inputs and constituent inputs
- Nutrient cycling for the nitrate transport equations
- Plant growth and land use management for plant stresses
- Aquifers, Channels, Reservoirs and wetlands
- Recall and point sources

Sources

Code at revision 62.0.1:

- Pathogens: src/path_ls_process.f90 (die-off, growth, wash-off), src/path_ls_runoff.f90, src/path_ls_swrouting.f90, channels src/ch_rtpath.f90; heavy metals: src/exco_read_hmet.f90, src/dr_read_hmet.f90
- Soil transport: src/salt_lch.f90, src/cs_lch.f90
- Salt chemistry: src/salt_chem_hru.f90, src/salt_chem_soil_single.f90, src/salt_chem_aqu.f90; solubility products in src/salt_data_module.f90
- Salinity stress: src/pl_waterup.f90, src/pl_biomass_gro.f90; options read in src/salt_plant_read.f90
- Salt uptake: src/salt_uptake.f90, src/salt_uptake_read.f90; road salt: src/salt_roadsalt.f90
- Selenium reactions: src/cs_rctn_hru.f90, src/se_reactions_soil.f90; sorption: src/cs_sorb_hru.f90; parameters: src/cs_reactions_read.f90
- Aquifers: src/aqu_1d_control.f90, src/cs_rctn_aqu.f90, src/cs_sorb_aqu.f90

Symbols and variables:

Symbol or setting	Variable
M_m, concentration	cs_soil%ly%salt, %saltc; cs_soil%ly%cs, %csc
θ_e	soil%anion_excl
β_N	hru%nut%nperco
solubility products (soil, aquifer)	Ksp11 to Ksp51, Ksp12 to Ksp52
k_{EC}, stress option, soil type, effect option	salt_tds_ec, salt_tol_sim, salt_soil_type, salt_effect
a, b	salt_stress_a, salt_stress_b
f_{salt}	pcom%plstr%strss
salt uptake switch	salt_uptake_on
selenium rates, inhibition, oxygen, shale	cs_rct_soil%kseo4, %kseo3, %se_ino3, %oxy_soil, %shale, %ko2a, %kno3a
K_d	cs_rct_soil%kd_seo4, %kd_seo3, %kd_born

Overview

Channels move water, sediment and dissolved and particulate constituents from upstream objects to the watershed outlet. A SWAT+ channel is defined by its geometry (width, depth, slope, length, side slope, Manning's n) in hyd-sed-lte.cha, from which a rating curve of depth, area, wetted perimeter, top width, velocity and travel time against flow rate is built at the start of the run.

Each day a channel:

1. adds rain falling on the channel surface to the inflow;
2. checks for overbank flow, traps sediment on the floodplain and fills floodplain wetlands;
3. computes bank and bed erosion;
4. routes the water through channel and floodplain storage;
5. removes transmission losses and evaporation;
6. runs in-stream water quality, pesticides, pathogens, salts and constituents;
7. computes water temperature.

When gwflow is active, the channel also exchanges water with the groundwater cells it crosses (see Aquifers).

Rating curve

The rating curve is computed for a trapezoidal main channel with a floodplain above bankfull. For a given flow rate it gives flow depth, cross-section area, wetted perimeter, hydraulic radius, velocity (Manning's equation), and travel time $TT = L/(3.6\,v)$ in hours, with L the channel length (km) and v the velocity (m/s).

Flow routing

Inflow is added to channel storage on each routing step. The routing method is set by `rte_cha` in `codes.bsn`.

Variable storage (`rte_cha = 0`, default)

$$q_{out} = \min\left(1,\ \frac{2\,\Delta t\,C_{st}}{2\,TT + \Delta t}\right) V_{stor}$$

Symbol	Meaning	Units	Input
q_{out}	outflow during the step	m³	
Δt	routing step	h	
TT	travel time at the previous step's outflow rate	h	
C_{st}	storage coefficient multiplier	-	`scoef` in `parameters.bsn`
V_{stor}	water in channel and floodplain storage	m³	

With a daily time step there is one routing step per day.

Muskingum (`rte_cha = 1`)

$$O_2 = C_1 I_2 + C_2 I_1 + C_3 O_1$$

$$C_1 = \frac{\Delta t - 2KX}{2K(1-X) + \Delta t}, \quad C_2 = \frac{\Delta t + 2KX}{2K(1-X) + \Delta t}, \quad C_3 = \frac{2K(1-X) - \Delta t}{2K(1-X) + \Delta t}$$

C_1 is not allowed to fall below 0, and the three coefficients are scaled to sum to 1. The storage time constant is a weighted mean of the travel times at bankfull depth and at 0.1 × bankfull depth (using the kinematic wave celerity, 5/3 of the velocity):

$$K = \frac{c_1\,K_{bf} + c_2\,K_{0.1bf}}{c_1 + c_2}$$

Symbol	Meaning	Input
I, O	inflow and outflow volumes at the start (1) and end (2) of the step	
K	storage time constant (h)	
X	weighting factor	`msk_x` in `parameters.bsn`
c_1, c_2	weights for bankfull and low flow	`msk_co1`, `msk_co2` in `parameters.bsn`

When Δt exceeds $2K(1-X)$, the step is divided into sub-steps so that the method stays stable.

Floodplain storage

After each routing step, the part of storage above the bankfull channel volume becomes floodplain storage and the rest stays as channel storage. Both drain through the routing equation.

Transmission losses and evaporation

Once a day, after routing, water is removed from channel storage:

$$TL = 24\,K_{ch}\,L\,W, \qquad E_{ch} = \eta\,E_o\,L\,W$$

Symbol	Meaning	Units	Input
TL	transmission loss	m^3	
K_{ch}	effective hydraulic conductivity of the channel bed	mm/h	`k` in `hyd-sed-lte.cha`
L, W	channel length and bankfull width	km, m	`len`, `wd`
E_{ch}	evaporation from the channel	m^3	
η	evaporation coefficient	-	`evap_adj` in `parameters.bsn`
E_o	potential evapotranspiration	mm	

Both are limited to the water in channel storage and use the full bankfull width, whatever the flow. Transmission losses are reported as channel seepage. In SWAT, transmission losses used the wetted perimeter and could recharge the shallow aquifer; in SWAT+ the loss leaves channel storage and is not routed anywhere unless gwflow is used.

Sediment

Overbank flow and floodplain deposition

The daily peak flow is estimated from the mean daily inflow with a peak rate factor that decreases with drainage area:

$$q_{pk} = r_{pk}\left[1 + 2.66\left(\frac{A}{100}\right)^{-0.3}\right]\frac{Q_{day}}{86400}$$

where r_{pk} is `pk_rto` in `sed_nut.cha` (default 1) and A the drainage area (ha). When the peak exceeds 1.5 times the bankfull flow rate, part of the sediment, organic N and particulate P is trapped on the floodplain:

$$\varepsilon_{fp} = \min\left(1,\ \frac{q_{pk}}{\bar{q}}\left[1 - \exp\left(-10^{-4}\,\frac{A_{fp}}{q_{pk}}\right)\right]\right)$$

with $\bar{q}$ the mean daily flow rate and A_{fp} a floodplain area of five channel widths times the channel length (m^2). Organic N and particulate P are trapped with the enrichment ratios `n_dep_enr` and `p_dep_enr`. Overbank water also fills the wetlands of HRUs that lie on the floodplain.

Bank erosion

Banks erode when the reach velocity, raised to account for flow around bends, exceeds a critical velocity that depends on bank cohesion (from clay content), vegetation cover and bulk density:

$$v_{cr} = c_{vcr}\,\log_{10}(2200\,D)\,\sqrt{0.0004\,(f_{bd} + 0.021\,\tau_{coh} + f_{veg})}$$

$$f_{veg} = 4500\,c_{cov}\,e^{-5D}, \qquad \tau_{coh} = \max\left(1000,\ -87.1 + 42.82\,cl - 0.261\,cl^2 + 0.029\,cl^3\right)$$

The bank retreat rate (m/day) follows a logistic function of v/v_{cr}, scaled by channel width and a duration factor that decreases with drainage area, and limited by an exhaustion rate of $0.0002\,W$. Eroded bank mass is retreat × depth × a quarter of the channel length × bulk density. Bank material adds organic N and P at the concentrations

`n_conc` and `p_conc`, with a share `p_bio` of the P as soluble P. A fraction `wash_bed_fr` of the eroded bank material is deposited in the channel; the rest moves downstream.

Symbol	Meaning	Input
D	bankfull depth (m)	`dp`
c_{cov}	bank cover factor	`cov_fact`
cl	bank clay content (%)	`clay`
f_{bd}	bulk density factor, $\max(0.001,\ 39.24\,\rho_b - 1000)$	`dry_bd`
c_{vcr}	critical velocity coefficient	12th value, labelled `eq_slp` in Editor files

Bed erosion

Bed erosion is computed when the velocity exceeds $0.293\sqrt{d_{50}}$ m/s, as $10^{-4}(v/v_{cr})$ m/day of bed lowering. It is reported in the channel morphology output but is not added to the sediment in the flow.

Water quality

In-stream nitrogen, phosphorus, algae, dissolved oxygen and carbonaceous BOD follow the QUAL2E kinetics (Brown and Barnwell, 1987), with rate constants from `nutrients.cha`. The calculation runs once a day for every channel with inflow, on the concentrations of the day's inflow.

Temperature and solution method

Every rate is given for 20 °C and corrected to the water temperature T_w:

$$k_T = k_{20}\,\theta^{\,T_w - 20}, \qquad T_w = 5 + 0.75\,\bar{T}_{air}$$

with T_w at least 0.1 °C and $\bar{T}_{air}$ the day's mean air temperature at the channel's weather station.

Process	θ
Algal growth and respiration; organic N hydrolysis; nitrite oxidation; organic P mineralisation; CBOD decay	1.047
Algal, organic N and organic P settling; CBOD settling; reaeration (also used for ammonium oxidation)	1.024
Benthic P and ammonium sources	1.074

The channel is treated as a mixed reach with a residence time τ, the travel time through the channel in days (at most 1). For each constituent with inflow concentration C_{in}, a constant source or sink M (mg/L/day) and a first-order rate K (1/day), the outflow concentration is

$$C_{out} = C_{in}\,e^{-a} + \frac{C_{in}/\tau + M}{1/\tau - K}\,(1 - e^{-a}), \qquad a = \left(\frac{1}{\tau} - K\right)\Delta t$$

with $\Delta t = 1$ day.

A first-order transformation from one constituent to another (for example organic N to ammonium) is converted into the equivalent constant term, so the mass removed from one constituent is added to the next.

Algae

Algal biomass A (mg/L) is computed from chlorophyll-a with `chla_alg`. The growth rate combines light and nutrient limitation:

$$\mu = \mu_{max}\, f_L\, f_{NP}, \qquad f_N = \frac{N}{N + K_N}, \qquad f_P = \frac{P_{sol}}{P_{sol} + K_P}$$

where f_{NP} is $f_N f_P$, $\min(f_N, f_P)$ or $2/(1/f_N + 1/f_P)$ for `q2e_alg` 1, 2 or 3. The light factor averages light over the water depth d:

$$f_L = 0.92\, \frac{T_L}{24}\, \frac{1}{\lambda\, d}\, \ln\!\left(\frac{K_L + I}{K_L + I\, e^{-\lambda d}}\right), \qquad I = \frac{f_{PAR}\, H_{day}}{T_L}$$

Symbol	Meaning	Input
μ_{max}	Maximum specific growth rate (1/day)	`alg_grow`
ρ	Respiration rate (1/day)	`alg_resp`
σ_1	Settling rate (m/day)	`alg_stl`
K_N, K_P	Half-saturation constants for N and P (mg/L)	`const_n`, `const_p`
K_L	Half-saturation constant for light (MJ/m²/h)	`lt_co` × 0.06
λ	Light extinction coefficient (1/m)	`lt_nonalg`
f_{PAR}	Fraction of solar radiation that is photosynthetically active	`slr_act`
H_{day}	Solar radiation of the day (MJ/m²)	weather
N	Ammonium plus nitrate in the channel storage (mg/L)	
T_L	Hours used as the day length (see the list below)	

The net growth $\mu - \rho$ is the source term for algae and settling at σ_1/d the first-order loss. Biomass is kept between 0 and 5000 mg/L. Algal growth takes up soluble P at `alg_p` per unit of biomass.

Nitrogen, phosphorus, CBOD and oxygen

Constituent	Sources and sinks	Inputs
CBOD	decay at κ_1; settling at κ_3/d	`cbn_bod_co`, `cbn_bod_stl`
Organic N	hydrolysis to ammonium at $2\beta_3$; settling at σ_4/d	`ptln_nh3n`, `ptln_stl`
Ammonium	gain from organic N hydrolysis; oxidation to nitrite at κ_2	`air_rt`
Nitrite	gain from ammonium oxidation; oxidation to nitrate at β_2	`no2n_no3n`
Nitrate	gain from nitrite oxidation	
Organic P	mineralisation to soluble P at β_4; settling at σ_5/d	`ptlp_solp`, `ptlp_stl`
Soluble P	gain $\beta_4 P_{org} + \sigma_2/d_{ch}$; loss to algal growth	`ben_disp`
Dissolved oxygen	reaeration $\kappa_2(O_{sat} - O)$; CBOD decay; α_5 per unit of ammonium and α_6 per unit of nitrite oxidised; a loss of σ_4/d	`air_rt`, `o2_nh3n`, `o2_no2n`

Soluble P is updated directly, $P_{out} = P_{in} + (\beta_4 P_{org} + \sigma_2/d_{ch} - \alpha_2\, \mu\, A)\, \tau$, with d_{ch} the bankfull depth of the channel and α_2 = `alg_p`. The oxygen saturation concentration (mg/L) depends on the water temperature T_K in kelvin:

$$\ln O_{sat} = -139.34410 + \frac{1.575701 \times 10^5}{T_K} - \frac{6.642308 \times 10^7}{T_K^2} + \frac{1.243800 \times 10^{10}}{T_K^3} - \frac{8.621949 \times 10^{11}}{T_K^4}$$

Reaeration uses `air_rt` directly; the stream reaeration formulas of QUAL2E and reaeration over dams are not computed. The inflow oxygen concentration is limited to between 0.001 and 30 mg/L before the calculation.

Behaviour in 62.0.1

Several parts of the in-stream calculation differ from QUAL2E:

- T_L is the day-length threshold for plant dormancy (`dorm_hr` in `parameters.bsn`, or a value of at most 1 hour derived from latitude), not the day length. At latitudes below 20° with `dorm_hr` = 0 it is zero or negative, the light intensity is set to almost zero, and algae do not grow.
- The extinction coefficient never exceeds `lt_nonalg`, so algal self-shading (`alg_shd_l`, `alg_shd_nl`) has no effect.
- Ammonium is oxidised at the reaeration rate `air_rt`; `nh3n_no2n` has no effect. Organic N hydrolysis uses twice `ptln_nh3n`.
- Algal uptake of ammonium and nitrate and algal oxygen production and respiration are computed but not applied.
- Sediment oxygen demand (`ben_bod`) is not taken from dissolved oxygen; the organic N settling term σ_4/d is subtracted instead. The benthic ammonium source (`ben_nh3n`) only enters the inflow concentration used to compute ammonium oxidation.
- The benthic P source is added as `ben_disp` (mg/m²/day) divided by the depth in m, which gives mg/m³ per day, but is added to a concentration in mg/L.

When the 24th value of `codes.bsn` is 1 (the column Editor files label `i_fpwet`; the `wq_cha` column is not used), a simplified exchange replaces the QUAL2E results for nitrate, organic N, soluble P and particulate P. Over the travel time TT (h), soluble N and P are converted to particulate forms:

$$\Delta N = \min\left(NO_3,\ \left[1 - e^{-k_N TT}\right] N_{org}\right), \qquad \Delta P = \min\left(P_{sol},\ \left[1 - e^{-k_P TT}\right] P_{sed}\right)$$

with k_N and k_P from `n_sol_part` and `p_sol_part` in `sed_nut.cha` (default 0.01).

Water temperature

When the channel's weather station has measured temperatures, water temperature is built from:

- the flow-weighted temperature of upstream channels;
- local inflows from the routing units draining to the channel (snowmelt, surface runoff and lateral flow) and groundwater, each given the mean air temperature over its lag period (`sno_lag`, `surf_lag`, `lat_lag`, `gw_lag` days) times a coefficient;
- heat exchange with the air, using dew point, wind speed and optional shading factors, over the travel time.

Parameters are in `temperature.cha`. Without measured air temperature, water temperature is $5 + 0.75\,T_{air}$.

Parameters

Parameter	File	Effect
`rte_cha`	`codes.bsn`	0 = variable storage; 1 = Muskingum
24th value (Editor column `i_fpwet`)	`codes.bsn`	1 = simplified soluble to particulate N and P exchange
`scoef`	`parameters.bsn`	Variable storage coefficient multiplier
`msk_co1`, `msk_co2`, `msk_x`	`parameters.bsn`	Muskingum weights and weighting factor
`evap_adj`	`parameters.bsn`	Channel evaporation coefficient
`wd`, `dp`, `slp`, `len`, `mann`, `side_slp`	`hyd-sed-lte.cha`	Geometry and roughness
`k`	`hyd-sed-lte.cha`	Bed hydraulic conductivity for transmission losses (mm/h)
`cov_fact`, `clay`, `dry_bd`	`hyd-sed-lte.cha`	Bank erosion resistance
`n_conc`, `p_conc`, `p_bio`	`hyd-sed-lte.cha`	N and P content of eroded bank material
`pk_rto`, `n_dep_enr`, `p_dep_enr`, `wash_bed_fr`, `n_sol_part`, `p_sol_part`	`sed_nut.cha` (optional)	Peak rate, floodplain enrichment, channel deposition and simplified nutrient exchange

Related pages

- Channel inputs and legacy channel inputs
- Channel, sediment and nutrient outputs
- Hydrology, Aquifers, Reservoirs and wetlands
- Pesticides, Salts and constituents

Sources

Code at revision 62.0.1:

- Daily channel driver: src/sd_channel_control3.f90
- Routing, transmission loss and evaporation: src/ch_rtmusk.f90
- Rating curve and Muskingum coefficients: src/sd_hydsed_init.f90, src/sd_rating_curve.f90, src/rcurv_interp_flo.f90
- Sediment and floodplain: src/sd_channel_sediment3.f90
- Inputs: src/sd_hydsed_read.f90 (`hyd-sed-lte.cha`, `sed_nut.cha`), src/sd_channel_read.f90 (`channel-lte.cha`), types in src/sd_channel_module.f90
- Water quality: src/ch_watqual4.f90, with the solution in src/wq_semianalyt.f90 and src/wq_k2m.f90 and the temperature correction in src/theta.f90. QUAL2E reference: Brown, L. C., and Barnwell, T. O. (1987), The enhanced stream water quality models QUAL2E and QUAL2E-UNCAS, EPA/600/3-87/007.
- Water quality symbols: μ_{max}, ρ = `ch_nut%mumax`, `%rhoq`; σ_1 to σ_5 = `%rs1` to `%rs5`; κ_1 to κ_4 = `%rk1` to `%rk4`; β_1 to β_4 = `%bc1` to `%bc4`; α_0 to α_6 = `%ai0` to `%ai6`; K_L, K_N, K_P = `%k_l`, `%k_n`, `%k_p`; λ = `%lambda0`; f_{PAR} = `%tfact`; T_L = `wgn_pms(iwgn)%daylth`; τ = `rcurv%ttime / 24`; d = `rcurv%dep`; d_{ch} = `sd_chd(jrch)%chd`.
- Temperature: src/ch_temp.f90, src/ch_read_temp.f90

Columns and variables (`hyd-sed-lte.cha` is read list-directed into `swatdeg_hydsed_data`, so position decides the variable):

Position	Editor column	Variable
3 to 8	`wd`, `dp`, `slp`, `len`, `mann`, `k`	`chw`, `chd`, `chs`, `chl`, `chn`, `chk`
9	`erod_fact`	`bank_exp`
10	`cov_fact`	`cov`
11	`wd_rto`	`sinu` (sinuosity, used for the bend radius)
12	`eq_slp`	`vcr_coef` (critical velocity coefficient)
13 to 16	`d50`, `clay`, `carbon`, `dry_bd`	`d50`, `ch_clay`, `carbon`, `ch_bd`
17	`side_slp`	`chss`
18	`bed_load`	`bankfull_flo`
19 to 23	`fps`, `fpn`, `n_conc`, `p_conc`, `p_bio`	same names

Other symbols: C_{st} = `bsn_prm%scoef`; X, c_1, c_2 = `bsn_prm%msk_x`, `%msk_co1`, `%msk_co2`; η = `bsn_prm%evrch`; r_{pk} = `sd_ch%pk_rto`; `rte_cha` = `bsn_cc%rte`. `codes.bsn` is read list-directed into `basin_control_codes`, so the 24th value (Editor column `i_fpwet`) sets `bsn_cc%qual2e`, and `wq_cha` sets `bsn_cc%wq`, which is not used.

Overview

Water that percolates below the soil profile recharges groundwater. Groundwater returns to the stream network as baseflow, moves up toward the root zone in dry periods ("revap"), and leaks downward out of the shallow system as deep seepage. SWAT+ offers two ways to simulate this:

- 1D aquifer objects (the default): each aquifer is a lumped storage defined in aquifer.aqu and connected to channels, reservoirs or other aquifers through aquifer.con. Recharge, baseflow, revap and seepage are computed once a day from a few parameters.
- gwflow (`gwflow = 1` in codes.bsn): the aquifer is replaced by a grid of cells. Each cell has a groundwater head, and water moves between cells by Darcy's law and exchanges with channels, reservoirs, wetlands, tile drains, canals and pumping wells.

In SWAT, every subbasin had a shallow and a deep aquifer. In SWAT+ an aquifer is an independent spatial object. A watershed can have one aquifer or many, and deep seepage from a shallow aquifer can be routed to a second (deep) aquifer object or allowed to leave the system.

1D aquifer

Recharge

Recharge is the water an aquifer receives from the objects routed to it: percolation from HRUs, routing units or other aquifers, and any channel or reservoir seepage directed to it in the connection files. The daily recharge depth is

$$w_{rchg} = \frac{V_{in}}{10\,A}$$

Symbol	Meaning	Units
w_{rchg}	recharge to the aquifer	mm
V_{in}	volume of water entering the aquifer on the day	m^3
A	aquifer area	ha

SWAT delayed recharge with an exponential lag (`GW_DELAY`). The SWAT+ 1D aquifer has no recharge delay: water leaving the soil profile is added to aquifer storage on the day it arrives. Recharge is added to storage first, and all outflows on the day are then computed from the new storage.

Water table depth

Storage is converted to a water table depth using the specific yield:

$$d_{wt} = d_{bot} - \frac{S}{1000\,S_y}, \qquad d_{wt} \geq 0$$

Symbol	Meaning	Units	Input
d_{wt}	depth from the land surface to the water table	m	initial value `dep_wt`
d_{bot}	depth from the land surface to the bottom of the aquifer	m	`dep_bot`
S	water stored in the aquifer	mm	
S_y	specific yield	m^3/m^3	`spec_yld`

At the start of the simulation storage is set from the initial water table depth, $S = 1000\,(d_{bot} - d_{wt})\,S_y$.

Baseflow

Baseflow (return flow) occurs only while the water table is at or above the threshold depth `flo_min`. It follows the recession equation used in SWAT:

$$Q_{gw,i} = \begin{cases} Q_{gw,i-1}\,e^{-\alpha} + w_{rchg}\,(1 - e^{-\alpha}) & d_{wt} \leq d_{flo} \\ 0 & d_{wt} > d_{flo} \end{cases}$$

Symbol	Meaning	Units	Input
$Q_{gw,i}$	baseflow on day i	mm	initial value `gw_flo`

Symbol	Meaning	Units	Input
α	baseflow recession constant	1/day	`alpha_bf`
d_{flo}	water table depth at or above which baseflow occurs	m	`flo_min`

Baseflow is limited to the water in storage. Because the threshold is a depth, a larger `flo_min` allows baseflow from a deeper water table. In SWAT the equivalent threshold `GWQMN` was a storage in mm.

The baseflow hydrograph goes to every object listed in `aquifer.con` with hydrograph type `tot`, split by the fractions given there. When channels are linked to the aquifer through the aquifer-channel link file, baseflow is spread over the linked channels from the outlet upward. The contributing channel length grows with the ratio of baseflow to `bf_max`, so `bf_max` is the baseflow at which every linked channel receives water.

Deep seepage

A fixed fraction of the day's recharge leaves the aquifer through its bottom:

$$w_{seep} = \beta_{deep}\, w_{rchg}$$

where β_{deep} is `rchg_dp` (fraction). Seepage is limited to the water left in storage after baseflow. It leaves the aquifer as its second hydrograph; routing it in `aquifer.con` with hydrograph type `rhg` sends it to another object, usually a deep aquifer.

Revap

When the water table is shallower than `revap_min`, water is removed from the aquifer in response to evaporative demand:

$$w_{revap} = \begin{cases} \beta_{rev}\, E_o & d_{wt} < d_{rev} \\ 0 & d_{wt} \geq d_{rev} \end{cases}$$

Symbol	Meaning	Units	Input
w_{revap}	revap	mm	
β_{rev}	revap coefficient	-	`revap`
E_o	potential evapotranspiration at the aquifer's weather station	mm	
d_{rev}	water table depth above which revap occurs	m	`revap_min`

Revap is limited to the water left in storage. In SWAT the revap threshold `REVAPMN` was a storage in mm and revap tapered off as storage approached it. In SWAT+ the threshold is a depth and revap is either on or off. Revap is removed from the aquifer and is not added back to the soil profile.

The daily order is recharge, water table depth, baseflow, seepage, then revap. When the aquifer is a point of diversion in the water allocation inputs, withdrawals are taken after recharge and before baseflow.

Nitrate and phosphorus

Nitrate in recharge water is added to the aquifer nitrate pool. Baseflow and seepage remove nitrate at the concentration of the aquifer water, $C_{NO3} = N_{aq}/S$. The pool also decays with half-life `hl_no3n`:

$$N_{aq,i} = N_{aq,i-1}\, \exp\left(\frac{-0.693}{t_{1/2} + 0.1}\right)$$

where $t_{1/2}$ is `hl_no3n` (days). A half-life of 0 removes almost all aquifer nitrate each day.

Soluble phosphorus in baseflow is the volume of water entering the aquifer on the day multiplied by the concentration `sol_p` (mg/L). It is a fixed concentration and does not draw on a phosphorus pool.

Pesticides, salts and constituents

Pesticides entering with recharge decay with the aquatic half-life of the pesticide. The concentration available for transport is the pesticide mass divided by a mixing volume made of baseflow, seepage, revap and a fraction of storage set by the pesticide groundwater mixing factor in `parameters.bsn`. Baseflow and seepage remove pesticide at that concentration. All aquifer pesticide is soluble.

Salt ions and other constituents are mixed through the aquifer water. Their concentration is mass divided by the groundwater volume, and baseflow and seepage remove mass at that concentration. See Salts and constituents.

gwflow grid model

Activation

Setting `gwflow = 1` in `codes.bsn` turns on gwflow. SWAT+ then reads the gwflow input files (`codes.gw`, `zones.gw`, `cells.gw`, `cellcon.gw`, `chancell.gw` and the optional files for tiles, reservoirs, floodplains, canals, pumping and phreatophytes), and the 1D aquifer objects are switched off. If `chancell.gw` is missing, gwflow is switched off and the 1D aquifers are used.

Cell water balance

Each active cell has a head h, a bottom elevation, a land surface elevation, a hydraulic conductivity K, a specific yield S_y, an ET extinction depth and an area A_c. The day is divided into flow time steps of length Δt (`time_step` in `codes.gw`). On each step the change in cell storage is the sum of lateral flows from neighbouring cells and the sources and sinks of the day:

$$\Delta V = \left(\sum_k Q_k + Q_{ss} \right) \Delta t, \qquad h^{new} = h + \frac{\Delta V}{S_y \, A_c}$$

The solution is explicit: all heads are updated from the heads at the start of the step.

Lateral flow between cells

Flow between a cell and each connected neighbour follows Darcy's law:

$$Q_k = K_f \, \frac{h_k - h}{L} \, b_f \, w$$

Symbol	Meaning	Units
Q_k	flow from neighbour k into the cell	m³/day
K_f	face conductivity, the harmonic mean of the two cells' K	m/day
L	distance between the cell centroids	m
b_f	mean saturated thickness of the two cells	m
w	face width, the square root of the smaller cell's area	m

Inactive cells and no-flow boundary cells pass no water. Constant-head boundary cells exchange water with their neighbours but keep their initial head.

Sources and sinks

- Recharge: percolation from each HRU or landscape unit is delayed and passed to the cells it overlaps: $R_i = (1-\delta)\, w_{perc,i} + \delta\, R_{i-1}$, with $\delta = e^{-1/\tau}$ and τ the recharge delay (days) of the cell's zone.
- Groundwater ET: ET demand not met by the soil profile is taken from the cell: all of it when the head is at or above the land surface, none when the head is below the extinction depth, and a linearly varying share in between.
- Channel exchange: each channel cell exchanges water with the stream through the streambed, $Q = K_{bed}\,(\Delta h / b_{bed})\, W\, L_{ch}$, where W is channel width, L_{ch} the channel length in the cell and b_{bed} the bed thickness. When the head is above the channel stage, groundwater discharges to the channel and Δh is head minus stage. When the head is between the bed and the stage, the channel leaks to the aquifer and Δh is stage minus head. When the head is below the bed, leakage uses the flow depth as Δh.
- Other exchanges: reservoirs, ponds, wetlands, floodplains, tile drains, canals, pumping (from water allocation or an external schedule) and phreatophyte transpiration are added when their input files are present.
- Saturation excess: when the head rises above the land surface, the excess water goes to the nearest channel.

gwflow can also transport heat and solutes (nitrate, phosphorus, salt ions and constituents) between cells and can track groundwater travel time. Inputs are described in gwflow inputs and outputs in gwflow outputs.

Parameters

Parameter	File	Effect
`gwflow`	`codes.bsn`	0 = 1D aquifer objects; 1 = gwflow grid model
`gw_flo`	`aquifer.aqu`	Initial baseflow (mm)
`dep_bot`	`aquifer.aqu`	Depth to the aquifer bottom (m)
`dep_wt`	`aquifer.aqu`	Initial water table depth (m)
`no3_n`	`aquifer.aqu`	Initial nitrate-N concentration (mg/L)
`sol_p`	`aquifer.aqu`	Soluble P concentration of baseflow (mg/L)
`bf_max`	`aquifer.aqu`	Baseflow at which all linked channels receive flow (mm)
`alpha_bf`	`aquifer.aqu`	Baseflow recession constant (1/day)
`revap`	`aquifer.aqu`	Revap coefficient
`rchg_dp`	`aquifer.aqu`	Fraction of recharge lost to deep seepage
`spec_yld`	`aquifer.aqu`	Specific yield (m^3/m^3)
`hl_no3n`	`aquifer.aqu`	Half-life of nitrate in the aquifer (days)
`flo_min`	`aquifer.aqu`	Water table depth at or above which baseflow occurs (m)
`revap_min`	`aquifer.aqu`	Water table depth above which revap occurs (m)

Related pages

- `aquifer.aqu` and connection files
- gwflow inputs and gwflow outputs
- Hydrology for percolation out of the soil profile
- Channels for baseflow in the stream network
- Aquifer, reservoir and plant output
- Water allocation for aquifer withdrawals

Sources

Code at revision 62.0.1:

- 1D aquifer daily routine: `src/aqu_1d_control.f90`
- Initialisation of storage, recession and nitrate decay factors: `src/aqu_initial.f90`
- Reader for `aquifer.aqu`: `src/aqu_read.f90`
- Types `aquifer_database`, `aquifer_data_parameters`, `aquifer_dynamic`: `src/aquifer_module.f90`
- Aquifer-channel links: `src/aqu2d_init.f90`, `src/aqu2d_read.f90`

- Hydrograph types `tot` and `rhg`: src/hyd_connect.f90
- gwflow activation: src/basin_read_objs.f90; inputs: src/gwflow_read.f90
- gwflow daily driver and solution: src/gwflow_simulate.f90, src/gwflow_lateral.f90
- gwflow sources and sinks: src/gwflow_rech.f90, src/gwflow_gwet.f90, src/gwflow_channel_exch.f90, src/gwflow_satexcess.f90, src/gwflow_tile.f90, src/gwflow_reservoir.f90, src/gwflow_canal.f90

Symbols and variables:

Symbol or column	Variable
w_{rchg}	`aqu_d%rchrg`
S	`aqu_d%stor`
d_{wt}	`aqu_d%dep_wt`
Q_{gw}	`aqu_d%flo`
w_{seep}	`aqu_d%seep`
w_{revap}	`aqu_d%revap`
N_{aq}	`aqu_d%no3_st`
`gw_flo`, `no3_n`, `sol_p`, `carbon`	`aqudb%flo`, `%no3`, `%minp`, `%cbn`
`alpha_bf`, `revap`, `rchg_dp`, `spec_yld`, `hl_no3n`	`aqudb%alpha`, `%revap_co`, `%seep`, `%spyld`, `%hlife_n`
$e^{-\alpha}$ and the nitrate decay factor	`aqu_prm%alpha_e`, `aqu_prm%nloss`
pesticide groundwater mixing factor	`bsn_prm%pestgwfact`
gwflow h, K, S_y, extinction depth	`gw_state%head`, `%hydc`, `%spyd`, `%exdp`
gwflow Δt, δ	`gw_time_step`, `gw_delay`

Overview

SWAT+ has two kinds of water body:

- Reservoirs are spatial objects in the routing network. They receive water from channels, HRUs, routing units, aquifers or recall objects, store it, and release it by rules written in a decision table. They are defined in reservoir.res, which points to records in hydrology.res, sediment.res, nutrients.res, initial.res and a release table in res_rel.dtl.
- Wetlands belong to an HRU. They store water on the HRU surface (including rice paddies and other impoundments), receive rain, irrigation and the HRU's runoff, and release water by the same kind of decision table. They are defined in wetland.wet and hydrology.wet and assigned to HRUs through the `surf_stor` column of hru-data.hru.

Both use the same release, sediment and nutrient calculations.

Reservoir water balance

A reservoir starts to operate in the year and month given by `yr_op` and `mon_op` in hydrology.res. Before that date, inflow passes through unchanged.

Each day:

1. Inflow is added to storage. If the reservoir is a point of diversion in the water allocation inputs, withdrawals are taken.
2. The release table is evaluated and the release computed (see below). The release is smoothed with lag factors.
3. Rain on the water surface is added, then release, evaporation and seepage are removed, each limited to the water available.
4. The surface area is updated from the new storage.

$$V_i = V_{i-1} + V_{in} + V_{pcp} - V_{out} - V_{evap} - V_{seep}$$

$$V_{pcp} = 10\,R\,A, \qquad V_{evap} = 10\,\eta\,E_o\,A, \qquad V_{seep} = 240\,K_{sat}\,A$$

Symbol	Meaning	Units	Input
V	water stored	m^3	
R	precipitation	mm	
E_o	potential evapotranspiration	mm	
η	evaporation coefficient	-	`evap_co`
K_{sat}	hydraulic conductivity of the reservoir bottom	mm/h	`k`
A	water surface area on the previous day	ha	

When gwflow is active, seepage is computed by gwflow as exchange with the groundwater cells under the reservoir.

Surface area

Surface area is a power function of storage, $A = \beta_1 V^{\beta_2}$. The coefficients are computed from the areas and volumes at the principal and emergency spillways (`area_ps`, `vol_ps`, `area_es`, `vol_es`):

$$\beta_2 = \frac{\log_{10} A_{es} - \log_{10} A_{ps}}{\log_{10} V_{es} - \log_{10} V_{ps}} \ (\leq 0.9), \qquad \beta_1 = \left(\frac{A_{es}}{V_{es}}\right)^{\beta_2}$$

If β_2 would exceed 0.9, it is set to 0.9 and $\beta_1 = (A_{ps}/V_{ps})^{0.9}$.

Release smoothing

The release from the decision table is blended with the previous day's release so that outflow does not jump:

$$Q_i = \alpha\, Q^* + (1 - \alpha)\, Q_{i-1}, \qquad \alpha = e^{-\lambda}$$

where Q^* is the release from the table and λ is the lag for rising releases or the lag for falling releases. The two lags are read from the last two columns of `hydrology.res`, which the SWAT+ Editor labels `shp_co1` and `shp_co2` (formerly area-volume shape coefficients). A lag of 0 gives no smoothing.

Release rules

Each reservoir names a release table. Names that begin with `ctbl_` refer to a conditions table; all others refer to a decision table in `res_rel.dtl`. The decision table conditions are evaluated as described in Decision tables, and every action that fires adds a release. Release options (the `option` of a `release` action):

Option	Release for the day
`rate`	constant × 86400 (constant in m^3/s)
`rate_pct`	constant % of the principal volume
`inflo_rate`	larger of (inflow + second constant × 86400) and constant × 86400
`inflo_frac`	constant × inflow
`ab_emer`	all storage above the emergency volume
`days`	(storage − lower bound) / constant days, where the lower bound is 0, or a fraction (second constant) of the principal or emergency volume
`dyrt`	(storage − lower bound) / constant days + second constant % of the principal volume, with the lower bound 0, the principal or the emergency volume
`dyrt1`	as `dyrt`, with the lower bound taken from a condition of the table
`inflo_targ`	inflow + (storage − target), with the target a fraction of the principal volume taken from a condition
`weir`	discharge over a weir from `weir.res`
`meas`	measured release from a recall file

Option	Release for the day
`natlake`	natural lake outflow, $k_r\, S_{eff}\, (S_{eff}/(S_{max} - S_0))^{1.5}$ with S_0 a fraction of the principal volume and S_{max} the principal volume
`nonirr-h06`, `irr-h06`	Hanasaki et al. (2006) release for non-irrigation and irrigation reservoirs, from the long-term mean monthly inflow and demand
`hydrop`	hydropower release: mean inflow scaled by a storage factor and a seasonal factor

Reservoir sediment

When the suspended sediment concentration is above the equilibrium concentration `nsed`, the excess settles:

$$C_{sed,i} = (C_{sed} - C_{eq})\, e^{-0.184\, d_{50}} + C_{eq}$$

where d_{50} is the median particle size in `sediment.res` (µm). Sediment leaves with the release at the settled concentration. When the concentration is at or below `nsed`, no sediment is released.

Reservoir nutrients

Organic N, particulate P, soluble N (nitrate, ammonium, nitrite) and soluble P are removed by settling. The fraction removed each day is

$$f = \min\left(1,\ (C - C_{min})\, r\, \theta^{T-20}\right)$$

Symbol	Meaning	Input
C	concentration of the pool	mg/L
C_{min}	minimum concentration for settling	`n_min_stl`, `p_min_stl`
r	settling rate; for organic N and particulate P it differs between the mid-year period (`mid_start` to `mid_end` months) and the rest of the year; soluble N and P have one rate each	`mid_n_stl`, `n_stl`, `mid_p_stl`, `p_stl`; positions 8 and 9 for soluble N and P
θ	temperature factor	`theta_n`, `theta_p`
T	air temperature	°C

For soluble N and P the fraction is multiplied by 0.2. After settling, the release takes nutrients in proportion to its share of the water.

Pesticides, salts and constituents in reservoirs are described on the Pesticides and Salts and constituents pages.

Wetlands and paddies

A wetland covers part of its HRU and stores water on the surface. Its principal and emergency volumes are the HRU area times the depths `dp_ps` and `dp_es` in `hydrology.wet`. When a weir is assigned in `weir.res`, the principal volume is the HRU area times the weir height, and the emergency volume is at least 1.2 times that.

Each day:

1. Rain and irrigation are added to the wetland.
2. The wetted area is updated from storage with the volume-depth coefficients of `hydrology.wet`. Rice paddies are treated as flat and cover the whole HRU.
3. Seepage into the soil is $24\, K_{sat}\, A$, with K_{sat} the `k` of `hydrology.wet` (or the conductivity set by a puddling operation). Seepage that the soil cannot take because it is saturated stays in the wetland. Nutrients move into the top soil layer with the seepage.

4. Release follows the decision table, or the weir equation for paddies, and becomes surface runoff of the HRU.
5. Sediment, nutrients, salts and constituents are updated as for reservoirs.

Evaporation from standing water is taken from the wetland first, as part of the HRU soil evaporation, before any water is taken from the soil.

Parameters

Parameter	File	Effect
`yr_op`, `mon_op`	`hydrology.res`	Year and month the reservoir starts operating
`area_ps`, `vol_ps`, `area_es`, `vol_es`	`hydrology.res`	Surface areas (ha) and volumes (ha-m) at the principal and emergency spillways
`k`	`hydrology.res`, `hydrology.wet`	Bottom hydraulic conductivity (mm/h)
`evap_co`	`hydrology.res`	Evaporation coefficient
`shp_co1`, `shp_co2`	`hydrology.res`	Release lags for rising and falling outflow
`nsed`, `d50`	`sediment.res`	Equilibrium sediment concentration (mg/L) and median particle size (µm)
settling rates, θ, minimum concentrations	`nutrients.res`	Nutrient settling
`dp_ps`, `dp_es`, `hru_ps`, `hru_es`	`hydrology.wet`	Wetland depths (mm) and HRU fractions at the principal and emergency levels
release table	`res_rel.dtl`	Release rules

Related pages

- Reservoir inputs and wetland inputs
- Decision tables and the `*.dtl` files
- Channels, Aquifers
- Aquifer, reservoir and plant output

Sources

Code at revision 62.0.1:

- Reservoir daily driver: `src/res_control.f90`
- Release options: `src/res_hydro.f90`; conditions tables: `src/res_rel_conds.f90`; weir release for paddies: `src/res_weir_release.f90`
- Initialisation, area coefficients and lags: `src/res_initial.f90`; reader: `src/res_read.f90`
- Sediment and nutrients: `src/res_sediment.f90`, `src/res_nutrient.f90`, `src/theta.f90`
- Types: `src/reservoir_module.f90`, `src/reservoir_data_module.f90`
- Wetlands: `src/wetland_control.f90`, `src/wet_initial.f90`; evaporation from standing water: `src/et_act.f90`

Columns and variables:

Column	Variable
`yr_op`, `mon_op`	`res_hyd%iyres`, `%mores`
`area_ps`, `vol_ps`, `area_es`, `vol_es`	`res_hyd%psa`, `%pvol`, `%esa`, `%evol`
`evap_co`, `k`	`res_hyd%evrsv`, `%k`
`shp_co1`, `shp_co2`	`res_hyd%br1`, `%br2`, copied to `res_ob%lag_up`, `%lag_down`
β_1, β_2	`res_ob%br1`, `res_ob%br2`
`nsed`, `d50`	`res_sed%nsed`, `%d50`; settling factor `res_prm%sed_stlr_co`
`mid_start`, `mid_end`, `mid_n_stl`, `n_stl`, `mid_p_stl`, `p_stl`	`res_nut%ires1`, `%ires2`, `%nsetlr1`, `%nsetlr2`, `%psetlr1`, `%psetlr2`
`chla_co`, `secchi_co` (positions 8 and 9)	`res_nut%nsolr`, `%psolr` (soluble N and P settling rates)
`theta_n`, `theta_p`, `n_min_stl`, `p_min_stl`	`res_nut%theta_n`, `%theta_p`, `%conc_nmin`, `%conc_pmin`
wetland `dp_ps`, `dp_es`, `k`	`wet_hyd%pdep`, `%edep`, `%k` (copied to `hru%wet_hc`)

Overview

Three object types put water and loads into the routing network without simulating the land surface that produced them, or change how much of a load reaches the next object:

- Recall objects release a time series of flow and loads read from a file: measured inflow at a gauge, a wastewater treatment plant, an industrial discharge, or the output of another model. A negative flow represents a diversion.
- Export coefficient objects release the same daily flow and loads every day of the run.
- Delivery ratios scale the loads leaving the elements of a routing unit, or the loads passing through a delivery ratio object.

Recall, export coefficient and delivery ratio objects are connected to the network through `recall.con`, `exco.con` and `delratio.con`, like any other spatial object.

Recall objects

Input structure

`recall.rec` (the file named in the `recall` line of `file.cio`) lists the recall objects. Each row has an id, a name, a time step code and the name of the data file:

Code	Data file content	Value used each day
0	sub-daily values	the values of the day's time steps
1	daily values	the value of the day
2	monthly values	the value of the month, on every day of that month
3	annual values	the value of the year, on every day of that year
4	none; the file name is the name of an export coefficient record	the export coefficient record

Each data file has a title line, a line with the number of years, a column header, and one row per day, month or year: day of year, month, day of month, year, object type, object name, and then the hydrograph values

```
flo sed orgn sedp no3 solp chla nh3 no2 cbod dox sand silt clay sag lag gravel temp
```

in m^3 for flow, metric tons for sediment and its classes, kg for nutrients, chlorophyll-a, CBOD and oxygen, and °C for temperature. The values are daily amounts, also in monthly and annual files: a monthly value is released on every day of the month, so it must be the daily amount for that month.

Rows before the simulation start year are skipped, and reading stops after the simulation end year.

Daily release

On each day inside the period covered by its file, a recall object sends the stored values for that day, month or year to the objects listed in `recall.con`. Outside that period a daily or monthly recall object releases nothing.

When a daily value has negative flow, the object takes water out of the network (a diversion). Nutrients are then removed in proportion to the diverted volume at the concentration of the source, as are salts and constituents when they are simulated.

Salts, constituents and pesticides

Salt and constituent loads for recall objects are read from separate salt and constituent recall files (see Salts and constituents). Pesticide loads for recall objects are read from `pest.com` when that file is present.

Recall database

recall_db.rec is a separate index that names, for each source, the organic-mineral, pesticide, pathogen, heavy metal, salt and constituent data. Water allocation uses it to set the quality of water from outside sources, treatment plants and water uses, and reservoir release tables use it to find measured releases.

Export coefficient objects

exco.exc lists the export coefficient objects. Each row has a name followed by the names of its records in the organic-mineral (exco_om.exc), pesticide, pathogen, heavy metal and salt files. Each row of exco_om.exc has a name followed by the same hydrograph values as a recall file.

The hydrograph of an export coefficient object is set once at the start of the run and released unchanged every day, so the values in exco_om.exc are daily amounts. An export coefficient can also be an element of a routing unit, in which case the routing unit's delivery ratio is applied to it.

Delivery ratios

Routing unit elements

Each element of a routing unit (an HRU, an HRU-lte, a channel or another routing unit) has a delivery ratio named in the `dlr` column of rout_unit.ele. The ratio multiplies the flow and loads the element contributes to the routing unit outflow.

`dlr` value	Ratio
`calc` or `0`	Flow is delivered in full. Sediment and nutrients are multiplied by $\min\left(1, \sqrt{t_{c,elem}/t_{c,ru}}\right)$, the square root of the ratio of the element's time of concentration to the routing unit's, for HRU and HRU-lte elements. For `sdc` channel and routing unit elements the ratio is 1. Other element types, including legacy channels, take the ratio last calculated for an earlier element (0 if there is none).
`full` or `1`	Everything is delivered in full.
any other value	0 for every variable, so the element delivers only its recharge. This includes the name of a delratio.del record: the names are looked up before delratio.del is read, so none is found.

The details are on Routing and landscape units.

Pesticide loads from routing unit elements are not scaled by the delivery ratio.

Delivery ratio objects

A delivery ratio object placed in the routing network multiplies all of its incoming hydrograph by the ratios of its delratio.del record and passes the result on.

Related pages

- Recall inputs and connection files
- Channels and Reservoirs and wetlands for receiving objects
- Water allocation
- Salts and constituents

Sources

Code at revision 62.0.1:

- Recall objects: src/recall_read.f90 (reads `recall.rec`, the data files and `pest.com`); daily release in the `recall` case of src/command.f90; diversions: src/recall_nut.f90, src/recall_salt.f90, src/recall_cs.f90

- Recall database: src/recalldb_read.f90, type in src/recall_module.f90
- Export coefficients: src/exco_db_read.f90, src/exco_read_om.f90, type in src/exco_module.f90
- Delivery ratios: src/dr_ru.f90, src/ru_read_elements.f90, src/ru_control.f90, src/dr_db_read.f90
- Hydrograph fields: `hyd_output` in src/hydrograph_module.f90

Variables:

Item	Variable
recall time step code, file, first and last year	`recall%typ`, `%filename`, `%start_yr`, `%end_yr`
stored values	`recall%hd(day or month or 1, year)`, sub-daily flow in `recall%hyd_flo`
`exco.exc` row	`exco_db` (`name`, `om_file`, `pest_file`, `path_file`, `hmet_file`, `salts_file`)
`exco_om.exc` row	`exco_om_name`, `exco` (type `hyd_output`)
element delivery ratio	`ru_elem%dr_name`, `ru_elem%dr`

Overview

A decision table is a set of rules of the form "if these conditions are met, do these actions". SWAT+ uses them wherever an operation depends on the state of the model instead of a fixed date: irrigation when plants are water stressed, planting when the soil is warm enough, fertilising only a share of the fields in a land use, releasing reservoir water by storage and season, or changing land use partway through a run.

There are four decision table files. They have the same format and are evaluated the same way.

File	Used for	Evaluated
lum.dtl	Conditional management of HRUs	daily, for each HRU whose management schedule lists the table as an automatic operation
res_rel.dtl	Reservoir and wetland releases	daily, for each reservoir or wetland that names the table as its release rule
scen_lu.dtl	Land use and management change scenarios	daily, for every HRU, for each table listed in `scen_dtl.upd`
flo_con.dtl	Flow control (water allocation, saturated buffers)	when the controlled object is processed

Structure

Each table has conditions, alternatives and actions.

- Conditions (rows) name a state variable, the object it is taken from, and a limit.
- Alternatives (columns) are rules. For each condition, an alternative gives a comparison operator (`<`, `>`, `<=`, `>=`, `=`, `/=`) or `-` when that condition does not matter for the rule.
- Actions (rows) say what to do. For each alternative, an action has an outcome `y` (do it if this rule is true) or `n`.

An example from `refdata/Ames_sub1/lum.dtl` irrigates with a sprinkler when the plant is water stressed, has accumulated more than 20 % of its heat units and has a leaf area index above 0.5, as long as less than 500 mm has been applied in the year:

```
name                  conds      alts      acts
irr_year_irr            4         1         1
var                    obj    obj_num          lim_var           lim_op  lim_const      alt1
irr_year               hru          0             null                -   500.00000         <
w_stress               hru          0             null                -   0.80000           <
phu_plant              hru          0          phu_mat                -     0.20000         >
leaf_area              hru          0          null                   -     0.50000         >
act_typ                obj    obj_num             name           option    const        const2
irrigate               hru          0         sprinkler     sprinkler_ilm    25.00000   20.00000  unli
```

An object number of 0 means the object the table is being evaluated for (the current HRU, reservoir or wetland).

Evaluation

Each time a table is evaluated:

1. Every alternative starts as true.
2. For each condition, the current value of the variable is compared with the limit. An alternative whose operator for that condition is not satisfied becomes false. A `-` leaves the alternative unchanged.
3. Each action is taken if at least one alternative that is still true has `y` for that action.

Any number of alternatives can be true at once, and an action shared by several alternatives is taken once.

Limits

The limit a variable is compared with is built from three columns: `lim_var`, `lim_op` and `lim_const`. For variables measured against a property of the object, `lim_var` names the property and `lim_op` combines it with the constant. For example, for `soil_water` the property can be `wp` (taken as 0), `fc` (profile water at field capacity) or `ul` (profile water at saturation), and `fc * 0.8` is written as `lim_var = fc`, `lim_op = *`, `lim_const = 0.8`. For reservoir volume, the property can be `pvol` or `evol`. For variables compared with a plain number, `lim_var` is `null` and only `lim_const` is used.

Condition variables

Group	Variables
Plant stress and growth	`w_stress`, `n_stress`, `p_stress`, `phu_plant`, `phu_base0`, `plant_gro`, `plant_name_gro`, `biomass`, `leaf_area`, `ground_cov`, `cur_yrs_mat`, `plant_com`
Time	`jday`, `month`, `year_rot`, `year_gro`, `year_cal`, `year_seq`, `year_start`, `days_plant`, `days_harv`, `days_irr`, `days_act`, `day_start`
Weather	`precip_cur`, `precip_next`, `p_pet`
Soil and land	`soil_water`, `soil_temp2`, `soil_carbon`, `p_lab_150`, `slope`, `hyd_soil_group`, `p_factor`, `k_factor`, `tile_drained`, `tillage`, `land_use`, `cal_group`
Water	`irr_year`, `irr_demand`, `irr_pou_dmd`, `tile_flo`, `channel_flo`, `aqu_dep`, `ch_order`
Water bodies	`vol`, `res_inflo`, `wet_depth`, `vol_wet`, `wtow_vol`
Probability	`prob`, `prob_unif`, `prob_unif1`, `prob_unif_lu`

Probabilistic conditions

Probability conditions make actions happen on a random day or on a random share of HRUs:

- `prob` compares a uniform random number with `lim_const`, so the action happens on that fraction of evaluations.
- `prob_unif` picks one random day in the window from day `obj_num` to day `lim_const` of the year, so the action happens once in the window.
- `prob_unif1` spreads the probability evenly over a window of `lim_const` days that starts on 1 January of the calendar year given in `obj_num`.
- `prob_unif_lu` spreads an operation over the HRUs of a land use (named in a `land_use` condition) during the window from day `obj_num` to day `lim_const`, so that the share of HRUs treated grows evenly through the window. This is how an operation such as “fertilise 30 % of corn fields in May” is set up.

Actions

Group	Action types
Crops and soil	`plant`, `harvest`, `kill`, `harvest_kill`, `till`, `fertilize`, `manure`, `fert_future`, `pest_apply`, `graze`, `burn`, `grow_init`, `grow_end`, `puddle`, `herd`
Irrigation and demand	`irrigate`, `irr_demand`, `manure_demand`, `wallo_dmd`, `set_dmd`, `irr_dmd`, `res_irr_dmd`, `fill_res`, `fill_wtow`, `fill_can`, `duty_reset`, `duty_fr`
Counters	`rot_reset`, `harv_reset`, `kill_reset`, `plant_reset`, `irr_reset`, `pheno_reset`
Land use and practices	`lu_change`, `hru_fr_update`, `cn_update`, `p_factor`, `contour`, `stripcrop`, `terrace`, `tile_install`, `septic_install`, `fstrip_install`, `grassww_install`, `grassww_uninstall`, `user_def_bmp`, `snow_change`, `chan_change`
Water control	`tileflo_control`, `tiledep_control`, `impound_on`, `impound_off`, `weir_height`
Releases	`release` with the options described in Reservoirs and wetlands

For management actions, `name` points to a record in the matching database (a plant, fertiliser, tillage implement, harvest operation, irrigation operation and so on), and `option`, `const` and `const2` give the amount or method. The `fp` column names an extra file record or object, depending on the action.

Related pages

- `*.dtl` input files
- Management for management schedules and automatic operations
- Reservoirs and wetlands for release options
- Plant growth and land use management
- Water allocation

Sources

Code at revision 62.0.1:

- Types `conditions_var`, `actions_var`, `decision_table` and the arrays `dtbl_lum`, `dtbl_res`, `dtbl_scen`, `dtbl_flo`: `src/conditional_module.f90`
- Readers: `src/dtbl_lum_read.f90`, `src/dtbl_res_read.f90`, `src/dtbl_scen_read.f90`, `src/dtbl_flocon_read.f90`; scenario activation: `src/cal_cond_read.f90` (`scen_dtl.upd`)
- Condition evaluation: `src/conditions.f90`, `src/cond_real.f90`, `src/cond_integer.f90`
- Actions: `src/actions.f90`; release actions: `src/res_hydro.f90`
- Call sites: automatic operations in `src/hru_control.f90`, scenarios in `src/time_control.f90`, reservoirs in `src/res_control.f90`, wetlands in `src/wetland_control.f90`

Columns and variables:

Column	Variable
`var`, `obj`, `obj_num`, `lim_var`, `lim_op`, `lim_const`	`d_tbl%cond%var`, `%ob`, `%ob_num`, `%lim_var`, `%lim_op`, `%lim_const`
`alt1` …	`d_tbl%alt(cond, alt)`
`act_typ`, `obj`, `obj_num`, `name`, `option`, `const`, `const2`, `fp`	`d_tbl%act%typ`, `%ob`, `%ob_num`, `%name`, `%option`, `%const`, `%const2`, `%file_pointer`
outcomes	`d_tbl%act_outcomes(act, alt)`
alternative still true	`d_tbl%act_hit(alt)`